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Structural Implications of Economic Liberalization on Agriculture and Rural Development

RuralStruc Program – Phase I

January 2007





The RuralStruc Program on the '***Structural Dimensions of Liberalization in Agriculture and Rural Development***' is a joint initiative of the World Bank, the French Cooperation (the French Development Agency, the Ministry of Agriculture and Fisheries, the Ministry of Foreign and European Affairs, the Agricultural Research Centre for International Development (CIRAD) and the International Fund for Agricultural Development). It is managed by the World Bank.

With a duration of three years (2006-2009), its objective is to propose a renewed analysis of the processes of liberalization and economic integration – going beyond the only commercial dimension – and their impacts on agriculture and the rural sector of developing countries. It also has the mandate to bring an updated vision of the situation of rural economies in terms of levels of wealth and diversification. The results obtained will make it possible to improve the dialogue between national and international partners and to provide orientations for the agricultural and rural policy debates.

The Program adopts a comparative approach across seven countries – Mexico, Nicaragua, Morocco, Senegal, Mali, Kenya and Madagascar – which correspond to different stages of the process of liberalization and economic integration. The Program's work is conducted with teams of national experts and researchers. Two phases were implemented: a first phase providing an overview of each country's dynamics (2006-2007), and a second phase comprising sectoral and regional case studies, supported by rural household surveys (2007-2009).

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EXECUTIVE SUMMARY

The colonial period agriculture and rural development policies favoured white settler agriculture. The indigenous populations were confined in reserve lands mainly providing labour to settlers. Land policies articulated during this time favoured the settler run large-scale agriculture with commodity boards instituted to cater for marketing and regulating agricultural production. Towards independence (early 1960s) land subdivision consolidation, registration reallocation and relaxing restrictions on commercial enterprises through the Swynerton plan permitted the entry of indigenous population into hitherto exclusive and lucrative agricultural enterprise. The post-independence economic experience in Kenya can be viewed in three stages: pre-liberalization era (period before 1980), liberalization era (1980-2002) and post-liberalization era (2003 and the period after)

On gaining independence, the new government inherited a colonial economic that emphasized import substitution policy propped with overvalued exchange rates. To achieve macroeconomic stabilization, the government imposed quantitative controls, high tariffs on competing imports, highly overvalued exchange rates, price and wage regulations, taxation of exports and requirement of 'no objection certificates' from domestic producers. The rapid economic growth realized during this time was attributed to the inward oriented policies and stable macroeconomic management. During this period agricultural funding and rural development was imbalanced with resources directed towards high potential areas and marginalization of low potential areas. The effects emanating from the first and second oil shocks and the coffee boom served to destabilize the economic trend.

The introduction of Structural Adjustment Programs (SAPs) in 1980 marked the start of liberalization in Kenya. The aim of SAPs was to shift the economy from an inward orientation to externally oriented policies that subject economic agents to competition. The SAPs required the following

- liberalization of prices and marketing systems
- financial policy reforms
- international trade reforms
- government budget rationalization
- divestiture and privatization
- parastatals reforms, and
- civil service reforms.

The early 1980s were marked by government reluctance to implement the SAPs. By 1985 the shortcomings of import substitution policy were evident leading to enactment of Session paper No. 1 of 1986 on Economic Management for Renewed Growth. This marked the turning point of economic policy from protectionism and import substitution policies to export promotion. This committed the government to movement away from import controls,

gradual reduction in tariffs, tariff harmonization and tariffication of quantitative restrictions. The system of export incentives encouraged included Manufacturing under Bond (MUB), general import duty and VAT exemption scheme, government financed export credit guarantees and a proposal of a fully functioning Preferential Trade Area (PTA)

Kenya was classified as an 'open' economy in 1993 by the World Bank (WB). The first wave of liberalization though not successful culminated into the second wave in which new export promotion strategies were instituted and unsuccessful ones abandoned. The whole period of SAPs implementation occurred at the backdrop of limited democratic space. The government implemented the reforms with little or no consultation from the private sector or stakeholders. The timing and sequencing of the reforms varied and the pace of implementation was uneven resulting into periods of progress and stagnation. The full opening of the economy in 1993 coincided with a rapidly changing political scene, embracing multiparty democracy and reduced external aid. Nevertheless, economic growth continued albeit in a declining manner. In 1995 the country signed onto the World Trade Organization agreement (WTO) in accession to General Agreement on Trade and Tariffs (GATT). Other pacts which followed later were the Common Market for Eastern and Southern Africa (COMESA), African Growth Opportunity Act (AGOA) and the reviewed African Caribbean Pacific-European Union (ACP-EU).

The onset of the liberalization saw the economy grow at a slow pace with real GDP and real agricultural GDP (AgGDP) growth rates declining. Real GDP grew from 32 billion in 1980 to 84 billion in 1990 and 103 billion in 2000. The annual average growth rate was estimated to be 17% and 2.2% in 1980/90 and 1990/00 respectively The country's' real GDP in 2005 was estimated at 1172.1 million Kshs signifying a 5.8% increase from 2004 (CBS, 2006).The real GDP has been on increase since independence.. The later decline in economic growth is attributed to the opening up of the economy which saw the exposure to external competition.

Agriculture and rural development

Economic growth in Kenya is still strongly pegged on agricultural contribution with an estimated 1% growth in agriculture resulting to 1.6% growth in the overall GDP. Despite this role, over the last two decades agricultural contribution to overall economic growth has declined from 34.18% in 1980, 28.38% in 1992 and finally 26.4% in 2003. The sector declined from a growth of 4.4% in 1996 to 1.5% in 1999 to a negative of 2.4% in 2002 but showed a minimal growth of 0.7% in 2003.

Agricultural and rural development cannot be divorced. Agriculture is the major livelihood source to the rural households in Kenya. About 80% of Kenya's population live in the rural areas and derive their livelihood from agriculture through crop and livestock production, forestry and exploitation of other natural resources. This population constitute over 70% of the Kenyan population living below poverty line. The decline in agricultural growth coincides with economic liberalization that opened the economy agricultural imports and decline in governmental support to agriculture. Reduced government support led to increase in production costs which reduced the country's comparative advantage by making the country's agricultural products relatively more expensive.

The effects of liberalization have been varied on different sub-sectors and economic agents. While some sectors of the economy have been casualties others have managed to perform

well under similar economic conditions. Sub-sectors such as coffee and sugar have retrogressed under liberalization with declining crop area, output, productivities and low world prices. Despite the general increase in area under coffee, production fell from a high of 128.3 thousand tonnes in 1988 to a low of 45.2 thousand tonnes in 2005 while unroasted coffee exports declined from 126,498 tonnes in 1986 to as low as 50,951 tonnes in 2005. The poor trend in coffee exports is owed to the decline in world coffee prices. In contrast tea and horticulture have had an increase in area, output, productivity, and international returns during this period. The sector export value was estimated at 42.29 billion in 2005 up from 22.8 billion in 1997. The area and output in tea had also increased significantly. Tea sector development is mainly attributed to the KTDA which has been in the forefront of fostering small-holder tea growth and the efforts of TRF which has fostered research on high yielding tea varieties. On the other hand the development of the private sector dominated horticultural sector is attributed to little government interference. The sector exports were estimated at 13.75 billion in 1997 and 44.56 in 2005. Despite the setting of quality and safety standards in the EU market, the sector has performed well.

Informal (off-farm) sector

The exit option for the declining agricultural performance over the period has been the informal sector. The sector has developed since 1992 and has employed an estimated 6.12 million directly and indirectly. This accounts for nearly a third of countries labour force. Most of the sector is rural based and with trade, service provision and manufacturing being significant sub-sectors. The springing up of the informal sector is largely attributed to the liberalisation of the communications sector and the cotton industry. The AGOA pact although attributed for the cotton industry losses, is also recognised as having created many jobs through sale of imported second hand clothes. Communications sector liberalisation opened the gates to the new communication technologies and service delivery.

Migration, urban concentration and rural depopulation

Both domestic and international migrations have taken place. Domestic migration has been mostly witnessed as rural-urban migration. Urban centres have increased simultaneously with the increase in urban populations. Urban population reached an estimated 5.4 million in 1999 up from 0.7 million in 1962. Urban centres increased from 34 in 1980 to 380 in 1999. The rate of rural-urban migration when taken as the rate of growth of annual successive urban population, sometimes surpassed the rural population growth rate hence resulting to rural depopulation and urban concentration. The rural urban migration has deprived the rural areas of agricultural labour force hence leaving rural areas with less skilled labour as it is hypothesized that the educated, skilled and productive are more inclined to migration. However, currently rural-urban migration trend is negative indicating that, either urban households are migrating back to the rural areas or the rural households have stopped migration to the urban areas. International migration declined since pre-independence period to the early 1990 when it was triggered by political agitation for multi-party system. This trend persisted to mid 1990s when it reversed. In 2005, estimates of international migrations indicate that majority of the Kenyans are in UK which accounted for 33%, and surprisingly Tanzania which accounted for 26% ahead of USA and the Southern America countries which accounted for 19%. The reasons of this observed international migration pattern is probably pegged on the stringency of entry rules and the anticipated job or educational opportunities.

Remittances

It is evident that migrants do not sever their relationship with rural areas. The transfer of funds between urban and rural inhabitants has been enhanced by reforms in the financial and communications services. Although, little information on the amounts remitted and it's specific uses is available, evidence exists to show the increasing importance of remittances in agriculture and rural development. Currently the Kenyan government has started to recognize the significance of remittances from abroad and is assessing the possibility of these remittances being substitute to Foreign Direct Investment (FDI) by creating an investment environment. While there may be limitation to international data, domestic remittances have been shown to increase by 1.98% from 1997to 2004

Trade

Kenya has been importing more than its exports even in the import substitution period. The trend of imports and exports became more pronounced after the economy was opened up and the reforms had taken place. Exchange rate fluctuations also became more pronounced and dictated the import-export trends. Trade liberalization led to increase in trade volume with higher imports than exports. The cover ratio declined over the period indicating increasing imports. On the same trend agricultural imports are showing increase relative to exports indicating the declining competitive strength of agricultural production. The extent of these effects to the country's comparative advantage however remains unknown.

Poverty and inequality

Under the economic reforms regime poverty prevalence has increased. Poverty increased from 48.8% in 1981/82 to 59.56% 2000 in within the rural areas, 29.3 in 1992 to 51.48 in 2000 and nationally it increased from 46.8 in 1981/82 to 56.78 in 2000. Economic reforms are clearly linked to poverty under the assumption that trade reforms reduce poverty and enhance human development through fostered economic growth. Trade reforms influence the enterprises, distribution channels and government expenditure thereby affecting poor households. The poor in Kenya are mainly found in rural and marginal areas where agriculture is the main livelihood source. Arable land resource is declining owing to the exploding population, labour force migration is leaving the rural areas depopulated and without human capital resource. Environmental degradation is pushing resources further away from the poor households reach trapping them in the environmental-poverty convolution. Decline in productivity occurred at the milieu of increasing population translating into increasing unemployment and consequently increasing poverty levels. Reduced revenue in agricultural sector especially in sugar and coffee planting areas has resulted into loss of livelihoods. It is evident that between 1997 and 2004 household income declined by 6%. The disparity in income changes witnessed in various agro-ecological zones is an indicator of the extent of income inequality. Further, the inequality gap has widened over the years as indicated by Gini coefficient of 0.52 in 1997 and 0.55 in 2000

Structure and evolution of agricultural marketing chains

Agricultural marketing in Kenya was dominated by boards which were responsible for controlling production and marketing of the output. Some of the boards were: Kenya Dairy Board (KDB) |Formed 1958, Maize and Produce Board 1950, later evolved to the National

Cereals and Produce Board, Tea Board 1951 and the Coffee Marketing Board 1946. Upon liberalization the roles and functions of these boards were refocused to make them more efficient and competitive in service delivery. The liberalization of the market saw the entry and exit of marketing agents with concentration and integration along the chain taking place. The dairy sector, maize, coffee and tea sub-sectors witnessed the entry of new market agents while those incompetent withdrew from the market. The boards partly maintained regulatory roles mainly to maintain standards and quality of produce. There was also emergence of alternative marketing chains especially in maize and dairy were it is estimated that these alternative chains control a significant share of the market. For instance, in dairy it is estimated that the informal chain control over 80% of the marketed milk.

Input supply chains such as those of maize seed and fertilizers were also liberalized. Increased use of these inputs is attributed to the extensive availability which was achieved after freeing the market.

Segmentation in production structures

Agricultural production in Kenya is characteristically regional. The country is segmented on the basis of climatic conditions each supporting a major agricultural production. Development segmentation was started during the colonial. This was perpetuated by post independence governments by that emphasised the development of the high potential regions at the expense of the rural arid areas. Economic reforms initiated in the early 1980s further segmented even the high potential areas triggering within region segmentation. Today, segmentation can be viewed in terms of agricultural production that is dictated by climatic conditions and public investment in infrastructue. Economic reforms has had different effects in different regions depending on the type of agricultural production in the area and resulting in economic imbalances, poverty and inequity.

Liberalization appears to have reinforced differential development that was initiated during the colonial period through classification of regions on the basis of crop potential and subsequent resource allocation. Liberalization effects on large and small-scale farmers does not present a clear picture but can be discerned more clearly along commodity lines.

While this study has identified structural changes that have taken place in the rural setting under the pre-liberalization, liberalization and post-liberalization eras, the analytical level cannot ascribe liberalization per se. Whereas it is feasible to match policy changes with changes in economic variables, the causality remains an empirical issue.

Manifestations of economic reforms and reshaping of the rural economies

While the vulnerable groups are these losing their livelihoods due to economic reforms, they are distributed within the country irrespective of the regions production potential. These are characterised by declining agricultural revenues that leads to declining education and health levels, declining food security and increasing poverty. It is shown that different regions exhibit varying levels of poverty that is consistent with the loss in agricultural income of the main agricultural products especially export crops.

The government has initiated several programs and policies to reverse these trends. These efforts include The Poverty Reduction Strategy Paper 2001 (PRSP) and the Economic Recovery Strategy for Wealth and Employment Creation (2003) that spells out strategies for rural economy reconstruction focusing on sectors that have a direct impact on the rural economy through industrialization, employment creation and decentralization of social infrastructure and services. Other strategies include The Ministry of Livestock and Fisheries Development Strategic Plan 2003 –2007, Strategy for Revitalizing Agriculture, 2004 – 2014, The National Development Plan (NDP), 2002-2008, and the Ministry of Agriculture Strategic Plan (2006-2010) that all seek to address the ailing agricultural sector.

This study was carried to elicit the structural implications of economic liberalization on agriculture and rural development. Several outcomes and challenges were identified aside from tracing agricultural policy evolution.

While economic growth declined, population increased over the reform period. Agricultural performance deteriorated and similarly did the household incomes as population increase continued to exert pressure on natural resources. Increasing urbanisation, internal and external migrations and the corresponding rural depopulation and human capital depletion pose new challenges.

Liberalization of markets has stimulated the interest and encouraged entry of multiple players in the many channels. Whereas this is seen as enhancing market efficiency, the problem of enforcing food safety and maintenance of quality standards is creeping into the system. Exit of players from the channels has led to concentration in the market and requires the development of contract farming and adherence to business ethics. Structures for competitive and ethical behaviour of economic agents have to be developed.

Segmentation initiated during the settler period has been perpetuated by the reforms. This can be witnessed in view of agricultural production that is shaped by regional climatic conditions and public infrastructure investments. New livelihood strategies, especially the non-farm activities have emerged due to declining farm revenue emanating from declining land and labour productivity, farm sizes and arable land per capita.

Vulnerable groups are distributed all over the country and these livelihoods are at stake due to the economic changes. Loss of these livelihoods is perpetuated by decline in agricultural income triggered by increasing input prices and declining per unit output. This has forced rural households to venture into alternative agricultural and non-agricultural enterprises. Rural economies are currently reshaping under the thrust of migration, remittances and the development of non-farm activities with the new configuration being dictated by government expenditure on agriculture and investment choices for domestic and international remittances.

INTRODUCTION

The evolution of economic policies in Kenya dates back to the colonial era (period preceding 1954) that was characterized by settler targeted government assistance and exclusion of indigenous populations from commercial agriculture. This was followed by the period spanning 1955-62 under which the Swynnerton Plan was implemented. This plan involved the first land reforms and opened up commercial farming to the indigenous Kenyan farmers. The period 1963 to 1982 or the early independence period saw the emergence of small-holder farmers and heavy government involvement resulting in the rapid growth of agriculture and rural sector. The liberalization era come into being in 1982 and continued to 2002. This was characterized by government withdrawal from service provision, deregulation of markets and increased private sector involvement. The post liberalization era began in 2003 under which the Economic Recovery Strategy for Wealth and Employment Creation (ERS-WEC), Strategy for Revitalizing Agriculture (SRA) are being implemented

The post-independence economic experience in Kenya can be categorised into three distinct stages:

- The pre-liberalization era (period before 1980)
- The liberalization era (1980-2002)
- The post-liberalization era (2003 and the period after)

The liberalization era can further be viewed in two stages: the early 80's to early 90's when the country's was in transition (transition period) and the period 1992 to 2002 when the country was viewed as having liberalised economy (liberalized economy). The transition period was typified by slow economic growth and rising poverty. The turbulent transition period was characterized by several events which either threatened or facilitated the course of liberalization. Among these were the introduction of the Structural Adjustment Programs (SAPs), the attempted 1982 coup, the world oil price drop in 1983, the 1984 drought and the mini-coffee boom of 1986. The government was also faced with the major challenge of capital formation to lay the foundation for structures that would support liberalization.

Trade liberalization was initiated through the Structural Adjustment Programs of the transition period. These consisted of a set of policies designed to make production and resource allocation more efficient. The broad aim was to shift the economy from the inward oriented import substitution policy to a more open outward orientated policy which would encourage exports and imports of goods and services. Another aim of the SAPs was to encourage privatization, public sector reforms and the support of the private sector. The beginning of SAPs in the country was marked by the signing of the Structural Adjustment Loan (SAL) in 1980. The emphasis of SAPs was on the promotion of the non traditional exports and market liberalization.

At the same time government embarked on vigorous export promotion strategy so as to increase returns from exports. This entailed reduction and restructuring of tariff, abolition of export duties, introduction of export retention schemes and Export Credit Guarantee Corporation. Incentives such as the Manufacture under Bond (MUB), green Channel and export compensation were also introduced

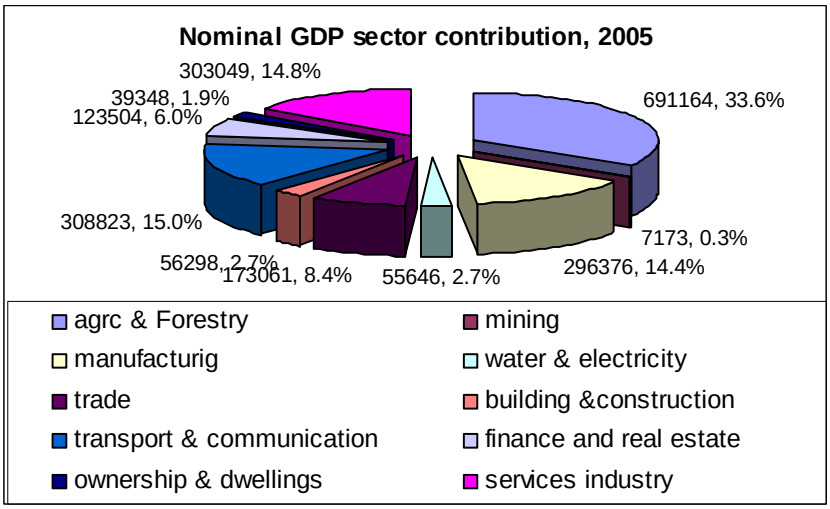
PART I -

PLACE AND ROLE OF AGRICULTURE IN THE NATIONAL CONFIGURATION

CHAPTER 1 - HISTORICAL BENCH MARKS: CHRONOLOGY OF EVENTS

Agriculture plays a pivotal role in the Kenyan economy contributing slightly over a quarter of overall GDP (Strategic Plan 2006-2010). The sector is currently estimated to employ more than 70% of the national labour force, and contribute more than 60% of total export earnings, 45% of the government revenue and provided 75% of the raw materials in the agro-based industry (Strategic Plan 2006-2010). In 2005, the sector recorded a growth of 6.9% in gross value added from 1.7% which was attributed to improved performance of the horticulture, dairy and cereals sub-sectors (CBS, 2006). The sector's economic importance has been declining over the period of economic reforms. The sector further contributes about 57% of national income both directly and indirectly (Strategic Plan 2006-2010). Agriculture is therefore currently still the most important sector in fostering the national economic growth. It is estimated that a 1% increase in the agricultural sector results into a 1.6% GDP growth in the overall economy (Strategic Plan 2006-2010). In 2005 the sector still dominated by contributing 34% to the GDP if combined with fisheries and forestry (Figure 1)

Figure 1: Nominal Sector contribution, 2005



Source: CBS Economic Survey 2006

1. Political Changes in Kenya

Kenya was a British colony prior to its independence. The struggle for independence mainly was characterised by the 1952 Mau Mau uprising that was essentially a resource conflict between the colonial government and the Africans with repression fuelling the struggle. There was formation of political and tribal outfits to help in the quest for freedom. Despite the oppression of the various personalities and sanctioning of the Mau Mau activities, Kenya gained independence in 1963. The immediate post-independence period was characterised by a single party system which articulated government's functions. The period staring 1980 saw the beginning of political changes with increased unrest that culminated in a failed coup in 1982. This period saw increased pressure on the government to open up the economy and political space. The reluctance of the government to implement economic reforms saw declining economic growth in this decade and by the late 1980s the agitation for multi-party system was intensified. The reluctant adoption of multi-party political system in 1992 came with little political and economic impact. Simultaneous political changes and economic reforms had a major impact on the economy resulting in drastic currency depreciation and high inflation levels.

2. Evolution of Agricultural Policies in Kenya

The pre-independence agricultural evolution can be traced back to the Second World War and categorized in three phases according to Brown,1960. These are; i) the recovery phase (1945-50) under which the main efforts were directed to repairing the damage caused by the policy of maximum production to cater for the second world war, ii) a planning phase (1951-1955) which laid plans for the long-term agricultural development. At the time, African farm development was encouraged and backed with considerable public spending on infrastructure and extension (Scott, et.al., 1995). iii) the rapid development phase (1955-60), in which production in both European and African farm lands was greatly accelerated. During the entire period (1945-1960), marketing boards were established to control production and marketing of agricultural products. Some of the boards were: Kenya Dairy Board (KDB) |Formed 1958, Maize and Produce Board 1950, later evolved to the National Cereals and Produce Board, Tea Board 1951 and the Coffee Marketing Board 1946. These and other marketing boards controlled commodity marketing and kept producer prices high. These boards later became an important source of fiscal imbalance and rent seeking. For instance, by 1987, the National Cereal and Produce Board had accumulated debt equivalent to 5 percent of GDP.

Post-independence agricultural policy focused on import substitution with an emphasis on strengthening small-scale farming based on a continuation of the marketing boards approach. Land reallocation and sub-division formed the core of the strategy. These policies quickly delivered increased agricultural output and contributed to overall economic growth.

The pursuit of liberalized market policies reduced government involvement in agriculture. Service provision started earnestly in 1980, but real commitment to implementation was realized by 1993. The withdrawal trend over the period was gradual and maintained in the current. The post liberalization policies articulated in the Strategy for Revitalizing Agriculture (SRA) of 2004, and the Economic Recovery Strategy for Wealth and Employment Creation (ERS-WEC) of 2003 have emphasised the improvement of agriculture through increased availability of working capital in the rural areas and commercial orientation.

CHAPTER 2 - MAIN AGGREGATES AND EVOLUTION

1. Demographic Characteristics

1.1. Population Growth

Kenya's population reached an estimated 33.4 million in 2005 (CBS, 2006) from 5.5 million in 1948. Table 1 presents the population trends from the census conducted in different periods. The population growth rate in 1948/62 was estimated as 3.2 increased to 3.4 in the subsequent census periods but declined to 2.9 during 1989/99 period. In the year 2004/05 the annual population growth rate was estimated to be 1.8 which was slightly lower than the previous 2003/04 1.9.

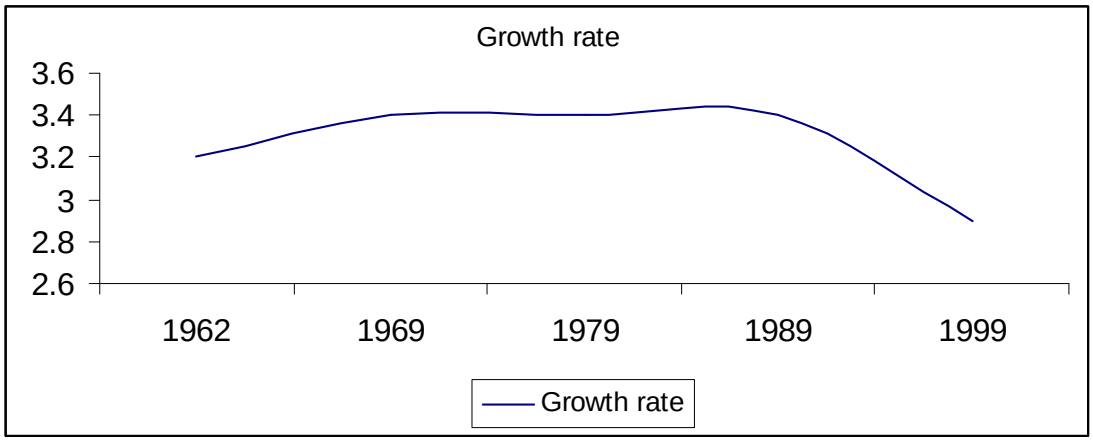
Table 1: Population Trends in Kenya 1948-1999

Census	Population	% increase	Growth rate
1948	5,497,599		
1962	8,636,263	57.1	3.2
1969	10,956,501	26.9	3.4
1979	15327,061	39.9	3.4
1989	21,448,774	39.9	3.4
1999	28,660,534	33.6	2.9

Source: Kenya 1999 Population and Census: Analytical Report on Population Dynamics Vol.III.

The decline in population can be attributed to the progressive government family planning programs aimed at curtailing population increase. There is substantive evidence increase in contraceptive use accelerated towards the late 1980s. The use of any method of contraception increased from 5.6% in 1978/78 to 27% in 1988/89, and then to 29.9% in 1998. The mean ideal family size over the whole period declined from 6.2 in 1978 to 3.8 in 1998 (CBS, 2002). The use increased almost fourfold within a period of 12 years and continued throughout the 1990s. Though there are discrepancies in the national population statistics and variation with FAO statistics, a consistent picture emerges that population growth has been declining. Whereas the absolute population total may be growing, growth rate has been on a declining phase (Figure 2).

Figure 2: Inter-censal Population Growth Rates

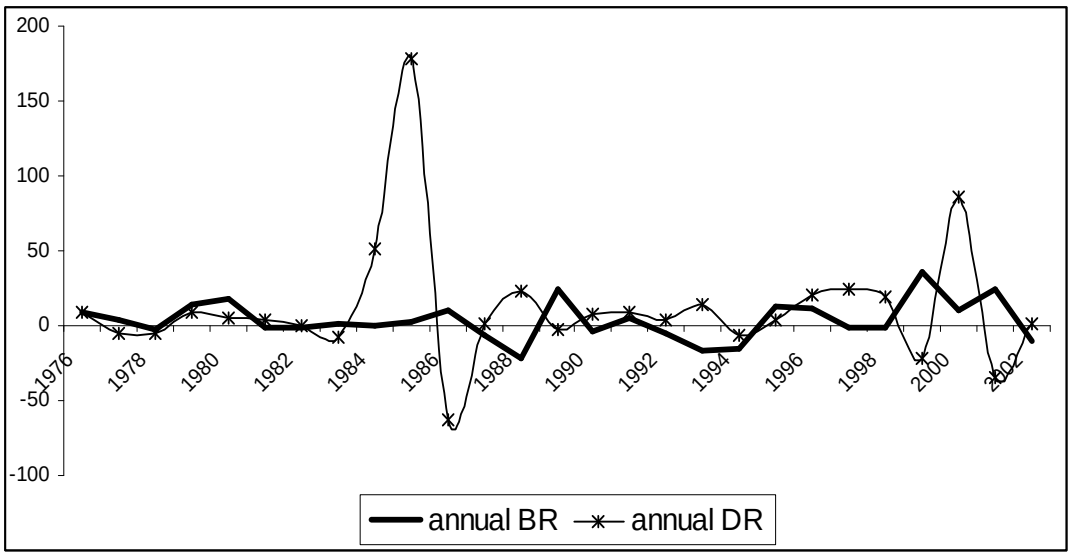


Source: Source: Kenya 1999 Population and Census: Analytical Report on Population Dynamics Vol. III

1.2. Demographic Transition

The trends of both the number of persons reported dead and the number of children born in Kenya has been on the increase with the number of births being higher than deaths. The slight decline in the number of births as from mid 1980s to 1990s could have been because of the family planning campaigns adopted by the government to curb population growth. During the same period the number of reported deaths increased particularly due to prevalence of HIV/AIDS (figure 3). The annual average death rates were estimated to be 10.7 percent surpassing birth rates which were estimated at an annual average of 4.8 percent between 1993/2002

Figure 3: Annual Birth and Death Rates in Kenya 1976-2002



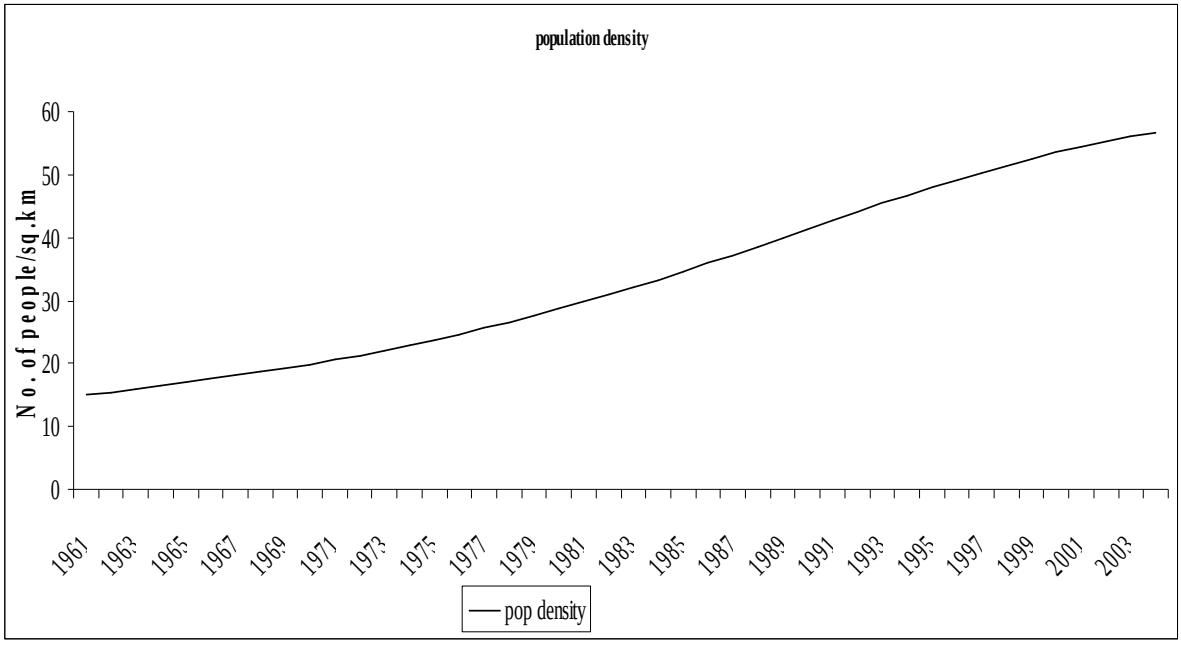
Source: Statistical Abstracts (Various Issues)

1.2.1. Population Density

The countries population density (No of people per sq. km) was 55.4 in 2003 56.4 in 2004 and 57.5 in 2005. The trend has been on a near constant increasing change over the period (Figure 4) There is however, a wide variation within the country in population densities with a density of above 230 persons per sq. km in high potential and as low as 3 persons per sq. km in arid zones (GoK, 1997-2001).

The country's total population can be categorized in three ways: i) agricultural and non-agricultural population; ii) urban and rural population and; iii) economically active population and non-economically active population. The latter categorization can further be separated to proportions of economically active in agriculture and non-agricultural economically active populations. Whereas total economically active defines the total labour force of a country or region, the economically active in agriculture defines the labour force in agriculture.

Figure 4: Population Density



Source: CBS Economic Surveys (various Issues)

Table 2: Total Urban Populations by Province in Kenya 1962-1999

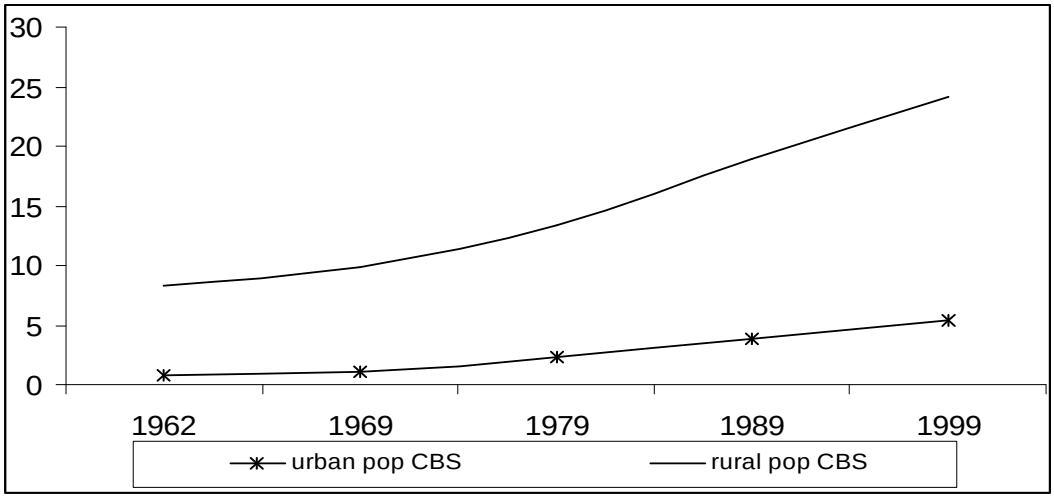
Province	Total Urban Population '000					Share % of urban pop.		Inter-censal Growth rate (%)	
	1962	1969	1979	1989	1999	1989	1999	1979-1989	1989-1999
Nairobi	343.5	506.3	827.8	1324.6	2087.7	34.1	38.9	4.7	4.5
Central	35.0	46.0	128.9	309.8	354.0	8.0	6.6	8.8	1.3
Coast	195.8	283.7	407.0	588.5	894.3	15.2	16.7	3.7	4.2
Eastern	28.7	38.0	233.3	354.4	265.3	9.1	5.0	4.2	2.9
Nyanza	28.1	43.8	207.8	352.5	423.2	9.1	7.9	5.3	1.8
R. Valley	112.5	148.6	341.7	672.2	940.3	17.3	17.5	6.8	3.4
Western	3.9	10.6	105.7	186.0	270.5	4.8	5.1	5.6	3.7
N. Eastern	-	63.5	-	90.7	125.6	2.3	2.3	3.6	3.3
Total urban pop (Millions)	0.7	1.0	2.3	3.9	5.4	100	100	6.7	3.8

Source: CBS Economic Survey, 2002

Increased rural-urban migration is associated with increased job opportunities, availability and access to better social amenities and infrastructure in urban areas. Urban areas in Kenya are defined as those areas with a concentration of people exceeding 2,000 (CBS, 2002). In the year 2004 the urban population in Kenya was estimated to be 13.2 million people (FAOSTAT) up from 0.7 million people in 1962 (CBS, 2002). Between 1962 and 1989, the number of urban centres increased from 34 to 134 respectively. However in 1999 the number of urban centres was categorised by boundaries, that is, whether core urban or local authority boundaries. On the basis of this categorisation there were 179 under core urban boundary and 201 under local authority boundary. Table 2 and Figure 5 present the urban population trends in each province within the country that has been on the increase in absolute terms in all provinces since 1962. However the share of urban population declined between 1989 and

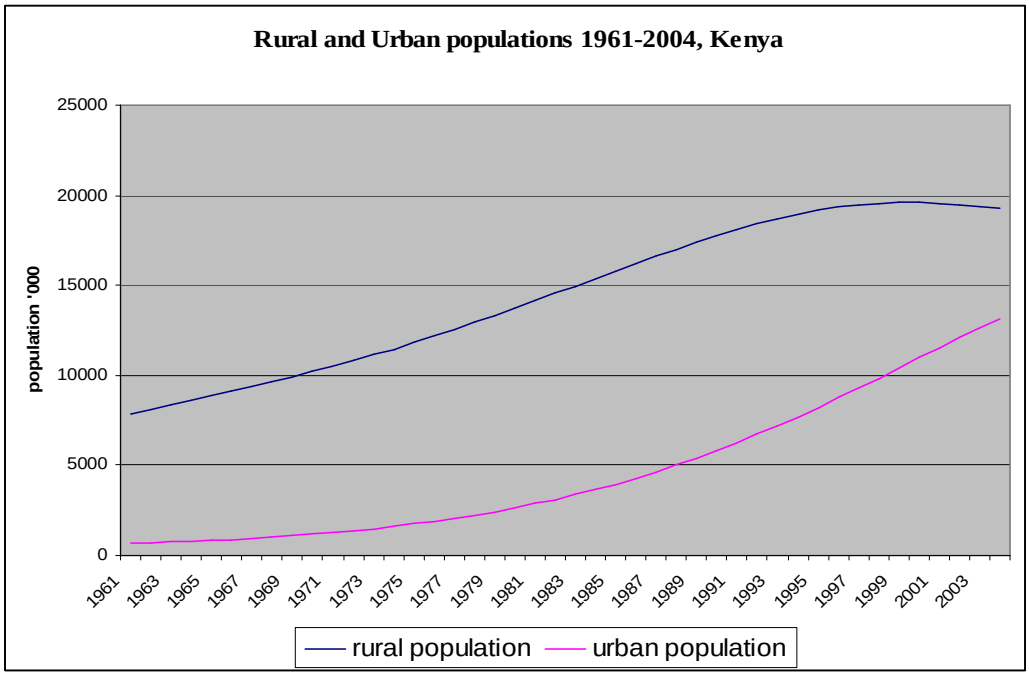
1999 for Central, Eastern and Nyanza provinces, while increasing significantly for Nairobi, slightly for Coast. Rift Valley, Western and North Eastern remained stable. The inter-censal growth rate for the total urban population declined from 6.7 to 3.8 in 1979/89 and 1989/99 respectively. The inter-censal population growth rates declined for all the provinces except for Coast province which increased from 3.7 to 4.2.

Figure 5: Trends in Urban and Rural Population (millions) 1962-1999



Source: CBS Economic Survey, 2002

Figure 6: Rural and Urban Population



Source: FAOSTAT

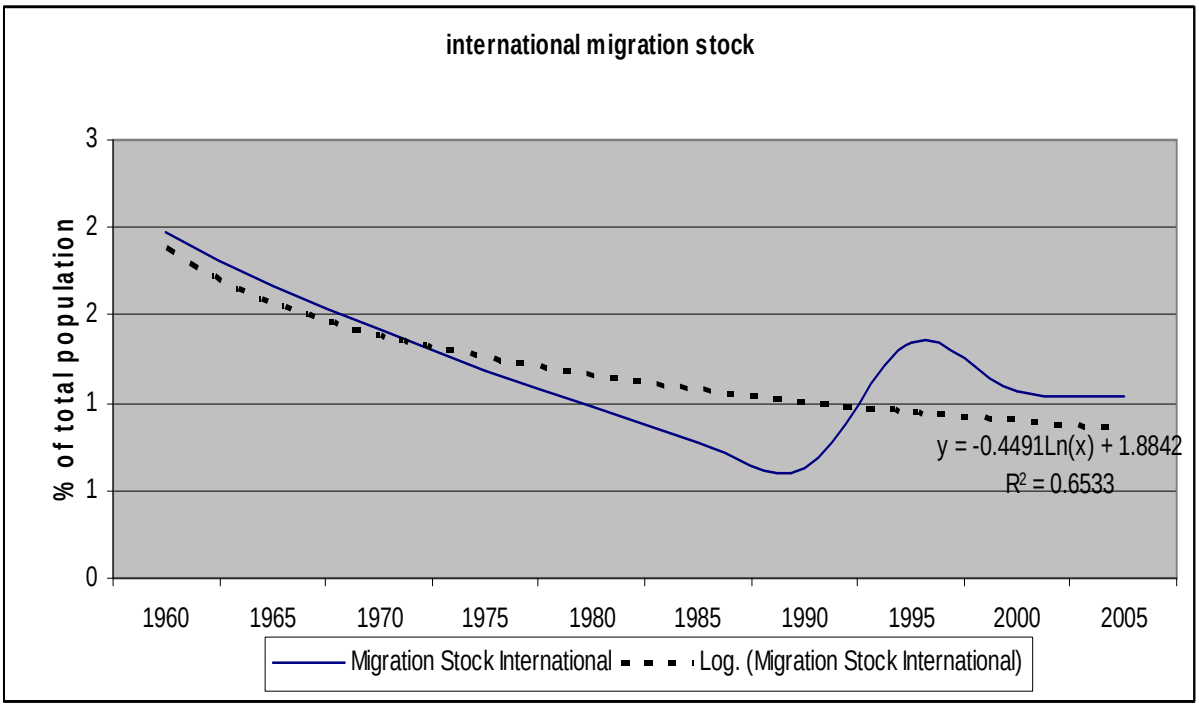
Available data shows that rural population has been on decline while the urban population has been increasing of late. Figure 6 shows the trends in urban and rural populations in Kenya according to FAO statistics. The decline in overall population growth is responsible for the declines in inter-censal urban population growth and the rural population decline.

The decline in rural population can be attributed to both demographic transition and migration while urban population absolute growth is attributed to rural depopulation. Rural depopulation (Rural-Urban migration1), here defined as the change in urban population holding constant demographic transition, is synonymous with the estimated urban population inter-censal growth rates.

1.2.2. Migration

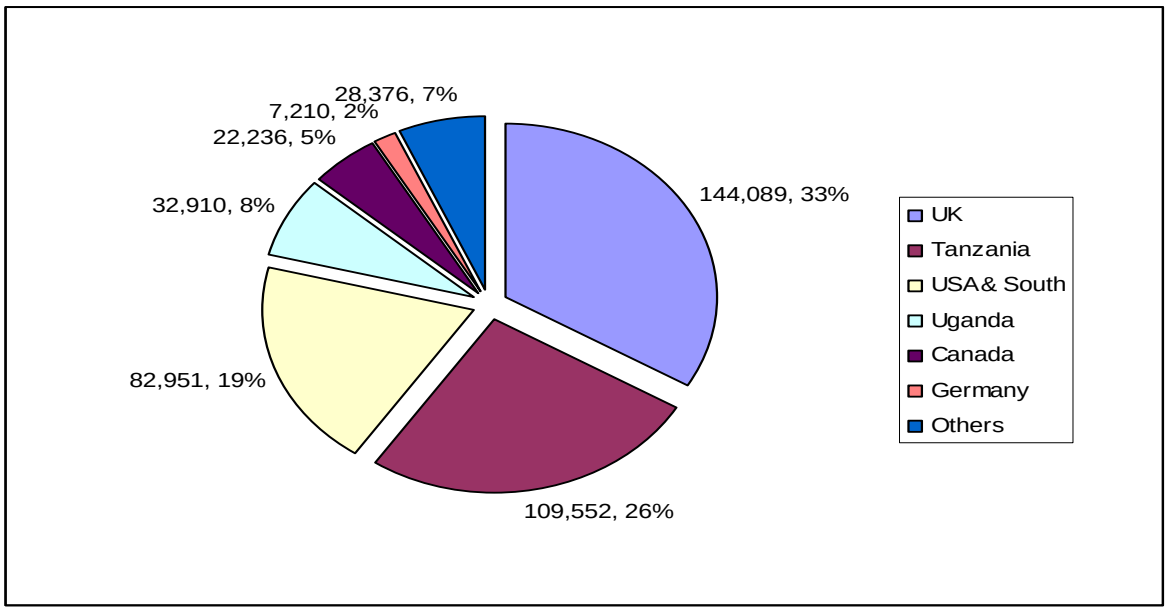
Migrants from Kenya to other countries were estimated to be 1.03% of the Kenyan total population (WDI, 2005). International migration from Kenya declined since 1960 until late 1990s (Figure 7). This can be attributed to the rapid and sustainable economic growth after independence and corresponding political stability in the immediate post-independence. This situation changed in the 1990s when increased international migration was observed partially attributed to the political repression. The period between late 1980s and 1993 was characterised by agitation for multiparty system. The ensuing political uncertainty and oppression led to migration to other countries in fear of political instability. Afterwards political stabilization, institutionalization of multiparty system and the start of economic growth was responsible for the decline in international migration

Figure 7: International Migration Stock



Source: World Development Indicators (WDI)

Figure 8: Migration Destinations



Source: Ratha & Shaw 2006

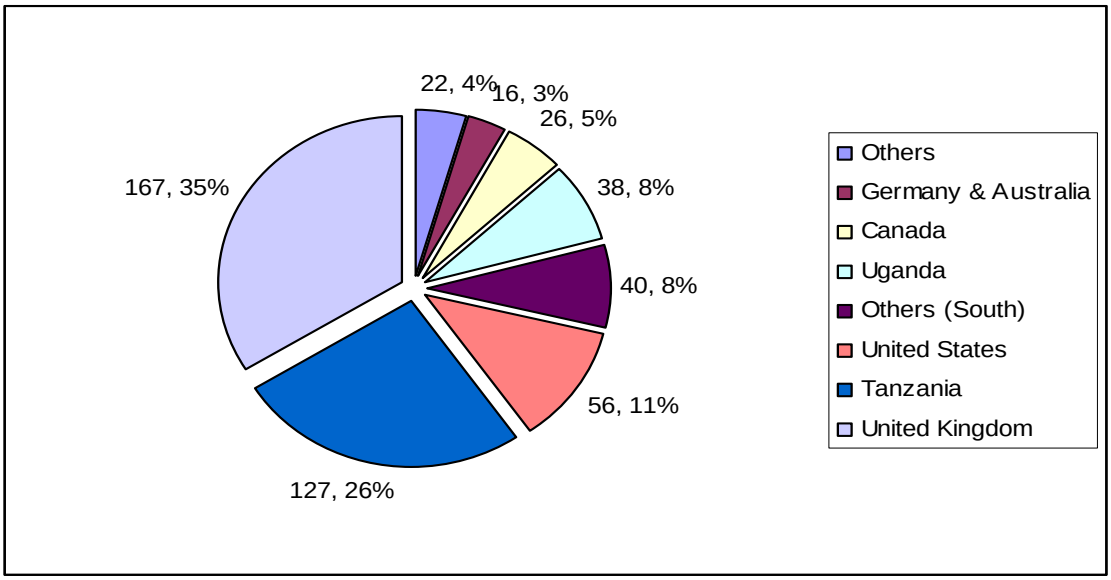
There were estimated 144,089 migrants from Kenya to UK in 2005. This accounted for 33% while migration to Tanzania accounted for 26% ahead of USA and the Southern America countries accounting for 19% (Figure 8). The migration to Tanzania can be attributed to the increasing economic growth in Tanzania which has presented job opportunities. Currently there are even reported cases of capital flight from Kenya to Tanzania owing to the less stringent environmental pollution rules.

International migration is mainly propelled by job opportunities and education. Remittances emanating from those who migrate to other countries are assumed to have impact on the economy. Towards the end of 2006, the Kenyan government put up its concern with remittance issue targeting to collect 715 million US dollars(EAS, 12 Dec ,2006) from the different countries. According to Ratha & Shaw 2006, Kenya received an estimated 494 million US dollars as remittances from other countries. Figure 9 shows the distribution of remittances to as from the various countries.

The annual change in urban population may indicate domestic migration (Rural-urban migration1) either into or out of the urban area holding constant the demographic transition. Although demographic transition accounts for a certain urban population increase, a large percentage of urban growth is owed to rural-urban migration. Rural-urban migration growth was 7.66% in 1963/80 6.84% in 1980/92 and 1.93% during the liberalization period 1993/02. Afterwards the growth rate was negative at an annual average of -2.09% (2003/04).

[1] Rural-Urban migration is estimated as U_{t-i}-U_t where U is the urban population. Estimation is only valid under the assumption that demographic transition is constant

Figure 9: Distribution of Remittances to Kenya



Source: Ratha & Shaw 2006

Table 3: Urban, Rural and Total Population Growth Rates

Period	Pre-liberalization 1961/80	Transition 1981/92	Liberalization 1993/02	Post-liberalization 2003/04
Urban population growth rate	7.66	8.11	5.04	4.37
Rural population growth rate	2.96	2.80	0.55	-0.52
Total population growth rate	3.26	3.69	2.28	2.00
Domestic Migration rate	7.66	6.84	1.93	-2.09

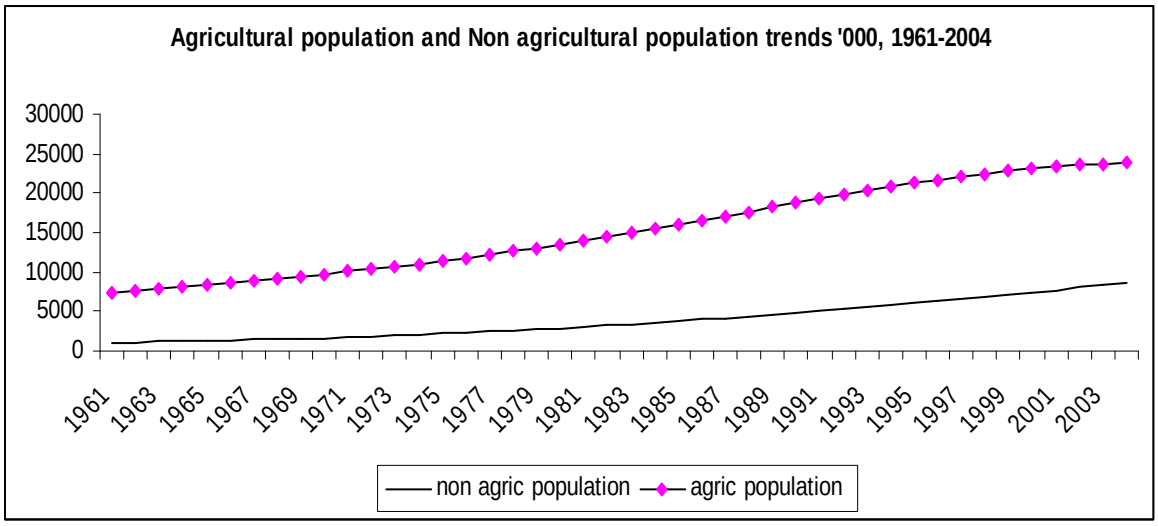
Source: Faostat

The migration rate is estimated from successive annual difference in the increase in urban population. Initially the trend rose faster than the growth in rural population resulting to rural depopulation and urban concentration. The decrease in migration rate indicates decline in rural depopulation and declining urban resources and appeal.

Rural depopulation is signified by the decline in rural population from 0.55 to -0.52, while the decline in rural-urban migration is signified by the decline in migration growth rate besides the decline in urban population growth rate in the liberalisation and post liberalisation era

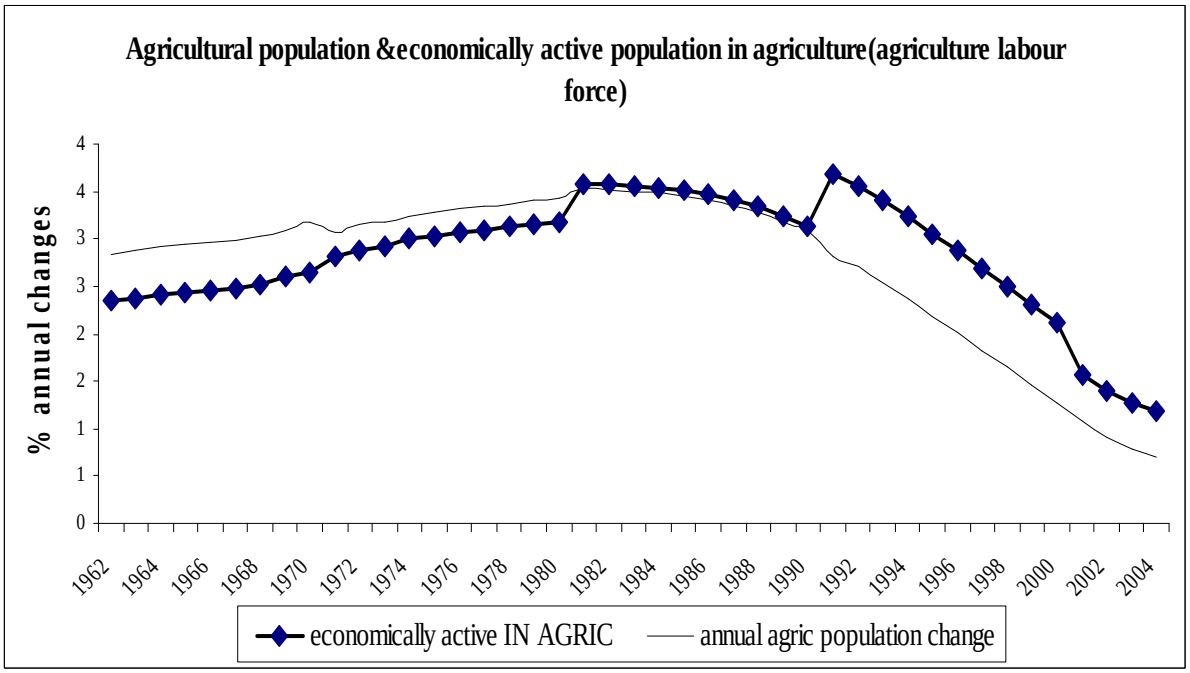
Classifying the population as agricultural and non-agricultural indicates that those people who derive their livelihood from agriculture are declining in number while the non agricultural population is on the increase (Figure 10). The number of people depending on agriculture in the year 1961 was estimated at 7.5 million with the non agricultural population at 1.07 million, currently the population is estimated at above 24 million and 8 million for agricultural and non-agricultural populations.

Figure 10: Agricultural and Non-agricultural Population Trends, 1961-2004



Source: Faostat

Figure 11: Annual Growth Rates in Agricultural Population and Agricultural Labour Force



Source: Faostat

The growth trend for the agricultural population has been on decline since early 80s also. This may be an indicator of the declining agricultural sector role or increasing role of other sectors such as manufacturing and construction.

The economically active in agriculture define the labour force in the agriculture sector. In the post independence period the number of people and workforce in the agriculture sector increased. While

initially agricultural population increased at an higher rate than the economically active population in agriculture, currently the trend has reversed although the growth rates are both on declining trend. The population depending on agriculture is thus currently on decline in a similar trend to the declining worth of the sector.

2. Place of Agriculture in the National Economy

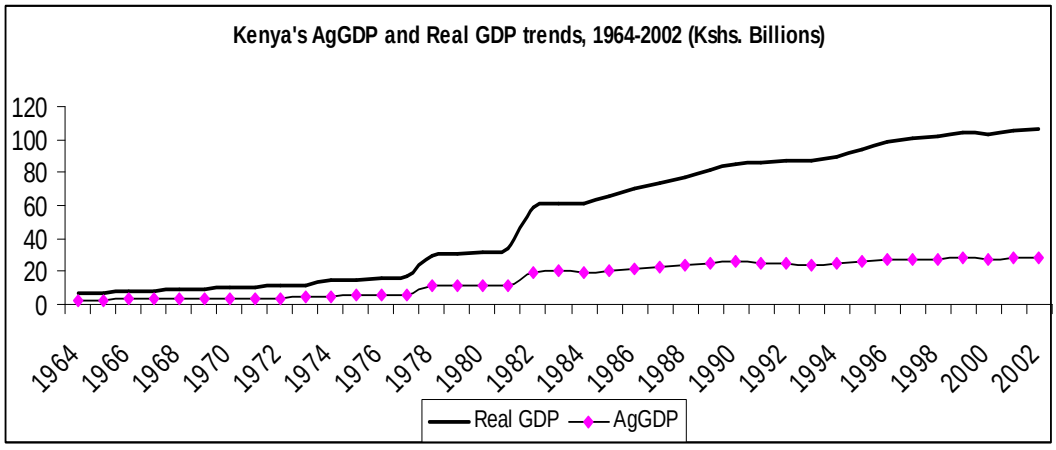
2.1. Gross Domestic Product

Kenya's economic growth can be best explained in the context of external shocks and the internal challenger under which the economy has to respond. Five phases are discernible: a rapid growth phase over 1964-73; an era of external shocks 1974-79, a period of stabilization and structural adjustment in the 1980s characterized by partial implementation of the SAPs; an era of liberalization and declining donor aid 1990-2002 characterized by the withdrawal of government from controlling of the economy (GoK, 2002-2008) and a post liberalization era 2003 onwards.

During pre-liberalization there was a strong growth in the first two decades after independence and weak or negative thereafter. The oil shocks experienced during pre-liberalization slowed economic growth. From early 1980s, implementation of the SAPs led to a relatively increased growth rate of GDP and AgGDP. The 1975/78 world coffee prices quadrupled and the country responded by increasing coffee exports six-fold while tea exports grew three-fold. This led to the GDP growth at that time. The GDP trends mirror the AgGDP trend (Figure 12) indicating the important contribution of agriculture to the economy.

The overall economic growth has been positive. Between 1963 and 1980, the economy grew at an average real growth rate of 11.3 percent, 10 percent in 1981 to 1992 and 2.0 percent from 1993 to 2002. The country's' real GDP in 2005 was estimated at 1172.1 million Kshs signifying a 5.8% increase from 2004 (CBS, 2006).The real GDP has been on increase since independence.

Figure 12: Real GDP Trend



Source: CBS Economic Surveys (Various Issues)

2.1.1. Inflation

The trend of inflation from a low of 2.7 in 1964-73 to levels of annual average rates of 12 in 1970s and 1980s (Table 4) was attributed to the expansionary monetary policies which were a response to the external shocks of oil crisis and coffee booms. The passive monetary policies under the oil crisis

and coffee boom were responsible for the rapid money supply in the 1970s while deficit monetization was the prime driver over the 1980-1995 period (GoK, 2002-2008).

The large financing requirements exerted continued pressure on inflation and exchange rates and strained trade and financial liberalization policies. Stabilization succeeded in bringing down the deficits in the latter part of the nineties (1% average) but rose again in 2000-01. Of the four agreements entered into with the IMF (1989, 1993, 1996 and 2000), three could not be completed, and Kenya could only benefit from 50 percent of the resources approved by the IMF (SDR648 million). Relations broke down in 1991, 1997 and 2001 for failure to maintain satisfactory macroeconomic progress. In all three occasions, impending elections made political considerations prevail.

Table 4: Money Supply and Inflation

	1964-73	**1974-79**	**1980-89**	**1990-95**	**1996-2000**	**2001-2004**
Inflation rate	2.7	12.1	12.3	23.1	8.9	6.9
Money supply growth %	-	20.7	12.0	27.2	13.2	9.28
GDP growth rate	6.6	5.2	4.1	2.5	2.0	3.3**

Source: National Development Plan, 2002-2008
*** averaged for the annual growth rates under SNA*

The wide spread drought between 1996 and early 97 led to a major shortfall in agricultural output and thus fuelled inflation. Other factors include frequent increases in petrol and petroleum related product price, slight increases in VAT (15-16%), and depreciation of the Kenya shilling following withholding of the donor support under the Enhanced Structural Adjustment Facility (ESAF).

2.1.2. GDP Diversification

The country's GDP is sourced from various sectors: agriculture and forestry; manufacturing; building and construction; trade, hotel and tourism industry, mining; electricity and water transport and communication; finance and real estates and the services industry. The percentage contribution of agriculture compared to other sectors has been on decline. While sectors such as transport and communication, services and trade have been growing, agricultural sector has been declining and even negative in some years.

While Agriculture's contribution to the real GDP declined during the three periods (pre-liberalization, transition and liberalization, 0.35%, 0.31% and 0.27% respectively), the share for the services sector increased from 0.19 in 1980/64 to 0.20 and 0.21 in 1992/81 and 2002/93 periods (Table 5)

Table 5: Sector Share to Real GDP in Selected Periods

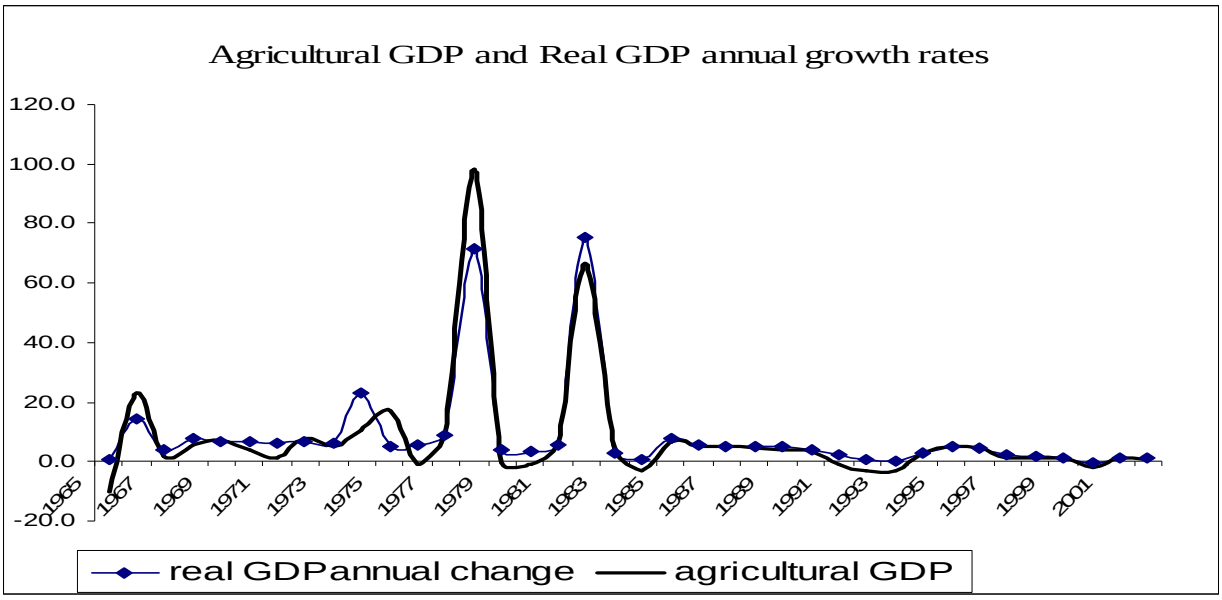
Year	**Agriculture. & Forestry**	**Building & Construction**	**Water & Electricity**	**Finance ownership /dwellings**	**Trade & hotels**	**Manu-facturing**	**Transport & comm**	**Services industry.**
1964/80	0.35	0.05	0.02	0.11	0.10	0.12	0.07	0.19
1981/92	0.31	0.05	0.02	0.15	0.11	0.13	0.06	0.20
1993/02	0.27	0.04	0.02	0.18	0.12	0.13	0.06	0.21

Source: CBS Economic Surveys (various Issues)

2.2. Agricultural GDP

Agriculture contributed 25.81% to the real GDP in 2004 less than 26.61% in 2003 (CBS, 2006). The trend of agricultural contribution to the real GDP has been on decline as indicated previously (table 5 above). The real GDP mirror agricultural GDP: growth rates have had similar trends independence as shown in Figure 13.

Figure 13: Real Agricultural GDP and Real GDP Annual Growth Rates



Source: CBS Economic Surveys (various Issues)

The first oil crisis in 1973/74 reduced the share of agricultural contribution to GDP from 34% in 1973 to 31 % in 1974. The declining share trend was however reversed by the coffee boom in 1976/78 which increased the share from 32.19% in 1976 to 37.36% in 1978 and nearly doubled the agricultural contribution towards the GDP in both real and nominal values (GoK, 1989-93). However, immediately the second oil shock in 1978/79 the share reduced the share to 35.71% by 1979 and onwards the share has been in decline and not even the mini coffee boom of 1986 had any significant effects. The agricultural GDP grew at an annual rate of 12.55% in 1963/80 at 8.18% in 1992/80 and 1.32% in 2002/93 periods.

In the last (1990-2000) decade the agriculture sector growth declined from 4.4% in 1996 to 1.5% in 1999 and to negative 2.4% in 2000 (ERS-WEC,2003).The declining competitiveness of the agricultural sector against the other economic sectors is partially attributed to the poor production technologies, fluctuating prices in the international markets, high production costs and imperfect agricultural information flow has led to low agricultural returns and consequently exit of productive labour force from agriculture.

However, in 2002 the sector experienced a minimal growth of 0.7% and 1.3% in the first half of 2003 (ERS-WEC, 2003). A recent assessment indicated that the agriculture sector grew by 6.7% in 2005 with more being expected from the sector (Strategic Plan 2006-2010).

2.3. Employment Structure

According to the 1998/99 integrated labour Force Survey (ILS), the country's total labour force was 77.4% (15.9 million) of the population aged 15-64 with the majority (57.95 of the active population)

in the 20-39 age (CBS, 2002). Currently there are estimated 17.1 million economically active people (FAOSTAT). The economically active population growth rate was 3.11% in 1961/80, 3.84% in 1981/92 and declined to 1.88 in 2003/04. Similar trend is evident in economically active in agriculture and non- agricultural population (Table 6).

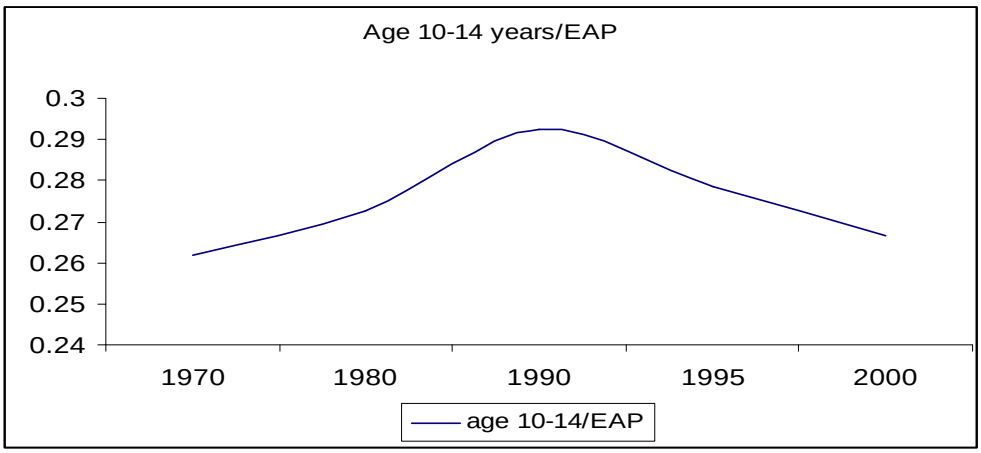
Table 6: Labour Force Growth Rate Trends

Period	Pre-liberalization 1961/80	Transition 1981/92	Liberalization 1993/02	Post-liberalization 2003/04
Economically active population	3.11	3.84	3.08	1.88
Economically active in agriculture	2.63	3.47	2.23	1.23
Non-agricultural population	5.48	5.19	4.17	3.25

Source: FAOSTAT

The 10-14 age cohort to the EAP ratio trend increased from 1970 but the trend reversed since 1990 to the current (Figure 14). This implies that there will continue to progressively be a smaller group entering the labour force and job creation in agriculture sector and other sectors.

Figure 14: Population Cohort Ages 10-14 Compared to Economically Active Population (EAP)



Source: FAOSTAT and WDI

Table 7 provides an estimation of the labour force entering the job market annually from labour force projections. Labour force projections from 1999 population census indicated that over 500,000 people will enter the job market annually. This slightly surpassed the governments' promise of creating 500,000 jobs annually. It was approximated that before 2000, more than one million. Between 2003 and 2004 an estimated 437,900 jobs were created (GoK, 2004-2005). This was below the targeted 500,000 jobs and below the 517,611 that entered the job market. This may be the reason as to why unemployment rate and poverty are on the rise.

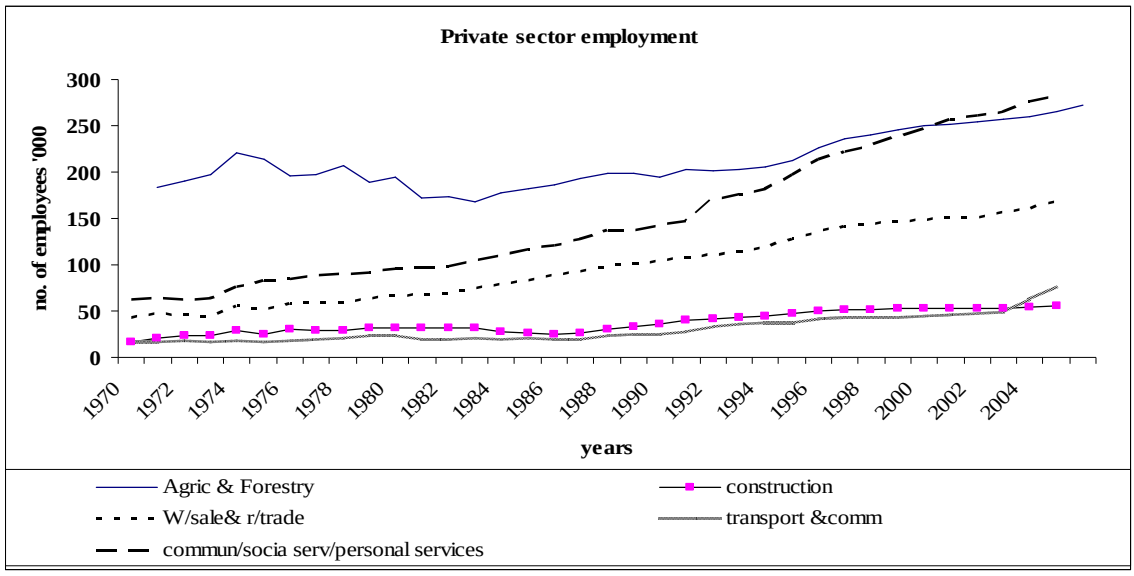
Table 7: Labour Force Projections, 15-64 years

Year	Labour force projections (number)	Cohort group (15-64 yrs)
1999	12,395,603	
2000	13,439,411	1,043,808
2001	13,933,416	494,005
2002	14,435,689	502,273
2003	14,945,839	510,150
2004	15,463,450	517,611
2005	15,988,074	524,624

Source: CBS (Central Bureau of Statistics), 2002. Kenya 1999 Population and Census: Analytical Report on Population Dynamics Vol. III. Ministry of Finance and Planning

The capacity of employment absorption by sector is shown by the trends of employment in the various sectors. While the growth for all sectors were positive, agriculture sector employment declined in the 1980s and later reversed as from mid 1990s. Employment in the non-agricultural sectors has been significant with the services sector surpassing agriculture as from 2001 (Figure 15).

Figure 15: Private Sector Employment

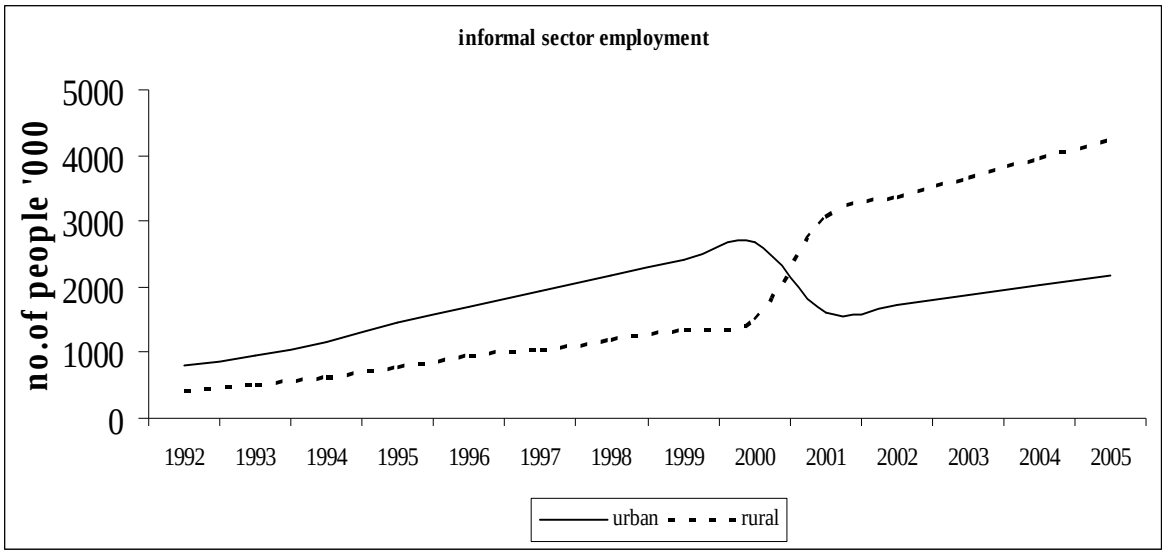


Source: CBS Economic Surveys (various Issues)

2.3.1. The Informal Sector

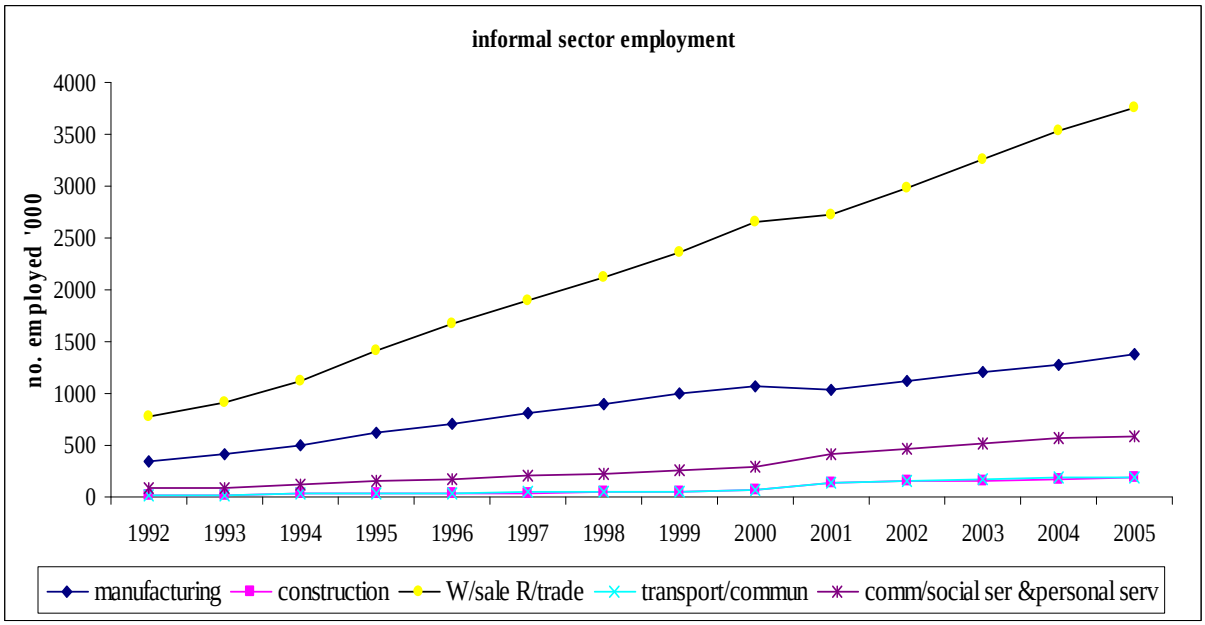
The informal sector has since 1992 contributed significantly to employment creation. In 1992 the informal sector was estimated to have employed 1.24 million people while in 2005, 6.12million people 2.17 million being in urban areas and 4.24 million in rural areas were estimated to be working in this sector. Figure 16 shows the trend in number of people engaged in the informal sector in both urban and rural areas. The percentage number to the total informal employment in urban area fell from 66.03 % in 1992 to 33.89 % in 2005 while in rural areas the percentage number of people engaged in informal activities increased from 33.97 % to 66.11% in 1992 and 2005 respectively. This can be attributed to low skill requirements for the informal jobs and available markets for low priced imported products that offer jobs in the rural areas.

Figure 16: Urban and Rural Informal Sectors Employment



Source: CBS Economic Surveys (1995-2006)

Figure 17: Informal Sectors Employment



Source: CBS Economic Surveys (various Issues)

Informal sector growth was fostered by the adoption of the session paper No.2 of 1992 on Small Enterprise and Jua Kali Development in Kenya. This paper led to the review of laws that impact on the performance of the sector and trade licensing reform thus leading to the introduction of single business permits. Growth of rural informal sector seems to have been favoured by PRSP strategy in 2001 hence surpassing urban informal sector employment. The PRSP envisaged rural enterprise development to alleviate poverty. The government focus seems to have shifted more to the rural areas as compared to the urban areas. Further this was perpetuated by the change of government in 2002. The new government promised creation of 500,000 jobs annually. This seems to have yielded since between 2003 and 2004 an estimated 437,900 jobs were created with 36,400 only being in the formal

sector (GoK, 2004-2005).During this same period retrenchment of civil servants prompted urban-rural labour movement in search of alternative livelihoods with the liberalisation of the telecommunications industry offering viable alternatives in the informal sector.

The informal sector is diversified to cover the manufacturing, trade, services and construction sub-sectors. Informal sector employment shows positive growth (Figure 17).The whole sale and retail trade sub-sector of the informal sector has had the highest growth rates. Propelled by increased access to rural financial services and ease of communication the sub-sector has continued to grow.

2.3.2. Rural and Urban Unemployment

In 1999, there were 1.8 million unemployed people with the number of unemployed females almost double the males (CBS, 2002). The unemployment sex ratio (ratio of unemployed males to unemployed females) was estimated at 0.49 giving a notable gender employment disparity. As shown in Table 88, urban areas absorbed 57.7 percent of the unemployed (CBS, 2002). It is hypothesized that unemployment has increased owing to the declining agricultural productivity in rural areas. Urban unemployment rate had risen to 25.1%, rural by 9.4% but less acute to the urban areas. The overall unemployment in the country was 14.6%.

Table 8: Sex Distribution of the Unemployed Persons, Aged 15-64 (numbers)

Region	Male	%	Female	%	Total	%	Unemployment rate %
Rural	332402	56.1	439540	36.4	77941	42.9	9.4
Urban	260063	43.9	768618	63.6	1028681	57.1	25.1
Total	592465	100	1208158	100	1800623	100	14.6

Source: CBS, Economic Surveys 2002

2.4. Household Income Structure and Incidence of Poverty

From the Welfare Monitoring Survey (WMS) done to evaluate the SAPs it is clear that rural poverty is on the increase. Overall poverty, rural and urban poverty incidences are on the increase as shown in (Table 9).Rural–urban migration rate although currently indicates signs of decline was responsible for the increasing urban poverty trends. Urban population increases led to social amenity constraints.

Table 9: Poverty Incidence by Province, National, Urban and Rural Percentages

Region	1981/82	1992	1994	1997	2000
Central	25.7	35.2	31.9	31.4	35.32
Coast	54.6	43.5	55.6	62.1	69.88
Eastern	47.7	42.2	57.8	58.6	65.90
Rift Valley	51.5	51.1	42.9	50.1	56.38
North Eastern	N/A	N/A	58.0	65.48	73.06
Nyanza	57.9	47.4	42.9	63.1	70.95
Western	53.8	54.2	53.8	58.8	66.11
Nairobi	N/A	26.5	25.9	50.2	52.56
Rural	48.8	46.3	46.8	52.9	59.56
Urban	N/A	29.3	28.9	49.3	51.48
National	46.8	46.3	46.8	52.3	56.78

N/A not available
Source: Gitu, 2004

The poor are also spread over all the regions of the country. There also exist poverty disparities in gender (Appendix 1). Female headed households constitute the larger proportion of the poor in nearly all the rural areas. This could be a reflection of the skewed access to resources in favour of male headed households.

This trend could be a reflection of deteriorating agricultural sector performance on which a large rural population depends. Rural household income is composed of net nominal income from crop, livestock and off-farm activities (business activities, salary and remittance income). Different agro-regional zones have different agricultural enterprises as favoured by climatic conditions. Between 1999 and 2004 off-farm income proportion increased from 24% to 53% while crops and livestock income proportion declined from 44% and 32% to 31% and 16% respectively. The same trend was observed on the regional basis with crops and livestock enterprises losing ground against the off-farm income sources (Table 10).

Table 10 Share of Net Income by Source and Region

	1997			2000			2004		
Agro-regional Zones	**crops**	**livestock**	**off farm**	**crops**	**livestock**	**off farm**	**crops**	**livestock**	**off farm**
Coastal Lowlands	11	17	72	33	4	64	20	8	73
Eastern Lowlands	34	29	37	37	13	50	20	8	72
Western Lowlands	27	37	36	33	17	50	27	13	60
Western Transitional	52	28	20	63	8	29	40	14	45
High Potential maize zone	52	30	18	37	26	37	33	22	45
Western Highlands	34	45	21	56	15	29	37	20	43
Central Highlands	47	34	20	54	15	31	35	15	50
Marginal Rain Shadow	19	51	30	15	23	62	21	15	64
Total	44	32	24	45	17	38	31	16	53

Source: Tegemeo Rural household Survey 1997, 2000, 2004

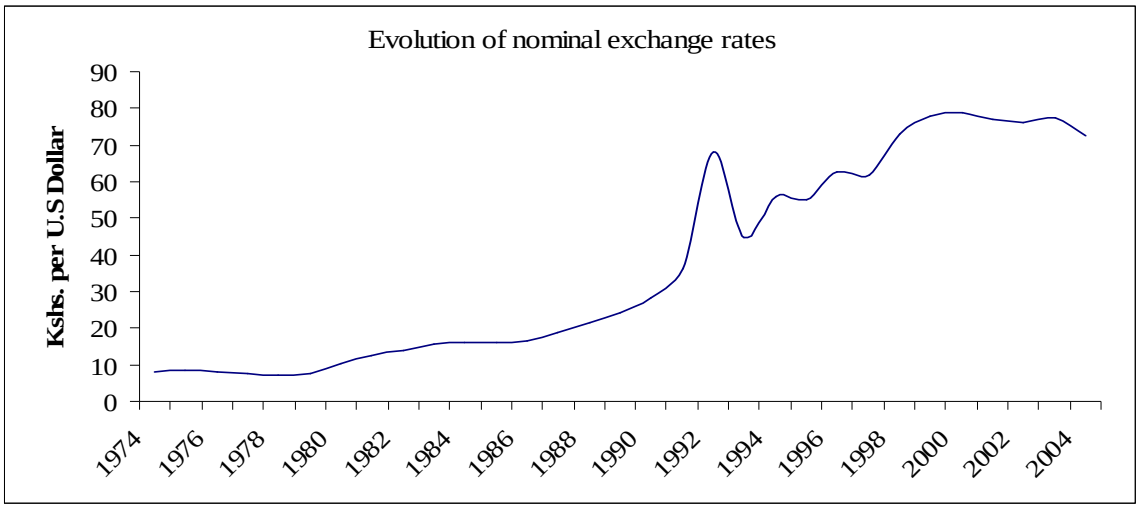
3. International Trade

3.1. Exchange Rates

The Kenyan shilling has lost value against major currencies especially the U.S dollar (Figure 188). In the period 1964 to 1973 the exchange rate of the Kenyan shilling to the US dollar averaged 7.1. The early 1980s to early 1990s saw the devaluation of the shilling in reaction to macroeconomic imbalances triggered by oil shocks, the coffee boom, the 1984 drought and the declining inflow of foreign direct investment. By 1994 the government implemented the floating exchange rate and repealed the Exchange Rate control Act. This led to the depreciation of the nominal exchange rate from the highly overvalued rates of the first decade after independence to an average exchange rate of 61.6 to the US dollar (GoK, 2002-2008). As the Kenyan shilling lost strength against the US dollar, the economy continued to be more 'susceptible' to imports from other countries. The exchange rate trend reflects the process of liberalization that as the government moved from import substitution, lifted import tariffs, dropped the currency devaluation policies and replaced the non-tariff barriers with equivalent tariffs, the economy became more outward oriented and imports out flowed exports.

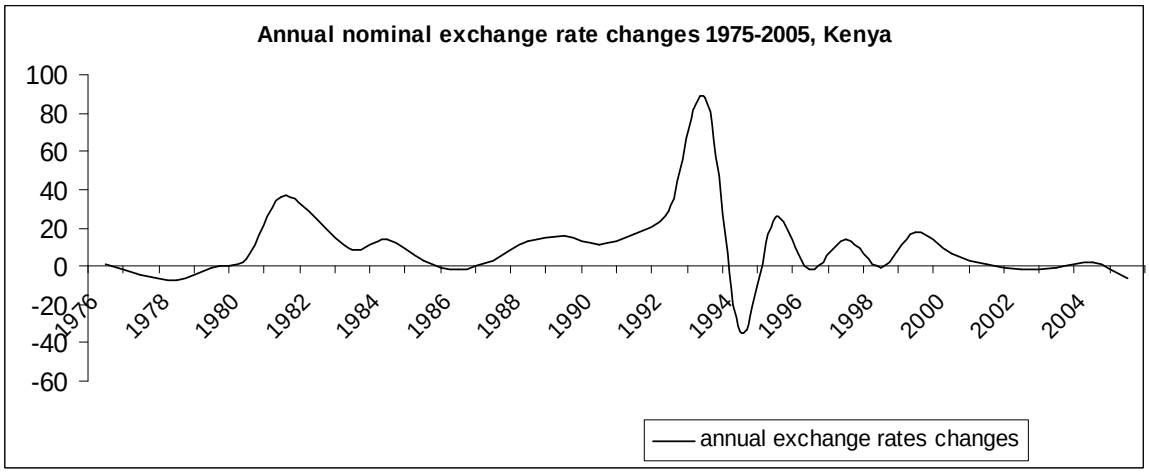
The annual change in exchange rates is presented in figure 19. Currency depreciation characterized the 1980s with the rate of depreciation peaking in parts of early 1993. This is mainly attributed to the political uncertainty that marked the onset of a multi-party political system. The exchange rate has been relatively stable since 2002.

Figure 18: Evolution of Exchange Rates



Source: CBS Economic Surveys (various Issues)

Figure 19: Annual Exchange Rate Growth



Source: CBS Economic Surveys (various Issues)

- **Trade Balance, Cover Ratio and Dependence**

Trade balance has been negative since independence increasing during the early 1970s, with the first and second oil shock in 1973/74 and 1979/80 respectively worsening the situation. The coffee boom 1976/78 reduced the negative trade balance within a short span of time. Since then currency depreciation, and falling world agricultural commodity prices have led to increased negative trade balance.

Several trade agreements signed between Kenya and other countries within Africa and overseas, the trade provisions such as those granted by the 1975 Lome convention, the regional agreements –EAC, ACP-EU, and WTO seem not to have had any significant effects to worsening trade balance. During the three phases of economic reforms the balance worsened with every. The drastic decline coincided with economic reforms that gradually led to the opening of the economy driving the balance of trade further down. In 2002, the countries balance of trade peaked to 142,518 million Kenya shillings.

Figure 20: Trade Balance

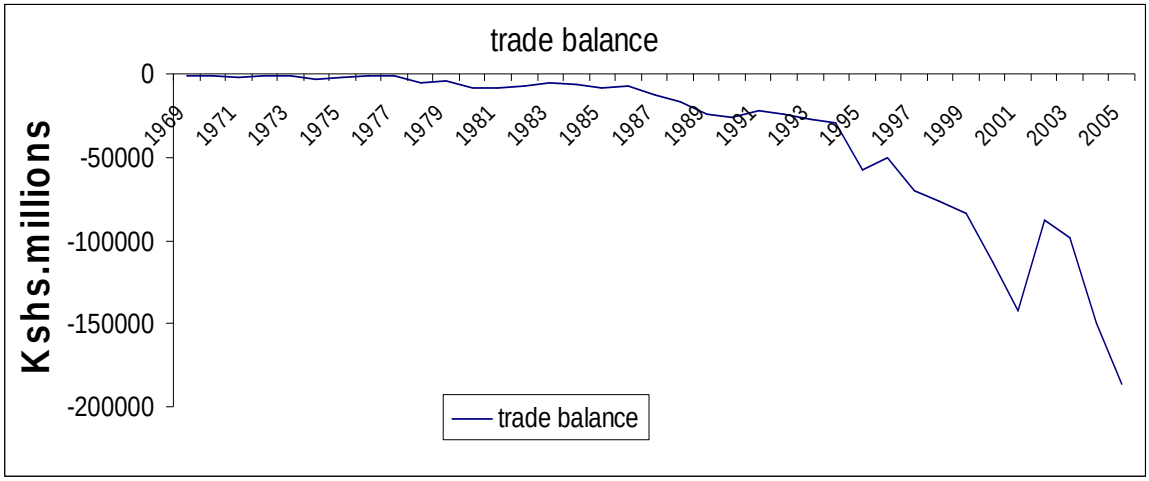
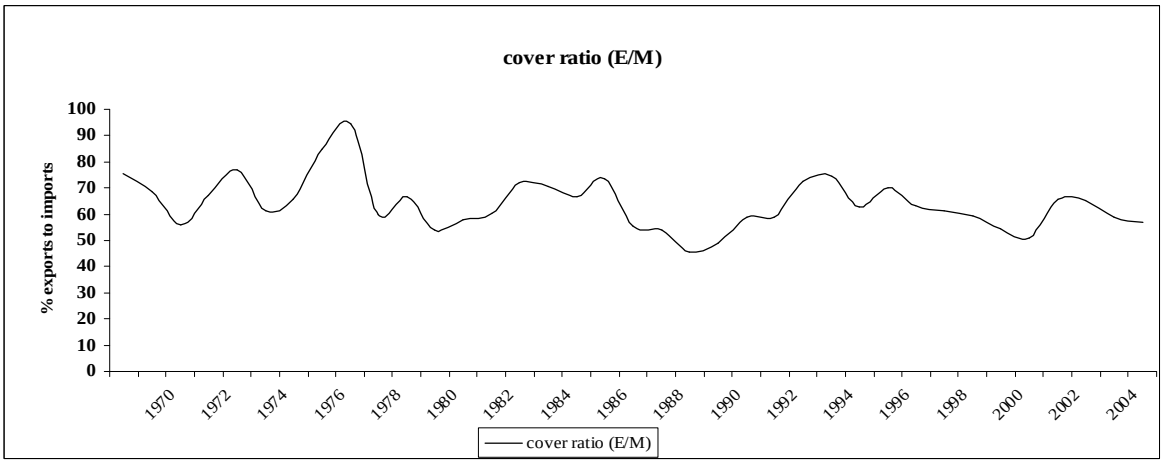


Figure 21: Cover Ratio



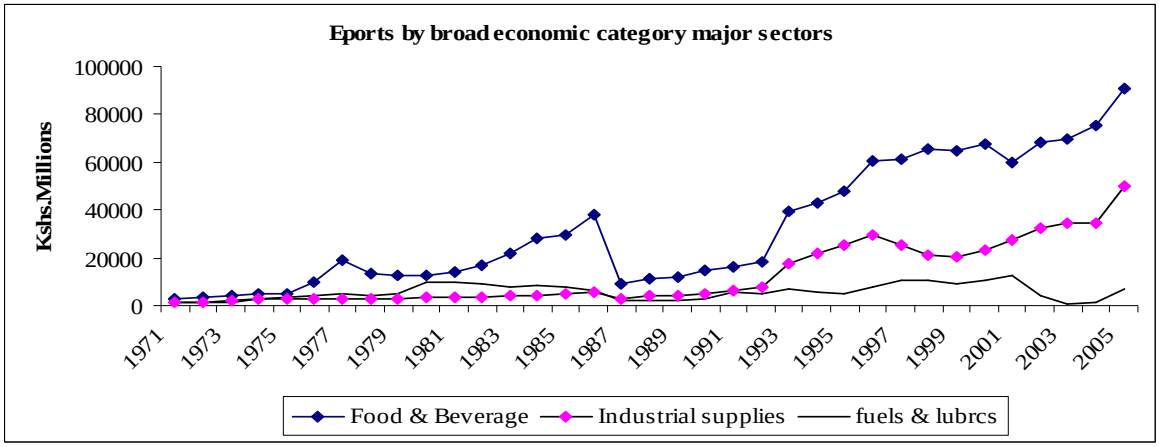
Source: CBS Economic Surveys (various Issues)

The cover ratio (Exports to Imports) has consistently been below one (Figure 21) indicating that the country has continuously imported more than exported. Only in 1977 did the ratio come close to one (94.4 %). This was prompted by the coffee boom which increased exports. The slight increase in cover ratio from 1990 to mid 1995 could be attributed to rapid currency depreciation. This led to increase in export quantities. However, despite the currency depreciation, the ratio still declined indicating increasing imports in 1996 to 2001. The election of a new government in 2002 appears to have positively influenced the cover ratio briefly through a strengthened currency and renewed hope in the economy.

3.2. Exports Diversification

Food and beverage products have comprised most of the exports from Kenya. This category constitutes more than half of total exports. In 2004 and 2005, the category contributed 0.47 of all total exports. Food and beverage grew rapidly during the pre-liberalization, transition, liberalization and post-liberalization periods reaching an annual average of 24.23%, 10.32%, 17.35%, and 10.37% respectively. The industrial supplies sector in these years showed an increase in the value of exports. On the other side, imports are dominated by industrial supplies and fuels and lubricants (Figure 22)

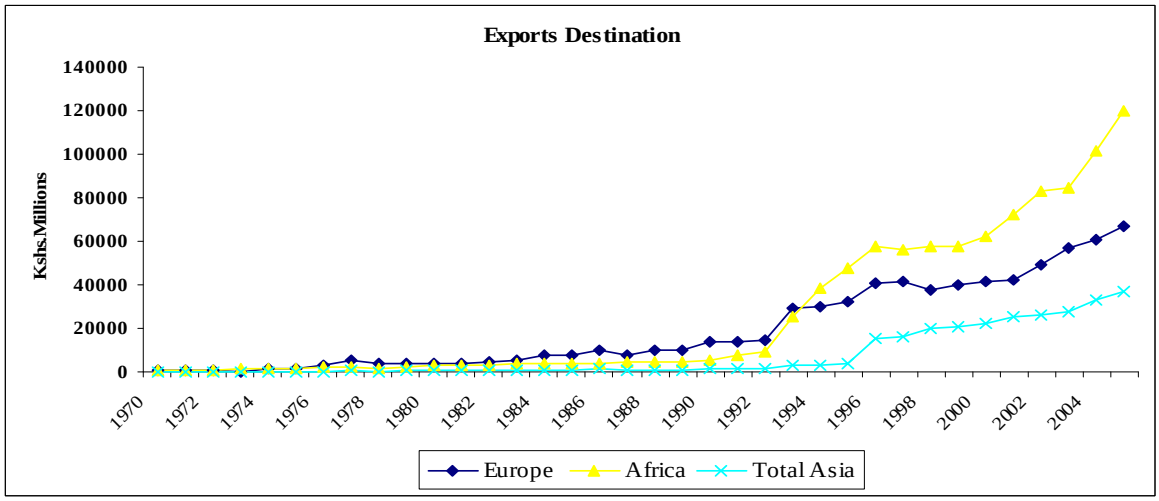
Figure 22: Exports by Broad Economic Category



Source: CBS Economic Surveys (various Issues)

There is a clear distinction between the pre and post liberalization periods as from early 90s'. During the pre-liberalization era inward oriented policies did not encourage exports growth and thus the growth rate was very low. Africa, Europe and Asia constitute the largest export market for Kenyan products at 49.3%, 27.5%, and 15.2% respectively in 2005 (Figure 23).

Figure 23: Export Markets



Source: CBS Economic Surveys (1974-2006)

The African destinations consist of all the African countries, the European markets destinations are constituted by majority Western European countries while the Asian markets include the Indian sub-continent. In 1993, the revival and transformation of East African Community in 1990 and the liberalisation of the Kenyan economy in 1993 opened export markets for the country. This coincided with phenomenal currency devaluation within the same period hence making Africa the leading export destination

3.3. Imports

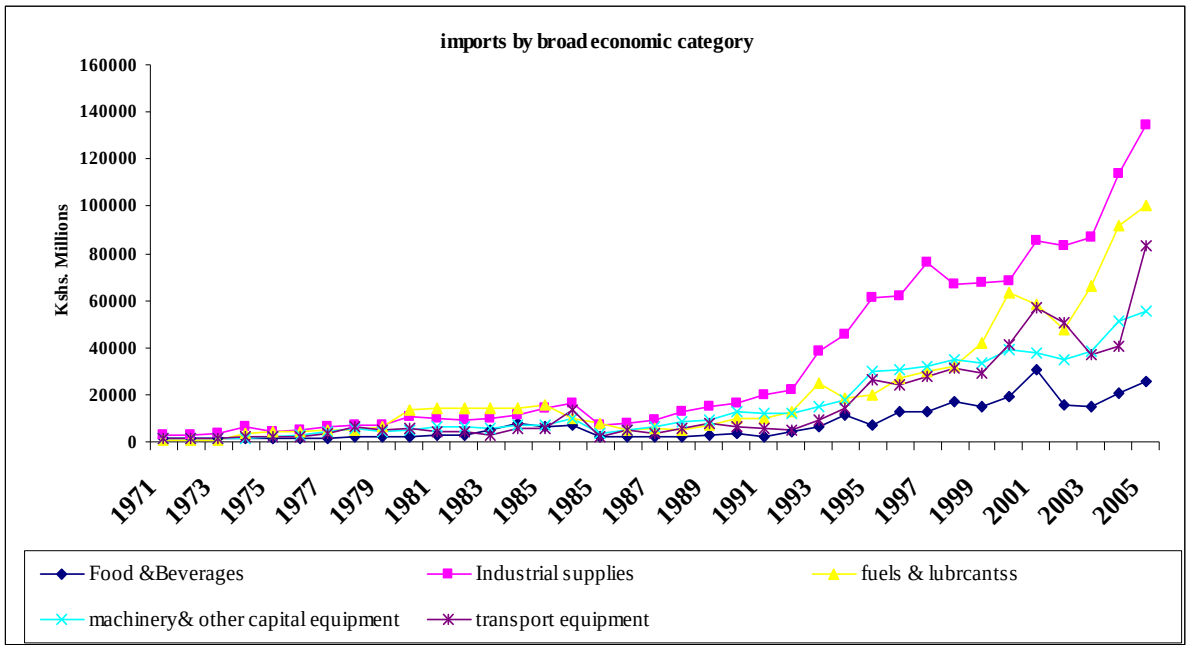
Imports grew at an average annual of 22.8% in 1970/80, 10.24% in 1981/92 17.56% in 1993/02 and 18.96% in the post-liberalization (2003/05) period. Even when the government maintained strict

import restrictions, industrial supplies and oil products could not be restricted since the economy could not do without them. In addition, the strong currency that existed during this period spurred imports from other countries while exports remained expensive. The signing of various international and regional agreements in the 1990s in a bid to open up the economy also led to increases in the flow of imports during the liberalization and post-liberalization eras.

- **Imports Diversification and Trends**

The trend of imports in all categories has been on the increase since the economy was opened up. The liberalization of the economy led to growth of imports with industrial supplies leading in value (Figure 24). The post-liberalization era immediately after 2002 indicates a strong and vigorous growth of imports.

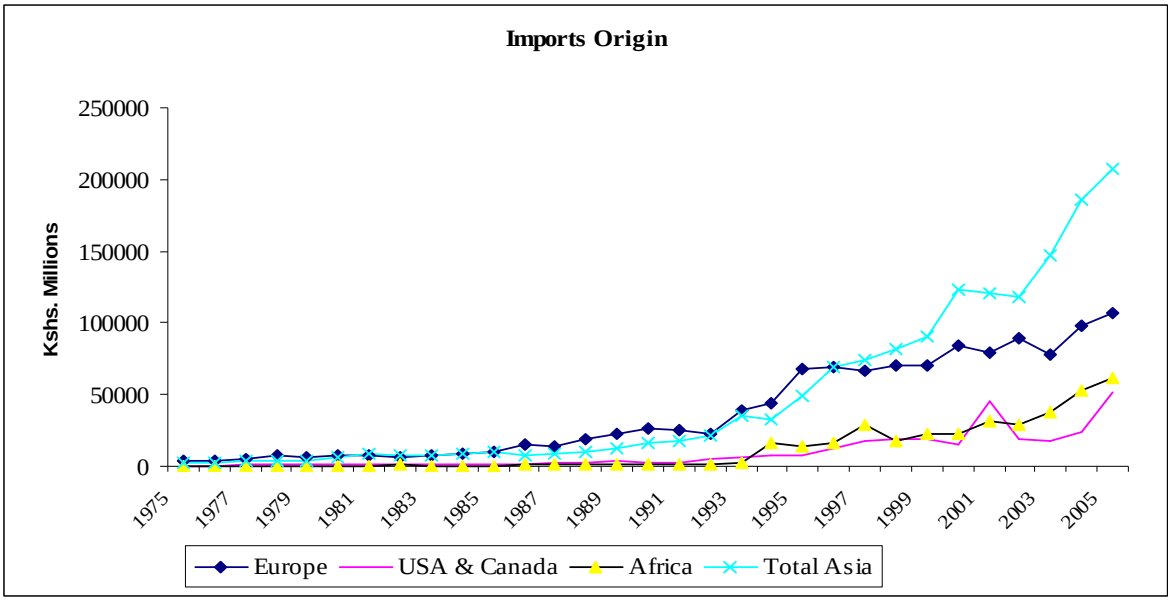
Figure 24: Imports by Broad Economic Category



Source: CBS Economic Surveys (various Issues)

The largest exporters to Kenya are the Asian countries (Figure 25). In the year 2005, 48.07% of all imports originated from Asian countries, 24.81% from the European countries, and 12.02% from USA and Canada, 14.26% from Africa 0.38% from Australia and Oceanic countries and 0.46% from other countries. It is evident that imports from Asian countries continue to soar especially after liberalization. The country's currency gain against the Asian countries currencies especially the Japanese yen relative to the depreciation against sterling pound and US dollar may have triggered this effect as from 1993.

Figure 25: Imports Origin

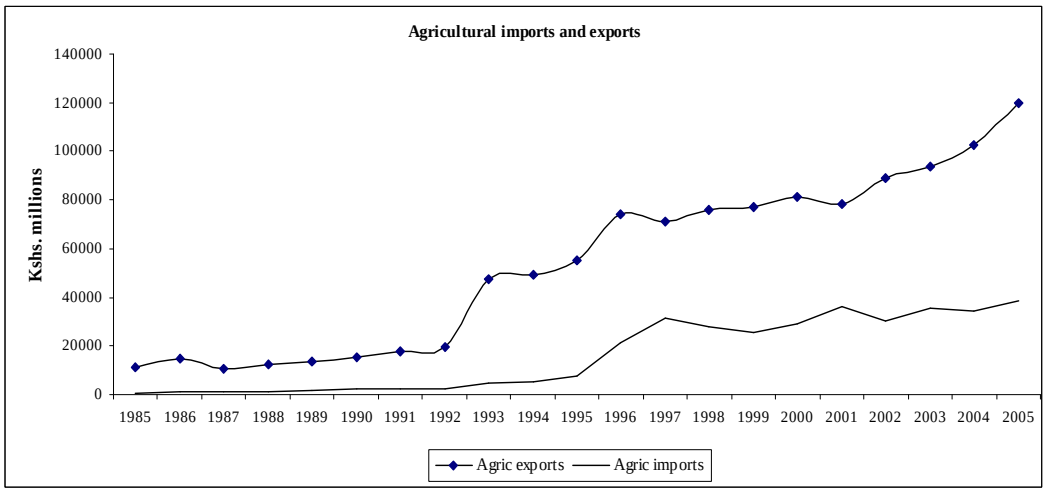


Source: CBS Economic Surveys (various Issues)

3.4. Agricultural Exports and Imports

Figure 266 shows the trend in agricultural exports and imports. The trends have generally been increasing over the period. Notably, exports imports, the vigorous growth in the agricultural exports and imports started immediately after 1992. The liberalization era during which the industry was exposed to external competition led to an annual average growth of 16.65% in exports while imports growth rate doubled the exports growth rate.

Figure 26: Agricultural Imports and Exports



Source: CBS Economic Surveys (various Issues)

Tea and horticultural exports increased while coffee exports declined (Table 11). Decline in coffee exports can be attributed to the decline in world coffee prices. The increase in tea and horticulture

value could not offset the value of agricultural imports and thus agricultural cover ratio continued to surge down indication the relative increase in agricultural imports.

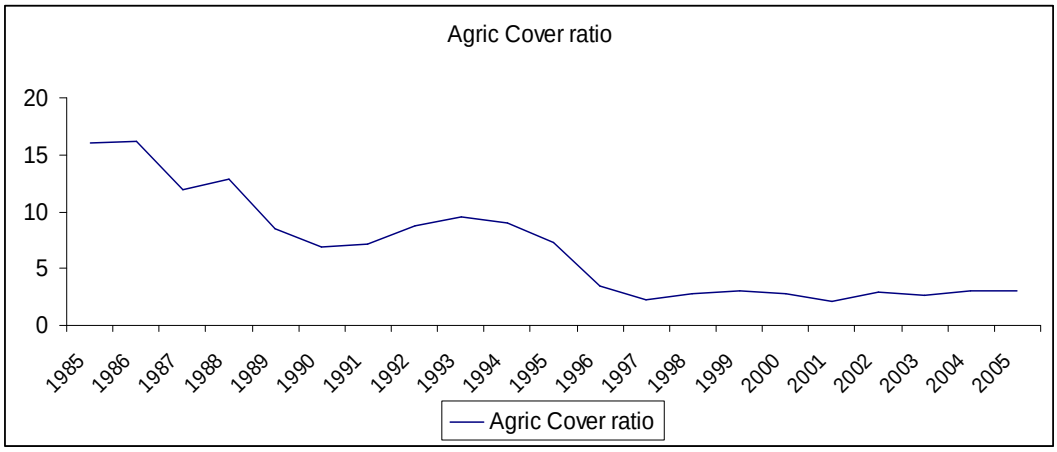
Table 11: Export Trends of Major Productions Tea, Horticulture, Coffee Kshs, Billions 1997-2005

Year	Tea	Horticulture	Coffee
1997	22.8	13.75	16.86
1998	32.97	14.94	12.81
1999	33.10	17.64	12.00
2000	35.2	21.22	11.70
2001	34.49	19.85	7.46
2002	34.38	28.33	6.54
2003	33.01	36.49	6.29
2004	36.07	39.54	6.94
2005	42.29	44.56	9.70

Source: CBS Economic Surveys (Various Issues)

The agriculture sector contribution to total exports has been declining and reached 47% in 2005 down from 69% in 1985. Agricultural trade cover-ratio (Figure 27) declined indicating surging agricultural imports against exports. The declining trend continued through the reforms period.

Figure 27: Agriculture Cover Ratio



Source: CBS Economic Surveys (various Issues)

3.5. Trade Agreements

Kenya became a signatory of East African Community (EAC) in 1967. This provided economic co-operation between Kenya, Uganda and Tanzania. With the collapse of the community in late 1970s' the country's exports market contracted and the government moved to re-organise the structures and institutions to ensure continuation of the performance of the functions of the community (GoK, National Development Plan 1989-1993). In 1983 Kenya became a member of the Preferential Trade Area (PTA) which acted as an instrument for collective action in trade liberalization. This coincided with the introduction of the SAPs. Trade liberalization in various African countries opened gates for Kenya's exports. Since 1995, coinciding with the accession to WTO, and further spurred by COMESA and re-emergence of the EAC, African countries have provided the largest market for Kenyan exports.

- **Common Market for East and Southern Africa (COMESA)**

COMESA was established in 1993 with a major focus of enhancing internal and external trade within the member countries through overcoming barriers encountered by individual member countries. It is now a 20 member country agreement with an estimated total population of 385 million people. COMESA region has perhaps led to the African continent becoming the leading destination for Kenya's exports. This may be attributed to the creation of a Free Trade Area in the COMESA region, in October 2000 which entailed the removal of all internal trade tariffs and barriers. Exports to COMESA have increased from Kshs. 51.4 billion in 1997 to Kshs.56.7 billion in 2000 representing an increment of 10.3%. Imports decreased from Kshs. 6.6 billion to 5.2 billion during the same period. COMESA has also facilitated the establishment of the African Trade Insurance Agency (ATIA) to insure investments against potential risk. Recently the COMESA member formed a customs union with a common external tariff. The most important markets for Kenya are the Uganda and Tanzania each making up of a third of the exports followed by democratic Republic of Congo making for 16% in the year 2003.

- **East African Community (EAC)**

This is a regional intergovernmental organization between Kenya Uganda and Tanzania. The 'new' EAC is a revival of the original EAC, a customs union that was established in 1967 but collapsed in late 1970s. The three member states had a combined GDP of $31.4 billion and a population above 90 million (Meredith, 2005) It entailed a customs union effective January 2005 with a common tariff at 0, 10, and 25 percent for raw materials, semi-processed and finished products respectively

Since the transformation of East African Co-operation into a community in 1990, Kenya's exports to the region have grown from 8% in 1990 to 24.4% in 2000.. Increased trade is expected following the transformation of East African Community into the East African Customs Union through elimination of internal tariffs and establishment of common external tariff.

- **IGAD**

The Intergovernmental Authority for Development (IGAD) endeavours to achieve regional co-operation and economic integration for the countries in the horn of Africa sub-region through the promotion of food security, sustainable environmental management, peace and security, intra-regional trade and improved infrastructure facilities. It also entails capacity building in the areas of conflict prevention and alleviation and mitigation of humanitarian crises

- **ACP-EU**

Kenya became a signatory of the African Caribbean and Pacific ACP-EU (the famous Lome convention) in 1975. The ACP-EU trade pact had both trade and aid provisions and was based on the colonial links between the ACP countries and their former colonial masters. The trade pact provided for non-reciprocative trade and aid between these two parties and almost 97% of ACP exports were allowed duty free access to EU markets. The trade agreements were renewed every ten years and aimed at increasing the export income of the ACPs, promoting industrialization and accelerating economic growth. The Lome IV convention expired in 2000 and was succeeded by the Cotonou Agreement that was extended to 2008 after which Economic Partnership Agreements (EPAs) will be effected. In this regard the ACPs will be required to sign Economic Partnership Agreements (EPAs) (Ronge, 2006). EPAs are designed for consistency with WTO provisions. Its negotiation in regional blocks and the requirement for reciprocity places Kenya in a precarious position. This is due to the prominent position Africa holds in Kenya's exports especially recognizing competition that will arise since these products are manufactures. In addition, the situation presents a contrast owing to the fact Kenya's regional partners export raw materials to the European markets.

- **AGOA**

The African Growth opportunity Act (AGOA) is trade agreement between US and designated 37 Sub-Saharan African countries including Kenya. The Act came into place in the year 2000 and was expected to be in place for 8 years but was amended to cover up to 2015. Under this Act, the high duties and restrictive quarter that have been imposed on exports of textiles have been lifted (GoK, 2002-2008)

- **WTO**

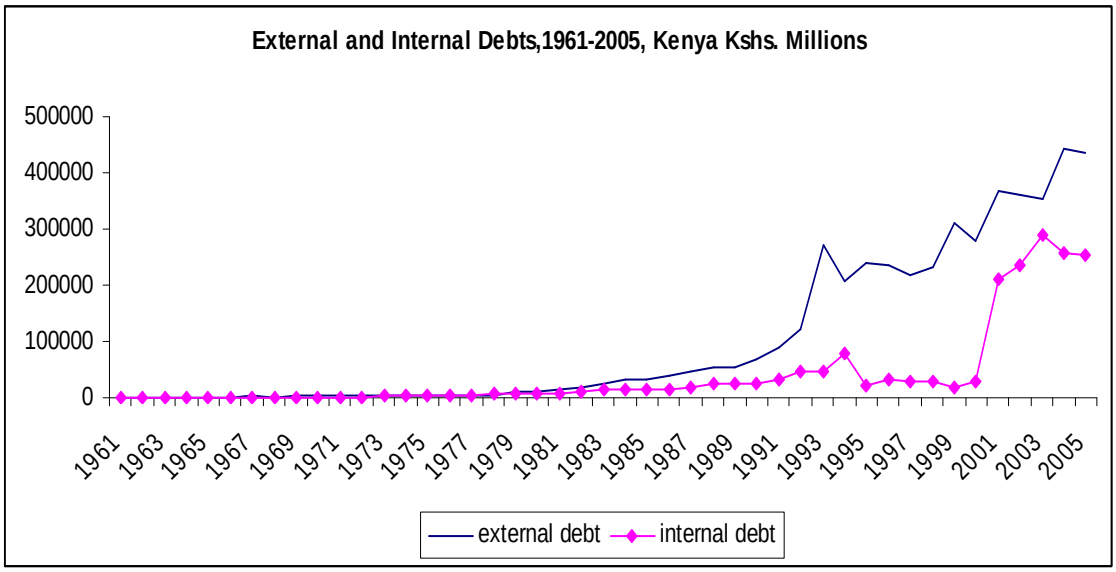
The World Trade Organization came into effect in 1995 as a successor to the General Agreement on Trade and Tariffs (GATT) to facilitate global free trade and provide forum and mechanisms resolving trade disputes. Kenya being a member since the inception of the Marrakech Agreement of 15th April 1994 signed in Morocco and completed in 31st December 1994 when accession to WTO was completed. Kenya is a signatory to all WTO agreements among them- the General Agreement on Trade and Tariffs (GATT), Agreement on Agriculture (AoA), General Agreement on Trade and Services (GATS), the Agreement on Textiles and Clothing (ATC), and the Agreement on Trade-Related Intellectual Property Rights (TRIPS).

4. Public Finance

The trend in external and internal debts is presented in Figure 288. In 2005 the total debt stood at 33.6% of the factor cost GDP up from 20.6% in 1964. The total debt rose to 689,003.98 million Kshs in 2005 from 1383.4 million Kshs in 1961. In 2005 the external debt rose to 434,453 million Kshs while the internal debt 254,550.98 Kshs from 1026.4 and 357 respectively in 1961 (Economic surveys Various Issues).

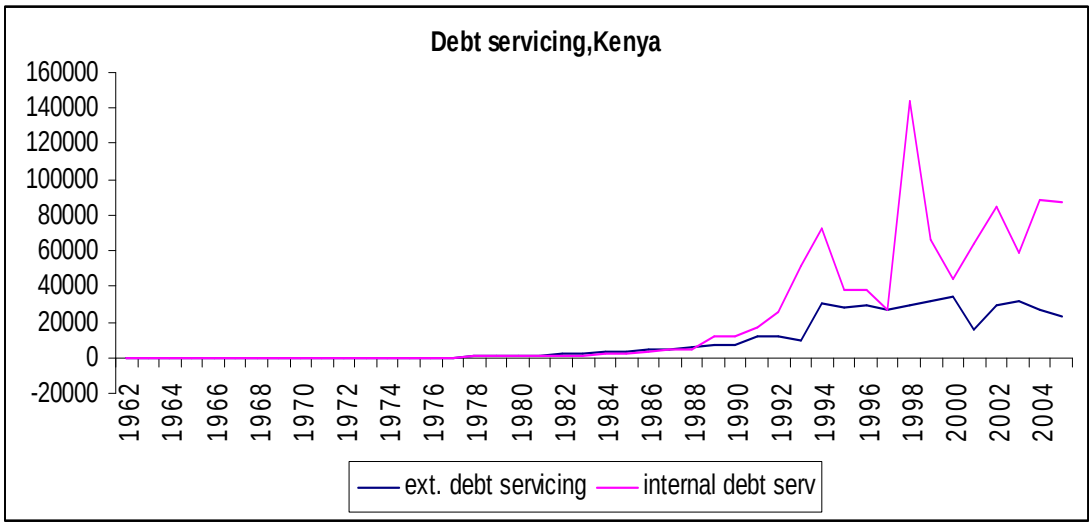
The increase in the debts coincided with the SAPs implementation and gradually grew further as the economy became more open.

Figure 28: External and Internal Debts, 1961-2005



Source: CBS Economic Surveys (1982-2006)

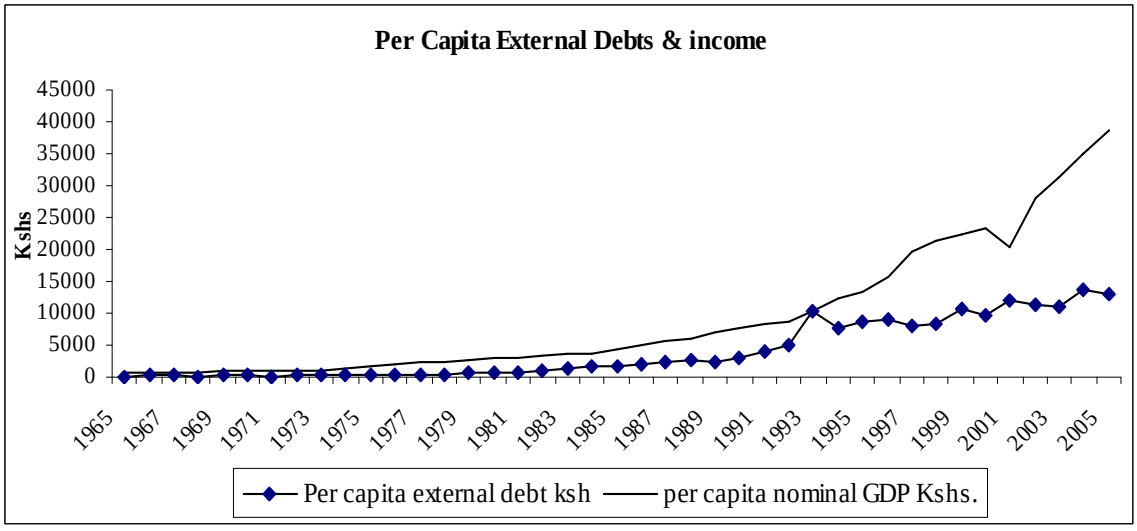
Figure 29: Debt Servicing in Kenya



Source: CBS Economic Surveys (various Issues)

Debt servicing for both external and internal debts has also been increasing (Figure 29). In the last three years external debt servicing has been on decline. In 2005 the economy incurred an expenditure total of 110,520.77 million Kshs in debt servicing for both external and internal debts. The external debt service amounted to 23404.19 million Kshs in 2005 up from 15 million Kshs in 1962. This implies that every Kenyan is incurring an expenditure of 700.72 Kshs annually to external debts service.

Figure 30: Per Capita External Debts & Per Capita Income



Source: CBS Economic Surveys (various Issues)

The per capita external debt service annual growth in 2004 was at 24% but went down to -3.7% in 2005. The total per capital debt accounted for 53.4% of the income per capita in 2005 up from 28.9 % in 1965. The pre-liberalization era saw both the external and internal debts rise. This was attributed to increased financial demand by the government to finance post-independence projects and provision of subsidies under the import substitution policy. The period starting early 1990s was characterized by

reduced foreign aid which only left the government with domestic borrowing that grew by more than fivefold in this period. The average annual internal debt growth rate during this period was estimated at 73%. The post liberalization era has seen the reemergence of external borrowing which is associated with the demand for funds for economic recovery and revitalization of agriculture. The growth of debts and their servicing coincides with the SAPs implementation. The growth in external debts started immediately when the country obtained its first loan for the adjustment programs in the early 1980s. In 1994, the per capita income (nominal) nearly equalled the per capita debt service (Figure 30). The liberalization era has witnessed marked but erratic growth in the external debt.

5. Environmental and Natural Resources

The country total surface area is 58,037,000 Ha (582,646 sq. km); the total land area is 56,914,000 Ha (571416 sq. km). This composes of the high potential 12%, medium potential 6%, low potential 74% and the other lands 9% (Table 12) The largest portion (45 %) of the high potential land lies in the Rift Valley province while the Eastern province harbours the largest portion of the medium potential land approximating 69%. North Eastern has the largest low potential land with 30.1%, Rift valley with 29% and Eastern 27%.

Table 12: Quality of Land ('000 ha) (fertility, erosion)

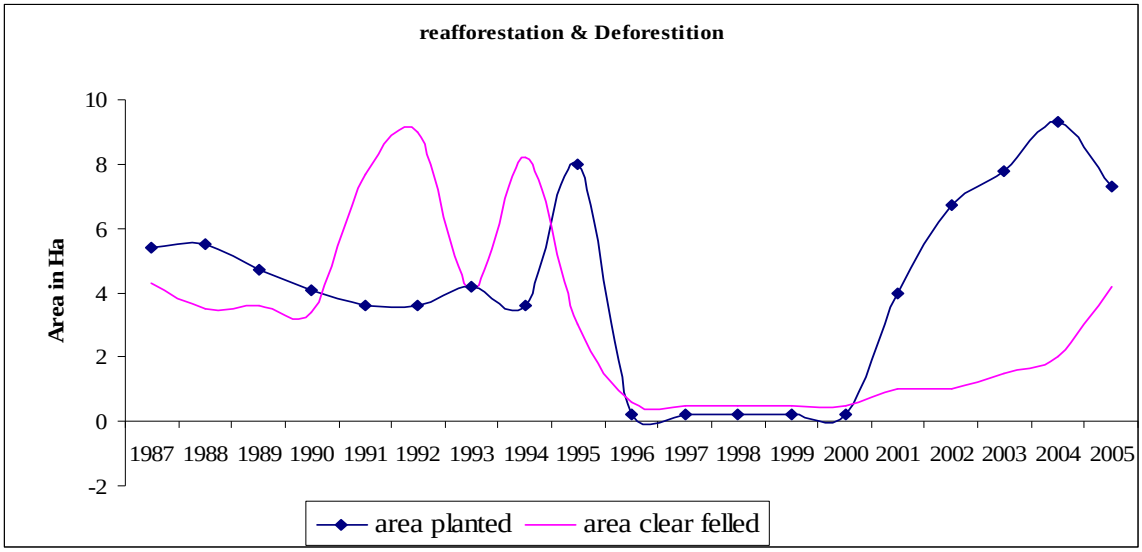
Province	High potential	Medium potential	Low potential	Total	All other land	Total land area
Central	909	15	41	965	353	1318
Coast	373	796	5663	6832	1472	8304
Eastern	503	2189	11453	14145	1431	15576
Nairobi	16	-	38	54	14	68
North-Eastern	-	-	12690	12690	-	12690
Nyanza	1218	34	-	1252	-	1252
Rift valley	3025	123	12220	15368	1515	16883
western	741	-	-	741	82	823
Total	6785	3157	42105	52047	4867	56914

Source: Statistical Abstract 2003

- **Forestry Resources and Deforestation**

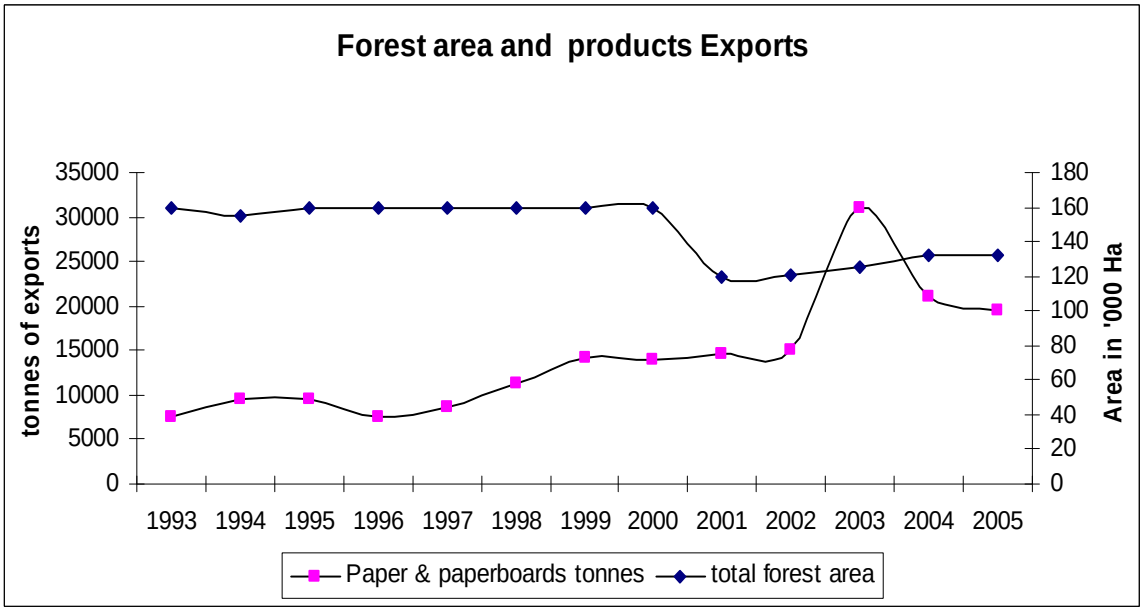
In 2005 the country's forest cover was estimated at 132,200 ha. This makes only 2.3% of the total land area in the country. Of this area, 9% of the forest cover was under indigenous trees, 75 % under exotic trees and 16% under exotic fuel wood. The area planted with trees increased tremendously from 2000 to 2004 but showed a decline afterwards. At the same time the area clear felled also increased but at a slower rate than the area planted (Figure 31). Forest products such as wood manufactures, paper and paperboards have earned the country foreign exchange; however, they have also contributed to depletion of forest stock through deforestation. An increase in the paper and paperboard exports was observed between 2002 and 2004 which explained the sudden decline in the area under forests in the country. The decline in forest cover coincided with the reforms era and highest level of forest products exports a phenomenon that occurred during the late 1990s (Figure 32). The total area under forest cover maintained a constant growth from 1993 to 1999, there was a sharp decline from 2000 to 2001.

Figure 31 Reforestation & Deforestation



Source: CBS Economic Surveys (Various Issues)

Figure 32: Forest Products Exports



Source: CBS Economic Surveys (various Issues)

CHAPTER 3 - CHARACTERISTICS AND STRUCTURES IN THE AGRICULTURAL SECTOR

Kenya's agricultural production is characterised by perishability, regional segmentation in production and seasonality. The fact that agricultural production in Kenya is regional and dependent on climatic conditions forms the basis of the segmentation hypothesis. This follows that if trade reforms favoured a specific agricultural sub-sector then owing to the regional nature of agricultural production, the effects were manifested regionally. The agricultural sector is dominated by primary production of a few commodities namely: cereals (maize, wheat and rice), traditional food crops (pulses, roots and tubers, millet and sorghum), industrial crops (sugar, pyrethrum, cotton, tobacco and sisal), exports crops (tea, coffee and horticulture) and livestock (milk, meat and eggs). Kenya's agricultural sector is characterized by smallholder mixed farming. Smallholders account for over 65 percent of the total agricultural output. Pastoralism is the main form of production in the Arid and Semi-Arid lands (ASAL) areas.

1. Some Structural Characteristics

- **Land Use and Structure and Land Holding Size**

Arable land is approximately 6.9 million hectares, of this smallholding, medium and large farm utilizes 39%, 46% and 15% respectively. About 3.2 million acres is subdivided into 3.5 million holdings with average ownership of 1.2 hectares (GoK, 2002). The arable land is devoted to crops and livestock production as follows: the dairy sector utilizes 2.4 million hectares or 47.7%, production of maize and beans utilizes approximately 23% and the remaining 31% is under export and industrial crops including wheat sorghum, millet, potatoes and other crops including horticultural commodities.

The main economic activity of the ASALs is nomadic pastoralism, which utilizes 24.2 million hectares approximately 50% while ranching and other livestock keeping utilizes 15.1 million hectares or 31% the remaining 9.1 million hectares or 19% is used for agriculture and agro-pastoralism. The ASALs constitute an estimated 84% of the total land surface, habouring around 25% of the population and supporting 50% of the total livestock (GoK, 2002-2008).

- **Agricultural Inputs**

Gradual decline in land holding sizes since independence has resulted in demand for intensive agricultural production technologies to increase scale returns to land and labour. Technological evolution has been embodied in agricultural inputs such as seeds and fertilizers. Increase in use of fertilizers, improved seed varieties use and use of mechanized labour is evident in agricultural production with significant effect on the agricultural sector and the rural areas. Technical evolution embodied in the use of factors of production (land, labour and capital), other production inputs such as high yielding seeds, fertilizers, agricultural technologies among others have changed overtime owing to the changing demand in agricultural products and the supply of these inputs to the various economic agents. While there has been increased use of fertilizers over the period under reforms, mechanized labour has been witnessed to decline relative to fertilizer and manual labour use. According to Tegemeo panel surveys, there was a 47% increase in fertilizer use between 1995/96 and 2003/04. The number of smallholders using fertilizers also increased over the period. The increasing trend in fertilizer consumption intensity has been maintained over the period of reforms. Inputs ratio analysis shows that fertilizers are the highly used inputs followed by manual labour and then mechanized labour. Fertilizer and improved seed use seems to have been greatly influenced by reforms that led to liberalization of the fertilizer market. From crop-fertilizer price ratios; it is evident that reforms in the fertilizer market have led to increased use fertilizers generally. Increase in

fertilizer prices coupled with increasing soil degradation in marginal areas has alienated poor farmers who cannot afford them. Further, the inefficient distribution channel has failed to supply some of these areas with fertilizers.

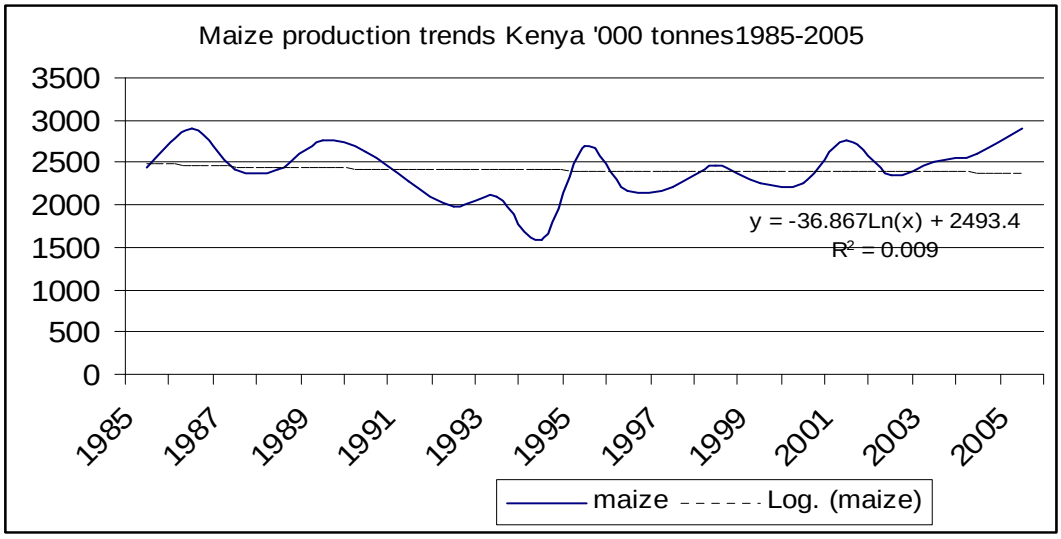
2. Agricultural Sectors

2.1. Evolution of Yields

2.1.1. Maize

Maize is the staple crop for the country and is grown country wide from low altitude to high altitude depending on the variety. Maize production has been fluctuating over the period with a general declining trend (Figure 33). The increase in maize output in the 80s was perhaps due to increased use of high yielding varieties and fertilizers. However the trend was afterwards not maintained. The yields have however shown a slight upward growing trend as from mid 1990s.

Figure 33: Maize Trends in Kenya, 1985-2005



Source: CBS Economic Surveys

2.1.2. Coffee

Coffee is grown between altitude 1,400 and 2,100 metres above sea level. The main production regions include Northern part of Nairobi, high plateau surrounding Mt Kenya, the Aberdare zone, West areas of Kisii, Nyanza, Bungoma, Kakamega, Rift Valley areas of Nakuru,Trans Nzoia and Taita Hills near Mt. Kilimanjaro (Nyangito, 2005). The coffee sub-sector in Kenya is one of the major employers and contributed about 5% of export revenues in 2003 (Nyangito, 2005). The sector is regulated by the Coffee Board of Kenya (CBK), that initialy monopolized coffee marketing. Regulation covered all stages of coffee production, processing and marketing. Coffee farming is mainly done by small-scale farmers organized into co-operative societies who account for 60% while 40% is done by large scale farmers at plantation level (Nyangito 2005). Cooperatives would provide credit to farmers during the planting season and the farmers were in turn required to deliver the produce to the cooperatives' pulping stations

The coffee sector has suffered from monopolistic practices, co-operative mismanagement, repeated droughts, falling international prices and weak infrastructure. In 2001, faced by the need to revitalize

the declining coffee sector (after a one third fall in exports in 1995/2000) the Coffee Act introduced competitive practices in the milling and marketing of coffee. In 2002, the government announced the end to the Coffee Board's marketing activities and the retrenchment of 90 percent of its staff, hopefully ending decades of state control over the coffee market. However, the 2002 coffee Act did not improve the plight of small holders leading to an amendment of the coffee act being introduced through the finance bill (2006) to streamline the sub-sector's activities and hopefully change the fortunes of the small holders

- **Coffee Production**

Co-operatives coffee area increased from 13,000 hectares to 128,000 hectares between 1964 and 2005, while the estates area under coffee increased merely from 32538 hectares to 42000 hectares in the same period. The near constant coffee area maintained by the estates indicates that policies articulated over this period of time did not favour the expansion of estate coffee. Increases in the total area under coffee has been driven mainly by smallholders.

The land policies involving land redistribution, subdivision of some estates, removal of restrictions that constrained Africans from planting cash crops and promotion of the smallholder agriculture since independence led to increase in smallholder/cooperative coffee area. The trend of reforms as adopted from early 1980s seems to have favoured co-operatives more than the estates by resulting to increase in area under coffee. The co-operatives coffee area grew steadily from independence but has substantially declined over pre-liberalization, liberalization, and post liberalization periods (Table 13)

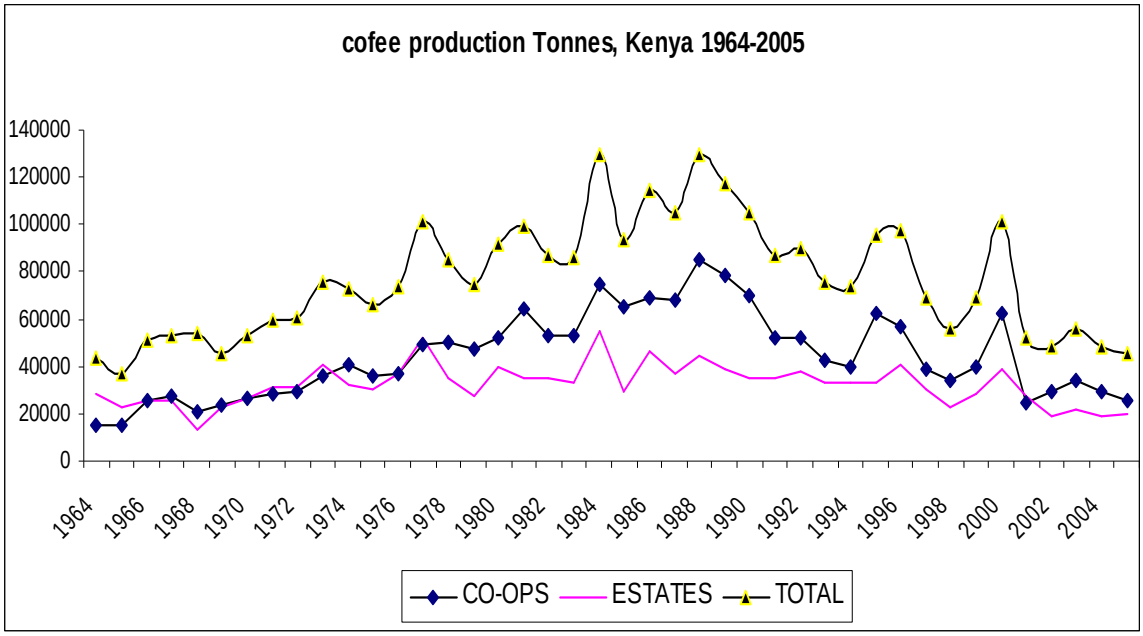
Table 13: Average annual growths in area and production 1964-2005

Period	Pre-liberalization 1964-1980	Transition 1981-1992	Liberalization 1993-2002	Post-liberalization 2003-2005
Area				
Co-operatives	12.62	4.66	0.58	0.19
Estates	-0.22	2.00	0.55	1.56
Total	5.45	3.82	0.57	.052
Coffee output				
Co-operatives	9.78	1.68	0.82	-3.04
Estates	6.52	3.88	-3.77	1.33
Total output	6.15	1.80	-2.14	-1.31

Source: Coffee Research Foundation (CRF) and CBS, 2006

The total national coffee output mirrors the co-operatives coffee output over the whole period with the co-operatives showing little variation in their output. The national total and co-operative output have had a phenomenal growth since pre-independence to the late 1980s when the trend started to decline (Figure 34). The trend exhibits clear effects of the 1976/78 coffee boom, the 1984 drought and the 1986 mini coffee boom for both the co-operatives and the estates. The two booms served to increase output in subsequent years with the drought reversing the trends. Coffee production has declined significantly from a peak of 129,000 tonnes in 1988 to 45,200 tonnes in 2005. This decline coincided with reforms. The decline in world coffee prices as from late 1980s marked the start of decline in coffee output for both co-operatives and estates. The co-operatives were adversely affected with their output declining from 84923 tonnes in 1988 to 25500 tonnes in 2005 while, the estates production declined from 44712 tonnes to 19700 over the same period. Following the WTO and COMESA inception in 1995 and 2000 respectively, coffee outputs were witnessed to increase in these two periods but followed by subsequent drops.

Figure 34: Coffee Production by Producer and at National Level, 1964-2004



Source: Coffee Research Foundation (CRF),2002, Economic survey 2006

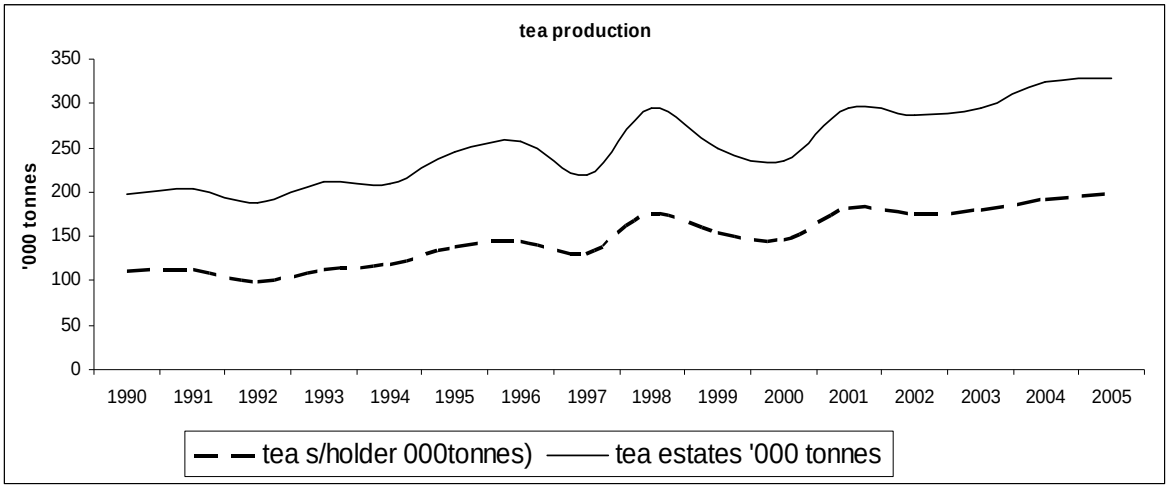
2.1.3. Tea

Tea growing regions in Kenya are found mostly in the Great Rift Valley, In the East of the Rift tea is grown in the slopes of Aberdare highlands, Nyambene hills. In the West of the Rift tea growing takes place in the Nandi Hills, highlands around Kericho, Mt. Elgon and the Kisii highlands. The altitude in these slopes range from 1500 to 2700 above sea level. These regions are endowed with an ideal climate for tea growing. The tropical, volcanic soils rich in nutrients give the tea a unique flavour and character. The rainfall ranges between 1200mm and 2700 mm annually.

Tea planting and production has expanded since 1963 from 18,000 tonnes and 21,488 hectares (Nyagito, 2001) to 328,500 tonnes and 141,300 hectares in 2005 (CBS, 2005). Small-holders production has risen from a mere 1.7% of the total amount of tea produced in 1963 (Nyagito, 2001) to 60.2% in 2005 (CBS, 2005, Nyangito, 2005). Figure 35 below presents the trend in tea production for small-holder and estate producers.

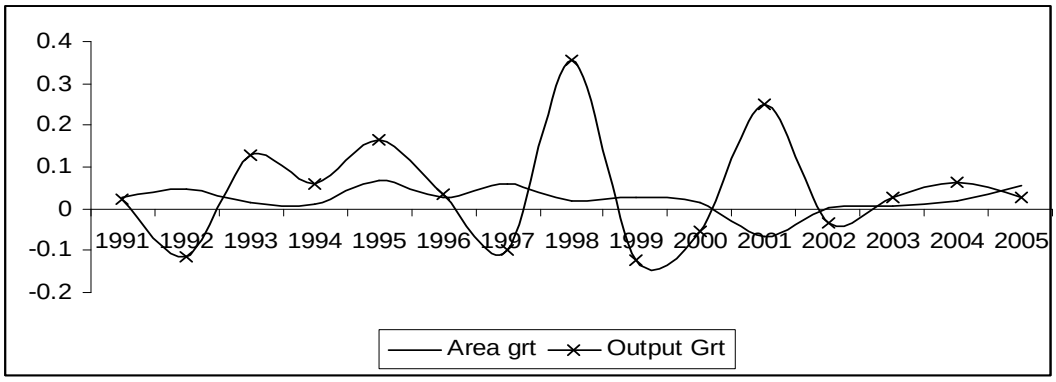
While liberalization may have improved small-holder tea productivity, there are indications that the reforms have been unfavourable to estate tea which declined drastically. The adoption of high yielding tea varieties and high input use, proper husbandry practices and improved competition in service delivery and marketing increased the small-holder tea output but with little variation in the area under small-holder tea (Figure 36).

Figure 35: Tea Production



Source: CBS Economic Surveys. (Various Issues)

Figure 36: Small-Scale Tea Area and Output Growth Rate 1991-2005



Source: CBS Economic Surveys, 1994-1996

The area under tea has increased overtime as shown in Table 14. The growth during the early years of independence was spurred by the land sub-division and reallocations policy adopted by the government. Onwards the increasing world tea prices coupled with good services by the Kenya Tea Development Agency (KTDA) (box 1) may have been responsible for the increase in the area. However, realizing that the future increase in output will not be based on putting more land under tea, the government through the Tea Research Foundation (TRF) embarked on vigorous research to develop appropriate tea. The growth rates in area under tea has been fairly constant with a drastic growth occurring in late 1980s and early 1990s. However the output has been steadily increasing indicating that research efforts on intensive rather than extensive production are taking effect

Table 14: Tea Area and Production

Period	Pre-liberalization 1974-1980	Transition 1981-1992	Liberalization 1993-2000	Post-liberalization 2001-2005
Area under tea '000 Ha	43.1	57.4	112.2	133.4
Output	25,288	100,765	250,387	315,600

Source: CBS Economic Surveys. (Various Issues)

Box 1: Kenya Tea Development Agency Ltd (KTDA)

Previously a state corporation, KTDA was incorporated as a private company in June 2000. KTDA Ltd is the management agency of the smallholder tea growers. The agency is responsible for the management of smallholder tea through provision of extension services, production inputs, green leaf collection, processing and marketing of processed tea on behalf of the smallholders. KTDA Ltd. currently manages 53 tea factories in the smallholder sub-sector serving over 400,000 growers.

2.1.4. Horticulture

The total hectarage under horticulture in 2004 was estimated at about 387,535 ha, out of which 99,497 ha were under vegetables, 128,484 ha were under Irish potatoes, 152,837 ha under fruits, 2583 ha under macadamia nuts, 1954 ha under herbs & spices, and 2177 ha under flowers (MoA, 2004).Horticultural sector has developed predominantly under the private sector with little government interference. Horticultural exports grew from 32000 tonnes in 1974 to 163200 tonnes in 2005 while the exports value grew from 3 million Kenya shillings to 38.8 billion Kenya shillings during the same period. The principal horticultural exports are cut flowers, French beans, garden peas, mangoes, avocadoes, processed and fresh pineapples and Asian vegetables. According to the 1993-2001 National Development Plan, horticultural production is dominated by small-scale growers who number about 1.8 million and constitute 80% of all the producers and account for an estimated 70% of the horticultural output (HCDA, 2002).

Table 15: Horticultural Export Quantities, Kenya 1994-2005

Year	**Flowers**		**Fruits**		**Vegetables**		**Total Export Quantities**	
	Quantity Mn MT	Area '000 Ha	quantity Mn MT	Area '000 Ha	Quantity Mn MT	Area '000 Ha	quantity Mn MT	Total Area '000 Ha
1994	-		-		-		65.2	
1995	29.4		13.9		28.5		71.8	
1996	35.2	1	16.9	95	32.7	83	84.8	179
1997	35.9	1	17.5	129	30.9	89	84.2	219
1998	32.5	1	10.3	135	30	92	72.8	228
1999	37	2	15.6	136	46.4	98	99	236
2000	38.8	2	15.4	136	45	89	99.2	227
2001	41.4		22.6		34.8		98.8	
2002	52.1		22.5		46.5		121.1	
2003	61		23.6		48.7		133.2	
2004	70.7		23		51.9		145.6	
2005	81.2		18.5		63.4		163.2	

Source: HCDA, National development Plan 2002-2008

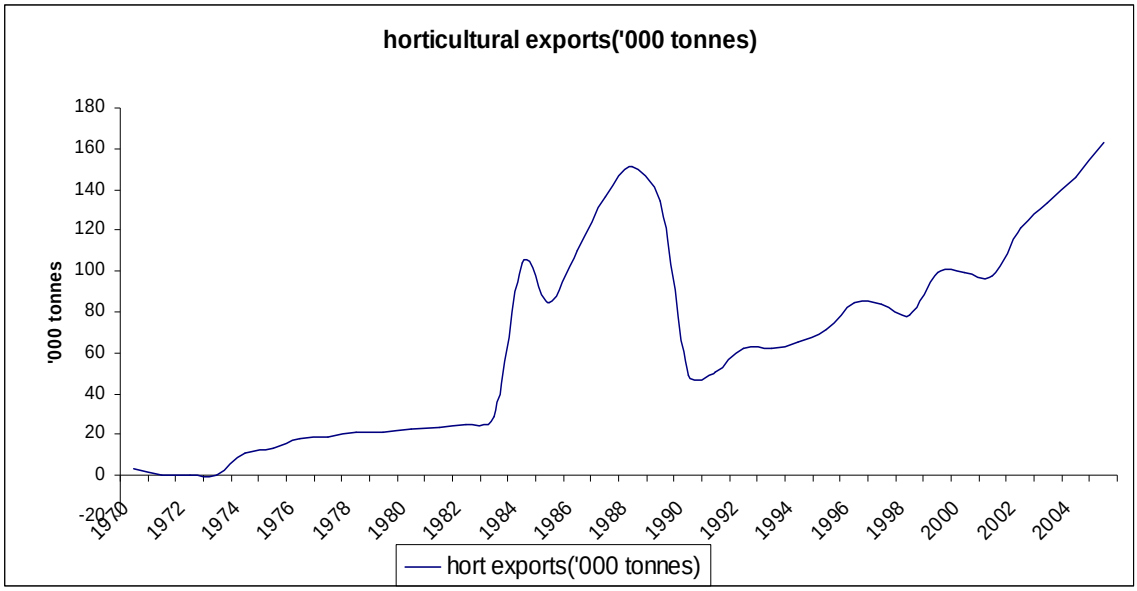
It is estimated that the industry supports more than a million workers in farms, packaging and export. However the sector's growth has been adversely influenced by transport infrastructure, inadequate airfreight and cooling facilities, fluctuations in electricity supply and international standards.

Despite these constraints the sector's has continued to grow with exports of vegetables, fruits and cut flowers rising. The exports value of cut flowers grew from 29.4 million tonnes in 1995 to 81.2 million tonnes in 2005 (Table 15).

Although there was an increase in the quantity of horticultural exports in the 1980s the value of the exports did not increase owing to the poor packaging which reduced the quality of the produce hence fetching poor prices in the export markets (CBS,1992).

The quantities of exports declined in the early 1990s due to the stringent quality standards which were put in place for world horticultural exports (Figure 37).This led to exit of small farmers who are, according to middlemen, unable to meet the high quality of produce demanded by buyers in Europe and North America. In the UK, the large supermarkets have captured the market for imported fresh vegetables (Dollan et.al IDS 1996). The supermarkets play a critical role in defining what is produced, how and by whom. The requirements they specify for cost, quality, delivery, product variety, innovation, food quality and safety systems help to determine the types of producers and processors are able to gain access to the fresh vegetable chain and the activities they must carry out.

Figure 37: Horticultural Exports (000'tones), 1970-2005



Source: CBS Economic Surveys. (Various Issues)

2.1.5. Sugar

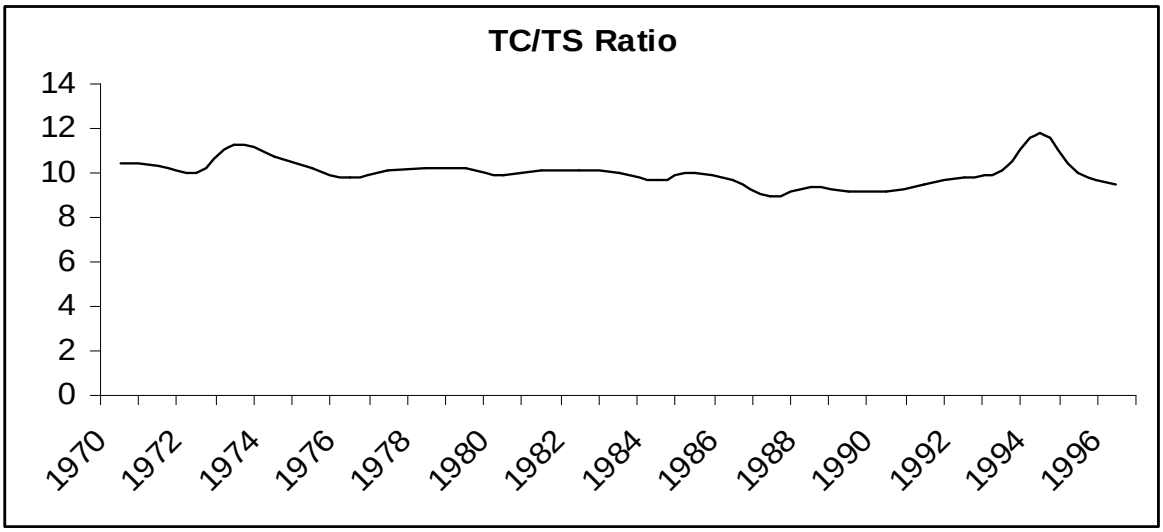
Sugar is grown mainly in the Western regions of the country. The Government has since then been widely involved in the expansion of sugar production through investments in sugar cane growing schemes and factories. Following independence the government involvement in sugar was based on various reasons: i) the need for self sufficiency, ii) foreign exchange, iii) savings as well as social and economic development through employment and wealth creation in rural areas. Cane production increased from 1.7 million MT in 1975 to 4.8 million MT in 2005. The area under cane grew rapidly from 46.8 thousand hectares in 1977 to 104 thousand hectares in 1988, declined between 1989 and 1993 and recovered to reach 144.8 thousand hectares in 2005. Initial policies of land subdivision and import substitution pursued by the Kenyan government led to protection of the domestic sugar industry. It is during this time that Kenya became a signatory of the famous 1975 Lome convention which granted Kenya an export quota to EU that was later withdrawn and distributed to other ACP countries due to Kenya's inability to supply.

Area under cane has grown from the early 1970s and over the liberalization era. The culmination of this was production deficit and consequently sugar imports. An index of production such as the ratio of cane crushed to sugar produced (TC/TS) can be used to assess the quality of the cane and processing efficiency under assumption of positive correlation. A computation of this index indicated

gradual decline since mid 1970s to early 1990s(Figure 38). The decline in this index indicated the drop in the quality of cane and poor milling efficiency which is a manifestation of poor machine maintenance and low quality varieties.

The result of this phenomenon has been widening gap between sugar demand and production in the country. While sugar production has been declining, sugar demand in the country has increased prompting sugar imports into the country. The widening of the deficit has been more pronounced since early 1990s when the economy was classified as open. Table 16 presents the trends of cane and sugar production over the reform period

Figure 38: TC/TS Ratio



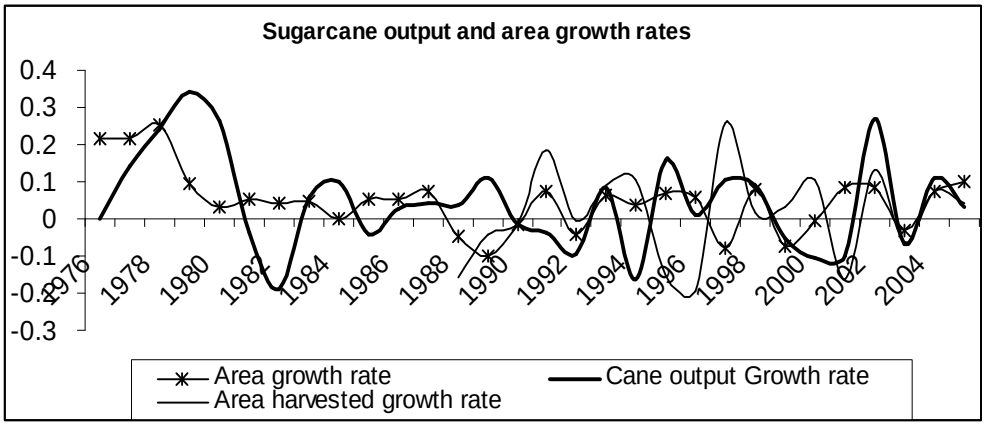
Source: CBS Economic Surveys. (Various Issues)

Table 16 Sugar and Cane Average Annual trends

Period	Pre-liberalization 1970/80	Transition 1981/92	Liberalization 1993/02	Post-liberalization 2003/05
Sugar production (tones)	187.9	310	405.8	484.5
Average consumption (tones)		414.8	592	676.4
Per capita sugar consumption (Kgs)		19.5	20.6	20.6
Average deficit (tones)		-33.77	-186.21	-196.96
Area under cane (Ha)	40,060	95,660	112,271	132,967
Cane output (Tonnes)		3711.6	4031.1	4555.3

Source: CBS Economic Surveys. (Various Issues)

Figure 39: Cane Output, Area Planted and Area Harvested Growth Rates



Source: CBS Economic surveys and Sugar year book

The area planted with cane, the area of cane harvested and output have fluctuated over the period due to the unfavourable climatic, management and trade agreements such as COMESA and ACP-EU and policies (Figure 39).

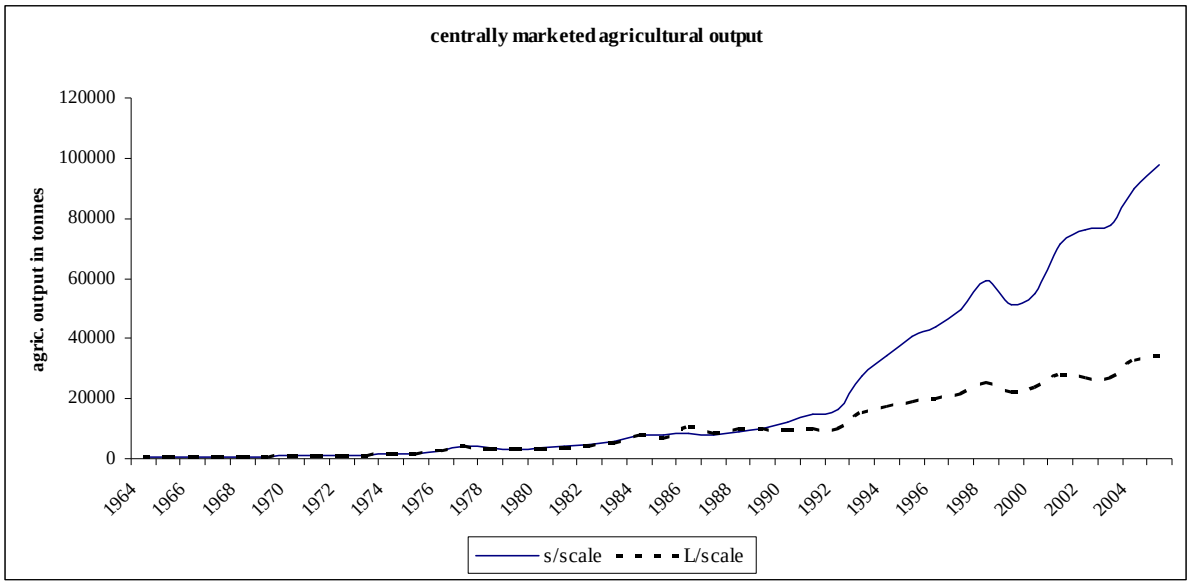
2.2. Commercialization and Transformation

Agricultural commercialization began during the settler period. During this period credit and marketing interventions were made by the government for production losses due to calamities and economic shocks. The formation of produce pooling farmer associations such as the Kenya Farmers Association (KFA) for wheat were made to enjoy economies of scale that emanated from this chain of marketing. Replication of these associations which came to be the post-independence government parastatals formed the basis of agricultural development. The government captures centrally marketed agricultural produce through these parastatals. Prior to reforms marketed produce indicated relatively equal market participation by both production segments (small-scale and large-scale). The variation in participation however, increased with increasing openness favouring the small-scale rather than large scale farmers (Figure 40).

Perhaps this phenomenon is attributable to the fact that liberalisation led to the repealing of the price control Act during this period consequently increasing the chances of small-holders' participation in the market, while the large scale farmers readjusted to recapture the lost market share. Table 17 presents the annual average growth rates over the period of reforms. It is evident that liberalization favoured the small-scale sector. However, the trend seems to reverse in the post liberalization with the large-scale growing slightly at a higher rate owing to the readjustment by the large-scale to suit competitive markets.

Agricultural production has not been stable since early 1980s when reforms were initiated. While it is the main source of economic growth, employment and foreign exchange, the sector has recorded negative growth rates over the reform period. Economic reforms were intended to spur growth and reduce poverty by enhancing production efficiency and consequently output. This has not been achieved with poverty increasing and agricultural productivity declining.

Figure 40: Centrally Marketed Output



Source: CBS. Economic Surveys. (Various Issues)

Table 17: Annual Average Growth Rate in the Centrally Marketed Agricultural Output

Period	Pre-liberalization 1965/80	Transition 1981/92	Liberalization 1993/02	Post-liberalization 2003/05
Total marketed agricultural output	13.44	12.4	16.06	8.74
Large-scale	12.62	11.02	11.76	8.94
Small-scale	14.92	13.90	18.19	8.69

Source: CBS Economic Surveys. (Various Issues)

- **Agro-food Production in Kenya**

Food production in Kenya is centered on maize which is the staple food in the country. Maize production over the reform period has shown a gradual overall decline while the production of potatoes and beans have exhibited a decline prior to cereals industry liberalization in 1992 but then the trend reversed afterwards. The overall consumption in maize has been higher than the production with deficit being corrected through imports.

CHAPTER 4 - PUBLIC POLICIES[2]

1. General Characteristics and Periodization of Public Policies

The first half of the transition period (1980-84) marked the first and failed attempt at economic adjustment partly because commitment to reform was limited. In the second half of the transition period (1985-1992), broader consensus was achieved and some trade liberalization measures were implemented. However, agricultural reforms lagged far behind. Implementation remained weak and commitment uneven with some sectors of the economy being earmarked as the priority for liberalization. In 1981 the government embarked on the replacement of non-tariff barriers with equivalent tariffs. One major flaw was that structural adjustment took place within an environment of severe fiscal laxity and weakened economy. The beginning of the 1990s marked a third wave of economic reform when interest rates were liberalized and exchange rates allowed to float. Fiscal adjustment followed in the 1994-1999 period. Limited progress was also made in reducing government intervention in the economy by privatizing state-owned enterprises, and liberalizing agricultural markets.

1.1. Import Liberalization

Import substitution formed the backbone of Kenya's economic policy in much of Kenya's post-colonial history. After independence, Kenya pursued an inward oriented policy of import substitution giving the domestic market the chance to thrive successfully. The government then shifted to strict import restrictions involving:

- Quantitative restrictions
- High tariffs on competing imports
- Exchange rate controls
- Controls on importation and licensing
- Controls on domestic prices and wages
- Taxation of exports and
- Requirement of "no objection certificates" (to be allowed to export) from domestic producers

A system of tariffs to protect local manufacturing had been in place since the 1940s and 1950s, and perpetuated thereafter, in different forms. Import duties averaged 3.4 percent of GDP in the 1972-77 (customs union) period, rose to 4.5 percent in 1978-1984, and fell back to 3.3 in 1985-92 and 3.6 percent in 1992-98.

In 1967, with the creation of the customs union with Uganda and Tanzania, protection was ensured through a system of import quotas, which, in addition to providing barriers to entry, was also a source of rent. The system was kept in place even after the fall of the regional customs union in 1977 when

[2]This area draws from Arianna, L., 2002. Kenya: Macroeconomic Evolution since Independence. WB

Tanzania closed its border and Uganda dealt with internal civil instability. Consequently Kenya's export markets were hard hit.

The fourth development plan for 1979-84 recognized the need for a more outward-oriented approach and called for the elimination of quantitative restrictions, and lowering and simplification of tariffs. In 1980-85, when items were allocated to restricted and unrestricted categories, the share of imports without quantitative restrictions doubled to 48 percent and average tariff reduced by 8 percent. Reforms, however, stalled and revenues from import duties rose rather than declined in the period. During the foreign exchange crisis of 1982-84, tariffs further increased by 10 percent. The 1984 drought that occurred during this period led to increase in imports as the government was obligated to source and supply adequate relief food. The 1989-93 National Development Plan recognized the need to reduce import tariffs and quantitative restrictions which were typified by import licensing and foreign exchange rationing as too restrictive and recommended a comprehensive review of the tariff structure with the aim of lowering the rates and achieve equitable levels of protection. Further it was recommended that the exchange rate movement be more market oriented as a means of rationing rather than the use of controls and licensing. The second stage of trade liberalization therefore started in 1987. This reform aimed at instituting automatic licensing for a large number of import items, and reducing tariffs. By 1991, only 22 percent of items representing 5 percent of imports were subject to quantitative restrictions. Tariff categories were halved to 12 and tariff spread reduced (the highest tariff falling from 135 to 60%). Production weighted tariffs fell from 62 to 46 percent in 1989-1992. A policy setback occured at the end of 1990 leading to the introduction of suspended duties on several products and increase in import duties on agricultural crops.

In 1993-94, the government removed import licensing and foreign exchange controls and implemented customs management reforms. The latter reforms were responsible for increasing revenues by at least one third more than can be explained by changes in trade, exemptions, and tariff rates. The elimination of import licensing was important to reduce rent-seeking activity. Liberalization was supported by a real exchange rate depreciation of 10 percent between 1987 and 1990 that reflected a 9 percent fall in the terms of trade.

In 1995 Kenya was among the founding members of the WTO and thus became a signatory to all WTO agreements among them the General Agreement on Trade and Tariffs (GATT), Agreement on Agriculture (AoA), General Agreement on Trade and Services (GATS), the Agreement on Textiles and Clothing (ATC), the Agreement on Trade- Related Intellectual Property Rights (TRIPS). These international agreements came in during the period of economic re-orientation and supported the countries bid to liberalize.

A new wave of trade reforms was triggered by Kenya co-founding and joining the Common Market for Eastern and Southern Africa (COMESA). In 2001, the government formulated a comprehensive tariff reform program to improve Kenya's external position, simplify duties and facilitate collection in view of joining the common external tariff of COMESA effective 2004. Licenses for export of agricultural products were removed, while others for minerals and precious stones were retained. The top import tariff was reduced from 40 to 35 percent (except for a 100 percent duty on sugar imports) and the number of tariff bands cut from 9 to 8. As a result the average unweighted tariff declined from 18 to 16.6 percent (2001/2000).

Tanzania and Uganda have half as many tariff bands, lower maximum tariff (25% and 15% respectively) and lower unweighted average tariffs. However, Kenya has a stronger revenue base than either of these two trading partners and import tariffs represent a smaller share in total Kenyan revenues. This implies that the fiscal cost of lowering tariffs is relatively lower for Kenya than for Uganda and Tanzania. Accomplishment of a uniform set of tariffs is estimated to cost Kenyan tax revenues no more than 5 percent, while it could affect Uganda's and Tanzania's by as much as 20 percent . Kenya has also most to gain from integration given that it is the most sophisticated of the three economies with the greatest regional export potential.

1.2. Exports Liberalization

During much of Kenya's post-independence history, a strong anti-export bias existed. The flat rebate export compensation scheme put in place in 1974 was ineffective and only benefited large exporters. It was only in the late1980s that some schemes started to be introduced to strengthen the incentive to export. In 1981 the government stopped requiring no objection certificates from domestic producers as a condition for export. In 1988, a manufacturing under bond (MUB) facility was instituted to allow customs authorities to waive import duties on imported inputs used in the production of export goods. In 1990, a duty/VAT import exemption was applied to firms with export contracts and past export performance and the Export Promotion Zones (EPZ) were approved.

The Lome convention came into being in 1975. The convention linked the African, Caribbean and Pacific (ACP) countries with the EU to form the ACP-EU agreement which had both trade and aid provisions. One of the trade provisions was the non-reciprocated trade access of the ACP country products to the EU markets with subsequent renewal every ten tears between 1975 to 2000 (Ronge, 2006). This was anticipated to increase the volume and value of exports to the EU markets.

In the 1989-93 National Development Plan, industrial and commercial sectors were accorded priority. The government embarked on vigorous export promotion strategy so as to increase returns from exports through reduction and restructuring of tariff, abolition of export duties, introduction of export retention schemes and export credit guarantee. At the same time, incentives like the MUB, green Channel and export compensation were recommitted to the government for implementation since they had not yet started operating.

The impacts of these measures however, appear negligible. For instance, MUB and EPZ only accounted for one percent of exports in 1993-1998. The duty exemption has benefited exporters for an equivalent 35 percent of total exports. Kenya experienced an export surge from 2 to 8 percent of GDP between 1988 and 1993 but this coincided with the sharp depreciation of the Kenya shilling, a fall in real wages, and the abolition of trade licensing and foreign exchange allocations. It is therefore not clear what proportion of export gains can be attributed to these measures.

The 1975 Lome convention expired in 2000 and was succeeded by the Cotonou Partnership Agreements which required ACP countries reciprocation to duty free EU market access as from 2008 with the countries being required to sign Economic Partnership Agreements (EPAs). This was preceded by the onset of the WTO in 1995 under which the PTA granted by ACP-EU pacts was challenged and the dispute granted exemption till 2007. The new trade agreements carry with them potential negative and positive effects if they come into place by 2008 and will be effective for the next 20 years.

1.3. Market Policies/Reforms

In 1985 Weaknesses of the import substitution strategy were clear and SAPs were not being implemented leading to the publication of Session paper No. 1 of 1986 on Economic Management for Renewed Growth. This paper emphasized a more outward oriented policy encouraging the exposure of the domestic economy to international competition and export of non traditional exports. Price controls were highlighted as the impediments or disincentive to investment and thus the government lifted price controls on various items by enacting the Restrictive Trade practices, Monopolies and Price Control Act (1988) and repealing The Price Control Act. This Act was intended to curtail monopoly, collusion on price fixing and the control of markets by large firms, factors which tended to suppress the competitive environment fostered by the government.

Market liberalization involving maize, sugar, tea, coffee and horticulture among others witnessed the entry and exit of different players at different levels of the value chain (production, value adding and marketing). For instance, there has been a noticeable entry and exit of small-scale and large scale

export horticultural products owing to the quality and safety requirements of the EU market introduced through EUREPGAP, WTO's ansd SPS.

1.4. Institutional/Parastatal Reforms

During independence the government inherited the pre-independence agricultural commodity boards which later formed the government parastatals. These controlled production, pricing and marketing of various agricultural commodities in the country.

The program of privatization, commercialization and deregulation undertaken by the government was aimed at increasing the role played by private sector, reducing the size of the public sector, increasing the efficiency of the government entities by exposing them to competition, and reducing the budgetary allocation and reliance on public funds. This was done under the condition that there would be no threat to state security, constitutional order or deviation from the pursuit of social or economic objectives.

By the 1980s, the government held interests in 250 commercial enterprises including marketing boards and held majority interest in at least half of them. They represented about a tenth of GDP but contributed little to economic growth. Productivity growth among parastatals was negative in the period. Taxed by economic inefficiency, they imposed a heavy burden on the budget and on public banks, which held the majority of their debt. They also contributed 28 percent to the trade deficit.

In 1991, the government embarked on rationalizing parastatals. Out of 250 33 were deemed strategic but five were earmarked for immediate restructuring. The government would divest from the remaining 207 enterprises. The medium term target was modest, with only 10 firms set for privatization and only 5 actually privatized by 1995. But privatization continued through the 1990s with a total of 165 enterprises being sold or liquidated. The economic impact of divestment was small because, with the exception of Kenya Airways sold in 1996, the remaining were small or medium enterprises. Furthermore, transactions lacked transparency, political support was weak, strategy unclear and institutional and administrative arrangements inadequate

In 2000, the government announced a renewed privatization strategy, which took into account previous experience and focused on increasing private sector participation in the enterprises previously deemed strategic.

1.5. Monetary and Fiscal Reforms

Government expanded rapidly in the 1970 and 1980s. Between 1972 and 1994, total government expenditures rose by 12 percent of GDP (to 32%) but contracted to 28 percent of GDP in 2001. Revenues showed a similar growth pattern through an increase in tax collection. Because of this unusually large tax base, Kenya is an exception in its ability to finance the greater part of its expenditures (and deficits) domestically.

The Capital Markets Authority (CMA) was established in 1990 to stimulate the development of long term debt, equity markets and ensure stability of the financial sector. To strengthen the capital base of weak financial institutions, three commercial banks and four Non-Bank Financial Institutions (NBFI) were licensed and became operational and further the CMA took steps to enhance the development of the stock market (CBS, 1992).

The Central Bank of Kenya (CBK) (Amendment) Act of 1996 granted the CBK operational autonomy and responsibility of managing money supply. The Act became an insulator to external interference and offered protection to the monetary policy (CBS 1998). Several incentives were put in place by the government to encourage the growth of capital markets. Amongst the incentives was "tax exempt status" for ten years for approved venture in capital funds, availing credit rating facilities to

corporate debt securities issuers and levelling the operational footing of various players such as licensed dealers and insurance companies. This was geared towards stimulating competition leading to increased liquidity and price stabilization as more players entered the finance market. The recognition of the role of Micro Finance Institutions (MFIs) as promoting development in Medium-Small Enterprises (MSEs) came in 1999 and led to the formation of a Micro-Finance department at Central Bank. An association of Micro Finance Institutions was also registered to liaise with the Central Bank on regulation, legal and accounting matters of the sector.

By 2000 policies employed in the management of the finance sector included restraint in monetary policy involving monetary market intervention through sales of government securities, mainly Treasury Bills to regulate domestic liquidity, close monitoring of cash ratios and sterilization of excess liquidity. The banking act was amended to give the Central Bank powers to remove errant directors, levy penalties and enforce regulation (CBS, 2000).

1.5.1. Interest Rates and Exchange Rates

Over the period 1964-1991 the government maintained a policy of fixing interest rates resulting in interest spreads that were very low and that rarely went above 5%. In 1991 the government decontrolled interest rates in a move towards liberalization of the financial sector (CBS, 1992). This led to the widening up of the spreads to over 13% and ensured that lending rates remained high while deposit rates remained relatively low (GoK, 2002-2008).

After independence the government adopted a rigid exchange rate policy but in the early 1980s the government devalued the shilling to achieve macro-economic stabilization resulting from the oil shocks, the 1971 exchange rate crisis and the 1984 drought. The year 1991witnesed the beginning of liberalization of foreign exchange control to encourage foreign currency inflows into the country. The central bank partially relaxed the exchange control act by drawing the clause covering declaration of foreign currency held by incoming travelers, introduced convertible Foreign Exchange Bearer Certificates (FOREX-Cs) and allowed tourism operators to be paid for their services in foreign currencies (CBS, 1992).

1.5.2. Money Supply and Inflation

The country transitioned from a low inflation environment in 1964-73 to high inflation rates in 1974-1995. This was consistently matched by expansionary monetary policies as the economy responded to external and internal shocks leading to a rapid growth of money supply in the 1970s and in the 1980-95 respectively (GoK 2002-2008).

The Central Bank of Kenya (Amendment) Act, 1996 helped limit government borrowing from the central bank and control of inflationary levels (CBS, 1998) and limited the extent to which the deficit could be monetized (GoK, 2002-2008).

2. Agricultural and Rural Development Policies

Agricultural and rural development policies have undergone major changes that can be assessed within the three major periods: pre-liberalization, liberalization and post-liberalization. These major periods can also be broken down to explain significant policy events. Brown (1960) identified three major phases of development which took place within the colonial- era; i) the recovery phase (1945-50, ii) a planning phase (1951-1955), and iii) the rapid development phase (1955-60), Production segmentation had already begun in the case where the pre-independence government perpetuated large-scale production. The transition to independence period (1960-1964) saw the government adopt a land policy in which land was subdivided and allocated to medium and small scale African farmers while preserving the large-scale farming sector which by then accounted for 85% of Kenya's export earnings. This continued to the early independence period.

During these phases the foundation for land policy, agricultural marketing and pricing were set. While the former was dictated by the Swynnerton plan, the latter was articulated through the formation of various production and marketing boards that formed government parastatals thereafter. Post independence agricultural and rural development saw great government involvement and intervention in every sector of the economy at both macro and micro levels. The articulation of land policy entailing subdivision and redistribution of land served to further segment production but this time with an emphasis on small-scale farming. Market control policies through crop marketing boards served to protect both consumers and producers from internal and external shocks. The post-independence period (1964-1980) was characterized by strong import substitution policy to protect the domestic economy from external forces. Such policies included quantitative restrictions, import tariffs, currency devaluations price controls and controlled movement of agricultural commodities. During the colonial era and the post-independence period, rural development and public investment focused on areas with maximum returns (high potential areas) with little or no emphasis on low potential areas. Although this could be justified economically, the resulting development disparity led to regional economic inequity resulting in human and capital flight and concentration in high potential regions. The land policy adopted during this period emphasized land adjudication, subdivision and reallocation. The liberalization era that spanned 1980-1992 started with the introduction of the SAPs in 1980. There was little commitment by the government to implement the requirements of the SAPs until the publication of Session Paper No 1 of 1986. The implementation of the SAPs involved the following:

- liberalization of prices and marketing systems
- financial sector reforms
- international trade reforms
- government budget rationalization
- divestiture and privatization
- parastatal reforms, and
- civil service reforms.

The implementation of these requirements saw a shift in the macro and micro economic climate which greatly influenced agricultural policy. Specific policies that had a major shift and led to corresponding shift in agricultural and rural development include: the food policy; the agricultural marketing and pricing policy; agricultural research and technology policy; inputs policy; exchange rates policy and foreign trade policy. Major reforms that were undertaken by the government included: price decontrols, foreign trade liberalization; decontrol of consumer and producer prices; decontrol of interest rates and foreign exchange rates including repealing the Exchange Control Act. The government also instituted a framework to reduce government spending, restructure the civil service, divest from economic activities that could be better undertaken by the private sector, provide incentives for foreign and domestic investment and better public financial management (GoK, 1997-2001). While there have been some positive effects due to liberalization, negative effects have been manifested in stagnated sub-sectors and rapid exit and entry of players.

As from early 1990s, the government realized the declining productivity trend in agriculture. This led to the inception of the District Focused for Rural Development (DFRD) which failed owing to inadequate institutional and legal structures, the Poverty Reduction Strategy Paper (PRSP), Economic Recovery Strategy for Wealth and Employment Creation (ERS-WEC) and the Strategy for Revitalizing Agriculture (SRA) were developed to stimulate economic growth and reduce regional disparity. These initiatives outlined government's commitment toward reversing the declining

agricultural trend and enhance its contribution to the economy. The SRA operationalized the ERS-WEC in the agricultural sector ministries. Several government strategies to revitalize agriculture and stimulate rural development as from 2001 are detailed in box 2

2.1. Financing of Agriculture and Rural Development

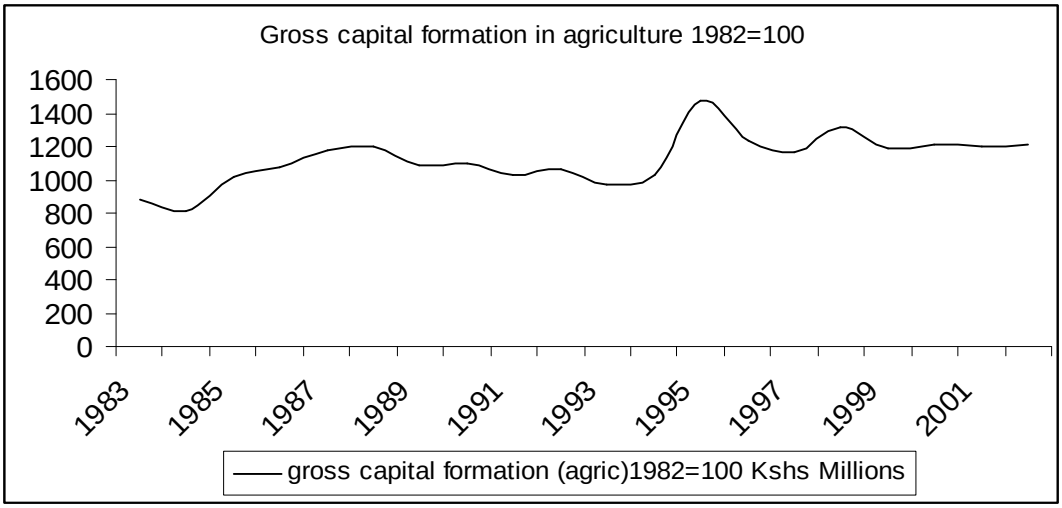
Capital formation in agriculture has been increasing. However, the period spanning mid 1980s to early 1990s was characterized by declining agricultural funding (Figure 41) as the government gradually withdrew from agricultural input subsidies and services. Expenditure on agriculture has declined tremendously since the inception of economic reforms. The expenditure declined from 10% proportion in 1980 to 3.5% in 2002 (Table 18).

Table 18: Agricultural Expenditure

Year	Agriculture Recurrent	Agriculture Development	Total Expenditure	Total public Expenditure	% Agric. Share to Public Exp
1980	904	1042	1946	19196	10.14
1985	1244	1552	2796	32568	8.5
1990	772	804	1576	56314	2.8
1995	4322	3410	7732	183408	4.2
1996	4590	6636	11226	202956	5.5
1997	4268	3488	7756	242610	3.2
1998	4868	4598	9466	272812	3.4
1999	4422	5316	9738	383408	2.5
2000	5906.6	2363	8269	270068.2	3.1
2001	7359.8	490.3	7850.1	307714.7	2.6
2002	7819.3	4112.6	11931.9	342450.7	3.5

Source: CBS Economic Surveys (Various Issues)

Figure 41: Gross Capital Formation in Agriculture, Kenya



Source: CBS Economic Surveys (Various Issues)

2.2. Development Strategies and Programs in Agriculture

Policies affecting agriculture consist of government decisions that influence the level and stability of input and output prices, public investment, costs and revenues, and allocation of research and development funds to improve farming and agricultural related processing technologies (Nyangito,

2001). Some of these policies affect agriculture more directly than others and constitute the agricultural sector policies affecting particular commodities and production techniques. These have included particularly quantitative controls, subsidies and taxes on inputs and outputs. Policies that affect agriculture indirectly are of two types. One set is macro policies that affect agriculture through macro prices such as general trade regimes, interest, exchange, and wage rates. The second set of indirect policies is concerned with investment decisions in provision of services such as research and development, education, health, transport, market infrastructure and institutions, which have a broad impact on agricultural sector productivity. These policies can be broadly classified into: pricing and marketing including marketing institutions, research and extension, land, credit and financial institutions including the role of co-operatives, infrastructure investment including transport and irrigation, food security and self-sufficiency, and agricultural input policies.

Several policy initiatives have been issued and documents have been prepared since 2001 to highlight the government objectives regarding sustainable growth and socio-economic development, and to build overall programmatic frameworks for their implementation. The most important policy documents are discussed below

- **The Poverty Reduction Strategy Paper (PRSP), 2001-2004.**

This document outlines priority areas and measures necessary for poverty reduction and economic growth. The Poverty Reduction Strategy Paper was prepared through a consultative process in all districts and involved the government, the private sector and the civil society. In it, government commits to priority actions in two broad areas: – creating opportunities for rural communities and the private sector to effectively carry out their activities in an increasingly competitive global environment, and accelerating policy and institutional reforms, particularly the large backlog of legislative and regulatory reforms. During the PRSP consultations, Agriculture and Rural Development (ARD) in general, received top ranking as the key sector through which to tackle the increasing level of poverty. The ranking within ARD (in ascending order) was crop development, rural water, livestock development, food security, lands and settlement, environmental management and fisheries (Agriculture Sector Brief and Horizon, 2015). Within agriculture, crop development, was the priority sector with poor extension services, inefficient rural financial systems, the poor state of rural infrastructure and poor marketing and distribution systems being identified as the main constraints. In the livestock sector the PRSP identified marketing systems and infrastructure, disease control and extension services as priority interventions. The challenge for the Government is to mobilize the necessary resources and build the institutional capacity to implement the proposed measures.

The causes of poverty were identified as: low agricultural productivity and poor marketing, insecurity, unemployment and low wages, bad governance, land issues, lack of infrastructure especially roads, cost of social services and education, HIV/AIDS epidemic and gender imbalance. The PRSP has addressed the removal of these causes and hopes that the extreme poverty level would be reduced by 50 percent by 2015 while the overall target is to reduce poverty prevalence to less than 30 percent by the same year. The strategies to achieve these targets in the agriculture sector would include: crop development through improved extension services, provision of credit to the small holder farmers, improve the rural infrastructure including roads for ease of transporting the farm produce to the markets, develop marketing linkages between producers and consumers through the provision of market information, and capacity building for the institutions charged with the implementation of the strategies.

- **The Economic Recovery Strategy for Wealth and Employment Creation (2003-2007) (ERSWEC)**

This document lays out the main policies of the new government elected in 2002. The Strategy intends to restore and sustain economic growth, generate 500,000 jobs per year to absorb over 2

million Kenyans who are currently unemployed, and reduce poverty. It lays out the main agricultural policies that are further elaborated on in two documents: Strategy for Revitalizing Agriculture (2004-2014) and the Strategic Plan 2003-2007 of the Ministry of Livestock and Fisheries Development.

- **The Ministry of Livestock and Fisheries Development Strategic Plan 2003 – 2007**

The Strategic Plan gives further insight into the government's priority interventions in the livestock sector. In the areas of disease control related to enhancing the export of livestock products, the Ministry proposes to develop and implement disease and pest eradication programmes, develop and operationalize disease free zones and set up risk analysis and Trace Back Systems to meet the requirements of international livestock markets. The strategy also calls for a streamlined legislative policy framework, review of the National Livestock Policy, the policy regime surrounding the provision of veterinary services, and the Wildlife Conservation Act that does not allow e.g. ostriches and crocodiles and other emerging livestock species to be domesticated. Liaison with various government departments will be used to enhance security in livestock producing areas (Agriculture Sector Brief).

- **Strategy for Revitalizing Agriculture, 2004 - 2014**

This is a joint strategy of the Ministries of Agriculture, Livestock and Fisheries Development and Ministry of Cooperative Development but the Livestock and Fisheries Development Ministry have already developed their own Ministry specific Strategic Plan described above. The Strategy for Revitalizing Agriculture (SRA) makes bold and potentially far-reaching proposals. It accepts the growth target for the sector of 3.1 percent, but does not say where the growth will come from. However, its analysis of the key constraints to Kenyan agriculture indicates that it is the beginning of a process that may more clearly define and achieve that target. The SRA recognizes low productivity as the key constraint in Kenyan agriculture. As a result, symptoms like high production costs and competition from imports are manifested. The productivity problem is decomposed into 3 components: extension, research and economic and financing problems. The extension problem manifests in the lack of awareness or use of existing productivity enhancing technologies, while the research problem as laid out in the SRA refers to non-existence of appropriate productivity enhancing technologies.

The economic and financing problem manifests in farmers aware of, but unable to meet the cost of available productivity enhancing technologies. This is attributed to some of the poor services they receive in terms of the policy, legal and regulatory framework, the input and output marketing services that result, and the poor access of the agricultural sector, particularly the typical small scale producer to different types of financial services.

- **The National Development Plan (NDP), 2002-2008**

The plan is a statutory policy document outlining the development policies and strategies to be pursued by the government and other development agencies over the medium term- a seven year period-and was launched more or less at the same time as the PRSP.

- **The Kenya Rural Development Strategy (KRDS), 2002-2017**

This is a longer-term framework document outlining a broad range of strategies for improvement of rural Kenya over the next 15 years. Considering food security promotion and attainment as the initial step towards poverty alleviation and equitable growth and development in rural areas, the KRDS is a road map for government, private sector, civil society (religious groups, NGOs, rural communities, CBOs) and other development partners. Several policy actions and interventions are proposed within the KRDS framework to facilitate the process of rural development with agriculture providing the stimuli, resources and market. Agricultural growth must serve as the catalyst for the broad-based

economic growth and development. Through forward and backward linkages to the non-farm economy, agriculture will generate raw materials, employment, income, larger market and growth in the rest of the economy (Horizon, 2015 and Agriculture Sector Brief).

- **Strategic Plan (2006-2010)**

This is a five year strategic management plan by Ministry of Agriculture targeting on the institutional, policy and civil service reform to achieve the governments set objectives through organizational restructuring, departmentalization in relation to prioritization and resource constraint

Conclusion

Until late 1970s Kenya's economy was still engulfed in protection policies and was therefore not open to external competition. Despite economic growth witnessed during that period, full economic potential could not be achieved. Reforms started in early eighties were initiated by the Structural Adjustment Programs (SAPs).

Whereas economic reforms have taken place, economic growth has declined over the period of liberalization. Agriculture the main sector in the economy has registered decline over the reform period. This decline in agricultural growth coincides with economic reform period that opened up the economy to imports as well as decline in government support to agriculture. Increase in production costs reduced the country's comparative advantage by making the country's agricultural products relatively more expensive and less competitive. Sub-sectors such as coffee, sugar have underperformed while tea and horticulture thrived before and after reforms. This has resulted to shift in income sources and a decline in overall household income triggering the increasing unemployment despite the new governments' efforts to create 500,000 jobs annually. Over the period, absolute population has increased although the growth rate has declined owing to the family planning programs put in place. The ratio of exports to imports declined indicating the increasing import trend despite the depreciation in currency and the various trade agreements. This was also reflected in the agricultural sector.

Amidst the increasing population, agricultural growth and the overall economic growth continues to decline. Population increase is translating into increasing annual labour force cohort entering the job market. This is exerting pressure on agriculture and the economy particularly in view of the dwindling employment capacity. New challenges have emerged in relation to the rapid urbanisation, internal and external migration with corresponding rural depopulation.

Varied trends in different economic variables have emerged over the period of liberalization. These trends particularly in the agricultural sector distinctively mirror the different periods undergone by the economy. The net effect of reforms in every sub-sector is unique, interlinked to other sectors and sometimes conflicting. This highlights the critical need for the reassessment of the effects of economic reforms so as to provide for an understanding of the possible trajectories and mitigating actions.

PART II -

THE STRUCTURE OF AGRICULTURAL AND AGRO-FOOD MARKETS AND THEIR EVOLUTION

CHAPTER 5 - AGRO-FOOD MARKETS

The perishable, seasonal and geographical nature of agricultural produce determines the extent and intensity of handling before reaching the consumer. Kenyan agricultural markets are characterized by trade involving primary commodities which undergo further processing within or outside the country. Most of the food commodities have short chains with less sophisticated handling. Only dairy industry which targets the local market has a relatively sophisticated handling chain due to the perishable nature of milk and milk products. Export marketing chains; primarily entail production and simple processing such as drying, pulping and milling. Little flavouring is done in the country. There is a lot of unexploited value addition prospects in the export value chains however these are constrained by tariff escalations.

Prior to 1980 domestic and international agricultural marketing was highly controlled through marketing boards to protect consumers and producers. The primary objective of agricultural marketing boards was to improve the income and prices accruing to producers. The secondary objective was to reduce variability in prices and bring about equity in marketing opportunities for producers. The degree of compulsion to which producers were subject, or legal powers, varied between different commodities and areas. With such boards in place it was possible to transform the agricultural output and input marketing system, determine commodity prices, the level and stability of food prices. The major agricultural board included the Tea Board established in 1951, the Coffee Marketing Board established in 1946 to market coffee, the Maize and Produce Control established in 1950 and later became the National Cereals and Produce Board (NCPB) to cater for all cereal products in the country and the Kenya Co-operative creameries, 1945 (Brown 1960) However, these created monopoly power, especially in agricultural processing industries, making reforms inevitable. The liberalization period witnessed gradual government withdrawal, greater private sector participation in commodity markets and provision of services. Economic reforms led to influx of new participants in the market enhancing competition. Price discovery was now largely left to the market forces. However government parastatals continued to have the advantage of the already established infrastructure and distribution systems to influence market prices. Even after liberalization and restructuring, these hitherto powerful entities enjoyed large market shares and indirectly determined the level of prices.

1. Major Food Commodity Chains: An Overview

The main food commodities in Kenya are composed of cereals. Maize is the staple food in the country supplemented by wheat and rice. The marketing chains of these commodities were hitherto dominated by the National Cereals and Produce Board (NCPB) until liberalization in 1992.

1.1. The Export Crop Marketing Chains

The main export commodities such as coffee and tea are dominated by marketing institutions that are mandated to sell the output to export markets. These crops are produced by both small and large-scale holders.

- **Coffee**

Almost 99% of Kenyan coffee is exported and the domestic market only consumes less than 1% of the total coffee produced in Kenya. The large-scale farmer sells parchment from his own factory directly to licensed millers and the Coffee Marketing Board (CMB), which sells it abroad. The small-scale farmer delivers ripe cherry to a cooperative society, which prepares it to parchment and sells to licensed millers. Overtime there has been development of middlemen between the small-scale producers and the main marketing channel and even some small-scale growers have acquired their

own parchment processing facilities. Following liberalization of the industry, five coffee millers have been licensed by the government.

- **Tea**

The local tea market absorbs only 5% of the total production (about 15,000 MT) while 95% of the locally produced tea is exported. Most of this tea is exported in bulk and only a small percentage is packaged for export. Tea from small-holders is sold as green leaf to KTDA for processing in their own factories or estates factories by agreement. KTDA markets tea on behalf of small-scale tea producers. The large estates produce made tea and sell through established commercial channels directly overseas or to the Mombassa auction. In the recent past, 85% of Kenyan tea was marketed through Mombassa auction, 10% exported directly by producer through their own private arrangements and the remaining 5% absorbed into the domestic market. The current existing market outlets are: Mombasa auction 75%, KETEPA (Kenya Tea Packers Ltd) 7%, Direct sales (overseas &local)-15% and Factory Door Sales-3%.

- **Horticulture**

About 96% of the horticultural produce is marketed locally while only 4% is exported (MoA, 2004). Small scale horticultural producers sell their produce to the domestic markets through various local market vendors such as the road side vendors, green grocers flower vendors or sell to medium-size or large exporters. The large scale farmers and contracted farmers sell their output to large exporters who export the produce to international markets.

1.2. Agricultural Commodity Marketing and Price Fixing

Kenya's agricultural pricing policy cannot be divorced from its marketing institutions. The pre and post-independence pricing policy was deeply rooted in the colonial central marketing system (boards).

In addition the previous marketing boards and parastatals who had already established relatively competitive market structures continued to perpetuate market monopoly owing to the economies of scale enjoyed. There are also regulatory responsibilities such as those of the KDB, TBK, and CBK; which are necessary to maintain quality standards and the competitive behaviour of involved firms.

Table 19: Participants in Marketing and Pricing of Agricultural Commodities in Kenya Before and After Liberalization

Commodity	Official marketing agent at producer level	Producer price determination prior to liberalization	After liberalisation
Maize, wheat, rice	NCPB, KFA (wheat)	Government	Market*
Sugarcane	Processing factories	Government	Market
Milk	Licensed processors	Government	Market
Coffee	Licensed millers	Determined by boards based on sales realizations	Market
Bulk tea	KTDA	Determined by boards based on sales realizations	Market
Packaged tea	KETEPA (Kenya Tea Packers Ltd)	Price controller	Market
Horticulture	Private trade, HCDA	Private trade	Market

Source: Jabara, C.L., 1985. Agricultural Pricing Policy in Kenya World development vol. 13 No.5 pp 614-626

Table 19 shows the participants in production, marketing and pricing of agricultural produce in Kenya before and after liberalization. While liberalization has brought about the entry of many players at every stage of the value chain in the commodity markets, producer and consumer welfare gains have yet to be discerned. Similarly, the privatization of government parastatals was intended to make them more efficient and enhance service delivery but these effects are still obscure. When the government withdrew from price regulation, market forces were left to determine equilibrium prices.

2. Presentation of Significant Chains

2.1. Food Marketing Chain

The food marketing chain was dominated by the National Cereals and Produce Board (NCPB) prior to liberalization. The institution evolved from the Maize and Produce Control Board in 1945 to Maize and Produce Board in 1950 and later to the Maize Marketing Board with operations conducted through the provincial marketing boards (Brown, 1970). The institution was charged with the responsibility of cereals marketing throughout the country. During the colonial period and in the post-independence era the board was charged with grain marketing controls which were used to raise local prices and stabilization of producer incomes, controlling the movement of cereals within the country and the export and import of maize and the national food security responsibility (Scott et.al., 1995). The import substitution policy adopted by the post independence government was perpetuated through price and movement controls which ensured producer and consumer protection. Thus through the board the government tried to maintain the farm gate prices between import and export parity prices.

The boards' objectives were pursued with varying success, with the short comings of maize marketing being acknowledged widely due to the inequitable and inefficient maize supply between surplus and deficit regions and the parastatals deficit that had become a government financial concern (Scott, et.al., 1995). NCPB was among the parastatals earmarked for restructuring to enhance its efficiency and reducing deficits through committing the institution to some performance contract by 1992.

Maize was sold by large-scale farmers to the NCPB through a network of buying centres throughout the country at a pan-territorial price and by small-scale to producers. The board accumulated stocks and transferred them from the areas of surplus to the deficit areas. Whereas large scale farmers could sell directly to the marketing board or a large agent, the small scale farmer was often compelled was compelled to sell at a lower price to a middlemen who in turn sold larger quantities to the NCPB buying centres. However, small producers in Kenya could also sell to a trader in a market in small quantities. Such traders then sold to an agent in the main collection centre. An alternative to NCPB is the informal market which mainly serves the rural areas. Differences in the official maize prices and the parallel market prices, and the unavailability of NCPB to the market participants have been largely sighted as the reason for existence of the informal markets.

Study by Nyoro et.al 1999 cited various marketing agents in the maize chain these were, assemblers, wholesalers, retailers, and dis–assemblers, posho millers and large-scale millers. During the peak period the wholesalers purchased an approximated 45% from the farmers and 61% during the slack period. Assemblers sold 76% of their total maize volume sold to wholesalers, 11% to retailers, and 5% directly to consumers in the months directly after harvest. But as the season advanced, assemblers sold only 55% of their volumes to wholesalers. Large-scale and small-scale mills accounted for 7% and 9% of assemblers' sales during the slack season, while dis-assemblers, retailers, and consumers accounted for a combined 29%.During the peak harvesting season, wholesalers sell about 51% of the total volume they handled to the large-scale millers, 13% to small processors mainly in the urban centers, 28% to the dis-assemblers in deficit areas, and 8% to retailers. Later in the season,

wholesalers sold a higher share of their maize (55%) to the dis-assemblers, 24% to retailers, and 11% direct to consumers. Notably, the volume of sales to large scale millers dropped off considerably as the season progressed, from 51% of wholesalers' sales to only 10% (to millers or NCPB (who generally sold to millers) combined). Large-scale millers tended to rely more on NCPB and imports for their supplies later in the year especially between June and November when domestic supplies are tight and prices rise up to import parity levels.

Dis-assemblers provide a link between the wholesalers and the consumers either directly or through the retailers and buy maize from the farmers by forming groups of 4 top 5 who then hired a lorry that transported the maize from the surplus region. Of the total volume purchased by the dis-assemblers during the peak period, 41% was from the wholesalers, 33% from farmers, and 26% from the assemblers. In the slack season, 55% of the maize traded by the dis-assembler was procured from wholesalers. Retailers and consumers accounted for 91% of the maize sold by the dis-assemblers with a bigger proportion (58%) being sold to retailers. The estimated CR4 (box 3.) for the assemblers, dis-assemblers and wholesalers were 0.22. 0.12 and 0.12 respectively. Most maize traders were usually more horizontally than vertically integrated. Survey results indicated that 33% of the retailers, 21% of assemblers and 18% of wholesalers and dissemblers were involved in the retail sale of maize meal. Many wholesalers were also assemblers. But for the majority of maize traders, the most important other business activities were not directly linked to maize trade (Nyoro et.al, 1999).

Maize market liberalization in early 1990s included the removal of the movement and price controls on maize trading, deregulation of maize and maize meal prices, elimination of direct subsidies on maize sold to registered millers and the liberalization of trade in both internal and external markets. Maize and maize meal prices, which prior to policy change, were set at pan-seasonal and pan-territorial levels were liberalized. Private traders were allowed to transport maize across districts without any hindrance whereas prior to this policy change, they were required to acquire movement permits for varying quantities of maize that was to be transported. The functions of the National Cereals and Produce Board (NCPB) were reviewed to allow it to focus on food security through acquisition and management of buffer stocks.

2.2. Dairy Marketing Chain[3]

2.2.1. Milk Marketing

Dairy marketing chain was controlled through the Kenya Cooperative Creameries (KCC) established by European dairy producers during colonial period and organized as a producer cooperative (Scott et.al., 1995) and regulated by Kenya Dairy Board (KDB) through the Dairy Industry Act enacted in 1958. KDB organized regulated and developed efficient production and marketing systems. License to process and distribute milk and milk products in Kenya was issued under the agency of Kenya Cooperative Creameries Limited (KCC). This gave KCC the monopoly/monopsony power over the dairy industry in Kenya. With the advent of liberalization, the government started to encourage formation of dairy cooperatives in the areas that were poorly served by KCC infrastructure. This was financed through the Rural Dairy Development Project (RDDP) to facilitate the establishment of processing facilities. In 1983 the government licensed two small-scale cooperatives to receive process and distribute milk and milk products for their members. During this time KCC handled 90% of the marketed milk (estimated 40% of the produced milk in Kenya) and the rest was held by two small-scale cooperatives, (Meru Central Farmers Cooperative Union (MCFCU) in Meru District and the Kitinda Dairy Farmers Cooperative Society (KDFCS) in Bungoma District) and other small private producers of pasteurized milk. Currently it is estimated that smallholder dairy production contributes over 56% and 70% of total and marketed milk production, respectively. The dairy industry in Kenya

[3] This area draws from A.M Karanja 2003. The Dairy Industry in Kenya: The Post-Liberalization Agenda

was liberalized in 1992. This led to emergence of new institutional arrangement in milk collection, processing and marketing. Liberalization encouraged opening up of the processing, distribution and marketing with informal marketing channels dominating at the farm gate level. These informal channels include hawkers, brokers, self-help groups as well as neighbours and business establishments like hotels. In total, the informal channel is estimated to control over 80% of the total marketed milk. Small-scale farmers have turned to informal channels of milk marketing neglecting the highly mismanaged dairy cooperatives which are an integral part of the formal milk marketing chain. Deregulation of prices encouraged competition in the informal and formal marketing channels. The prices in the informal marketing chain are 22% higher than in the formal marketing chain.

2.2.2. Dairy Market Structure

Since liberalization of the dairy industry, a total of 42 milk processors have been licensed by KDB. In 1999, 34 of these processors were operational with the rest having closed down. By the end of 2001, only 22 of the processors were in operation, currently there are 26 processors in operation (Table 20). The decline in the number of milk processors and mini dairies is mainly due to the inability of new firms to compete. This phenomenon has resulted in milk producers losing out on their deliveries and has raised fundamental issues on business ethics and the need for protection of industry players. Consequently, legally enforceable contracts in the industry will have to be instituted. Presently, the KDB has acted as an arbitrator but without legal instrument. Some of the processors have completely exited from the milk business due to severe competition while others have been taken over. These trends indicate that the milk processing market is consolidating and could benefit from economies of scale. However, as more and more firms exit milk processing, the market is becoming more concentrated. The dairy industry 2-firm and 4-firm concentration ratios (CR) in 1999 were estimated to be 42% and 65%.The four firms which are private limited companies are by order of their market share Brookside, Premier, Spin Knit and Meru Central dairies.

By 2001 the 2-firm and 4-firm concentration ratios had increased to 64.5% and 79.5%, respectively. The largest four firms in 2001 remained the same but the order was changed with Meru Central overtaking Premier. Moreover, there was a major increase in the market share for the two biggest firms from 42% to 65%.

Table 20: Licensees 2002 -2006

	2002	**2003**	**2004**	**2005**	**2006**
Processors	31	27	23	19	25
Mini Dairies	109	84	76	62	65
Cottage industries	42	56	47	62	50
Cooling plants1	0	0	3	16	19
Milk bars	177	841	650	818	987
Producers2	111	350	325	485	609
Total	470	1358	1124	1462	1755

1-licence introduced in 2004
2-licensed to sell milk directly to consumers
Source KDB

Box 2: Concentration Ratio

The concentration ratio is expressed in the terms CR_x, which stands for the percentage of the market sector controlled by the biggest x firms. For example, CR_3 = 70% would indicate that the top three firms control 70% of a market. CR_4 is the most typical concentration ratio for judging what kind of an oligopoly it is. A CR_4 of over 50% is generally considered a tight oligopoly; CR_4 between 25 and 50 is generally considered a loose oligopoly. A CR_4 of under 25 is no oligopoly at all. We would add that a CR_3 of over 90% or a CR_2 of over 80% should be considered a super-tight oligopoly. The problem with this measure is that CR_4 does not indicate what the relative size of the four largest companies is. It may be that a CR_4 of 80 means that one company controls 50% of the market, while the others have 10% apiece. That's a very different market structure than one where every firm has a 20% share.

2.2.3. The Dairy Marketing Value Chain

The milk value chain involves collecting, cooling, processing and distribution. Most of the produced milk in the country is sold through the informal channel which is characterized by selling of un-pasteurized milk. Figure 422 and Figure 433 shows the value chain before and after liberalization. The informal chain involves small-scale milk vendors who target urban markets. Milk producers also sell milk directly from their farms. A large proportion of milk collected by co-operatives and self help groups is sold to processors while the surplus is sold through the informal channel. The formal chain involve collection of milk from the producers either through the co-operatives or agents and transporting it to the cooling plants or to the processors. According to a survey done by Tegemeo institute in 2002, five major rural milk outlets were identified during the field surveys. These are Brokers, Traders/hawkers, Transporters, Co-operatives and Farmer groups and Processors. In the smallholder zones covered by the nation-wide survey, the main milk outlet at the farm level for both the morning and evening milk was local sales to neighbours and business establishments.

The informal market outlets are the most dominant, accounting for over 76% of the total milk sold. The co-operatives, self-help groups and direct sales to processors were the formal milk marketing channels, which absorbed around 24% of the milk sold. As shown above, 30% of the milk collected by co-operatives and self-help groups is sold in the informal market. It is estimated that the informal market absorbs at least 80% of all the milk sold by smallholder farmers in the country. Milk sold to neighbours and the one hawked by the farmers had the highest price while milk sold directly to processors was paid the least price. This price differential may explain why most smallholder farmers prefer the informal marketing channels. Milk brokers and hawkers account for 20% of the daily milk produces while cooperatives control only 13%.

Milk pricing at every stage of value addition is determined by the transportation and overhead costs. Transportation costs are the most pronounced in the value chains in the country owing to a dilapidated and poor road and communication infrastructure.. Currently some cooperatives have integrated vertically by acquiring cooling and processing equipment. Contractual agreements between processors, co-operatives and the dairy farmers exist but they are hardly enforced.

Figure 42: Dairy Marketing Value Chain Before Liberalization

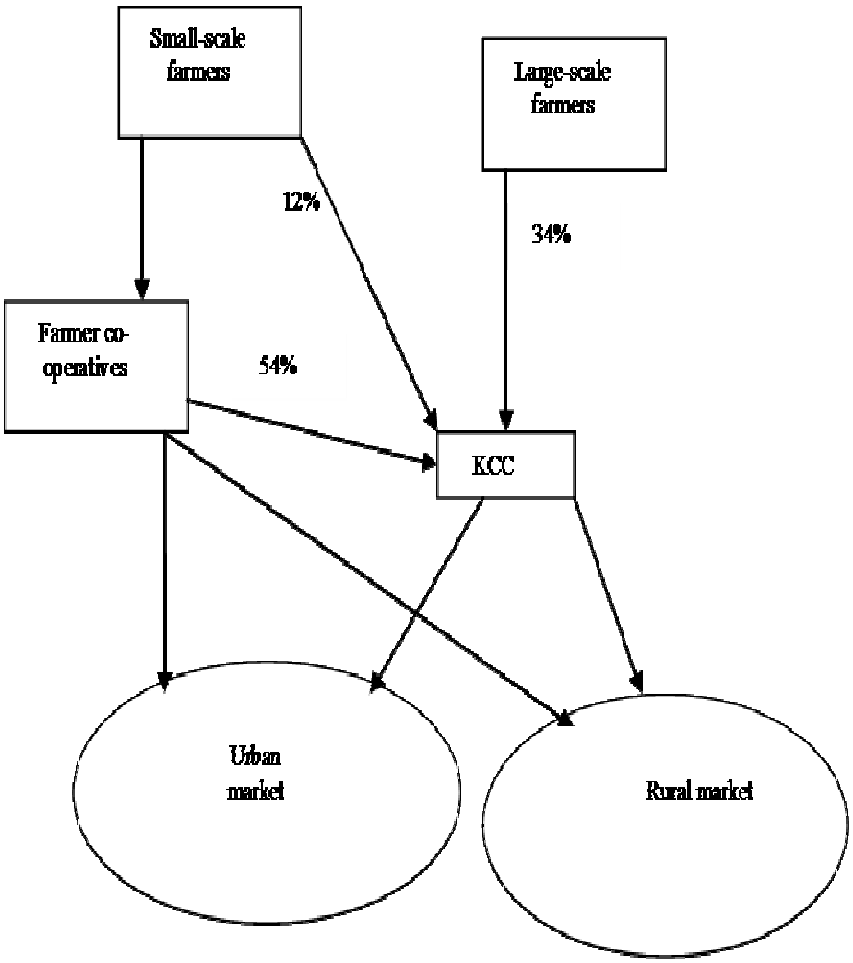
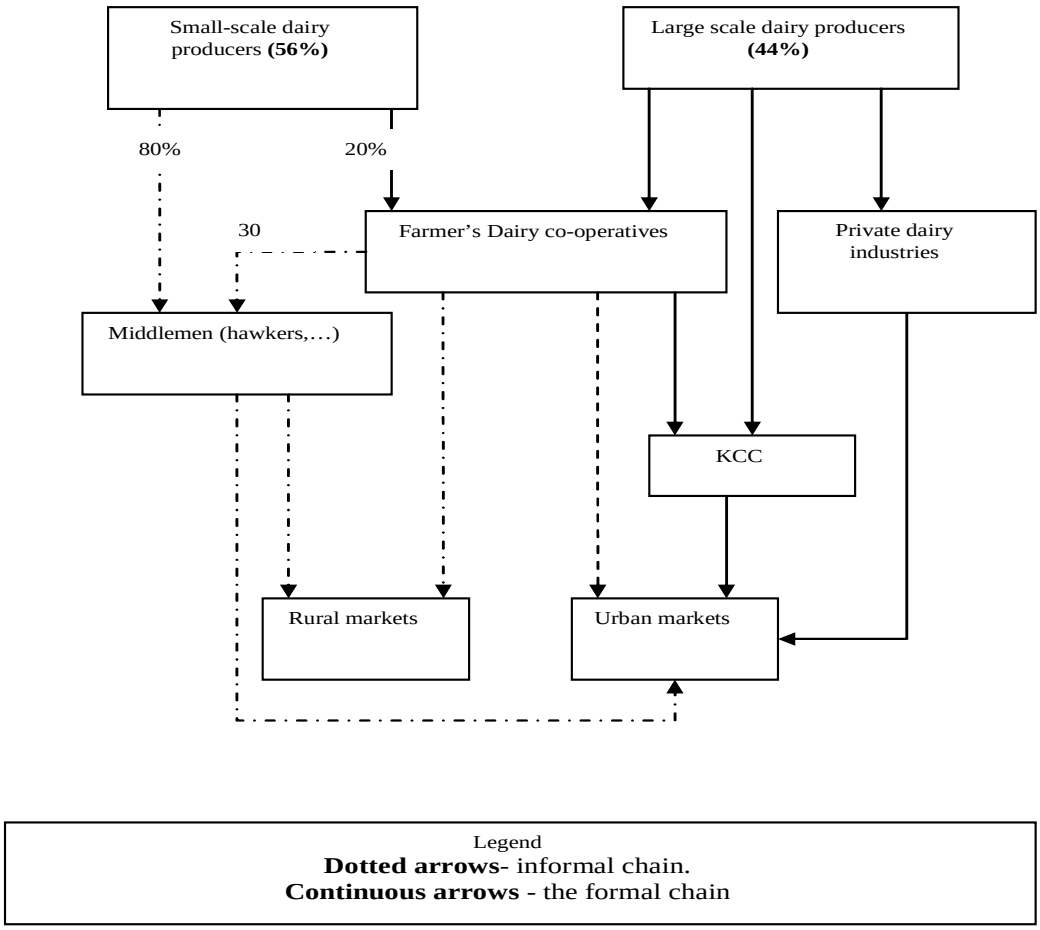


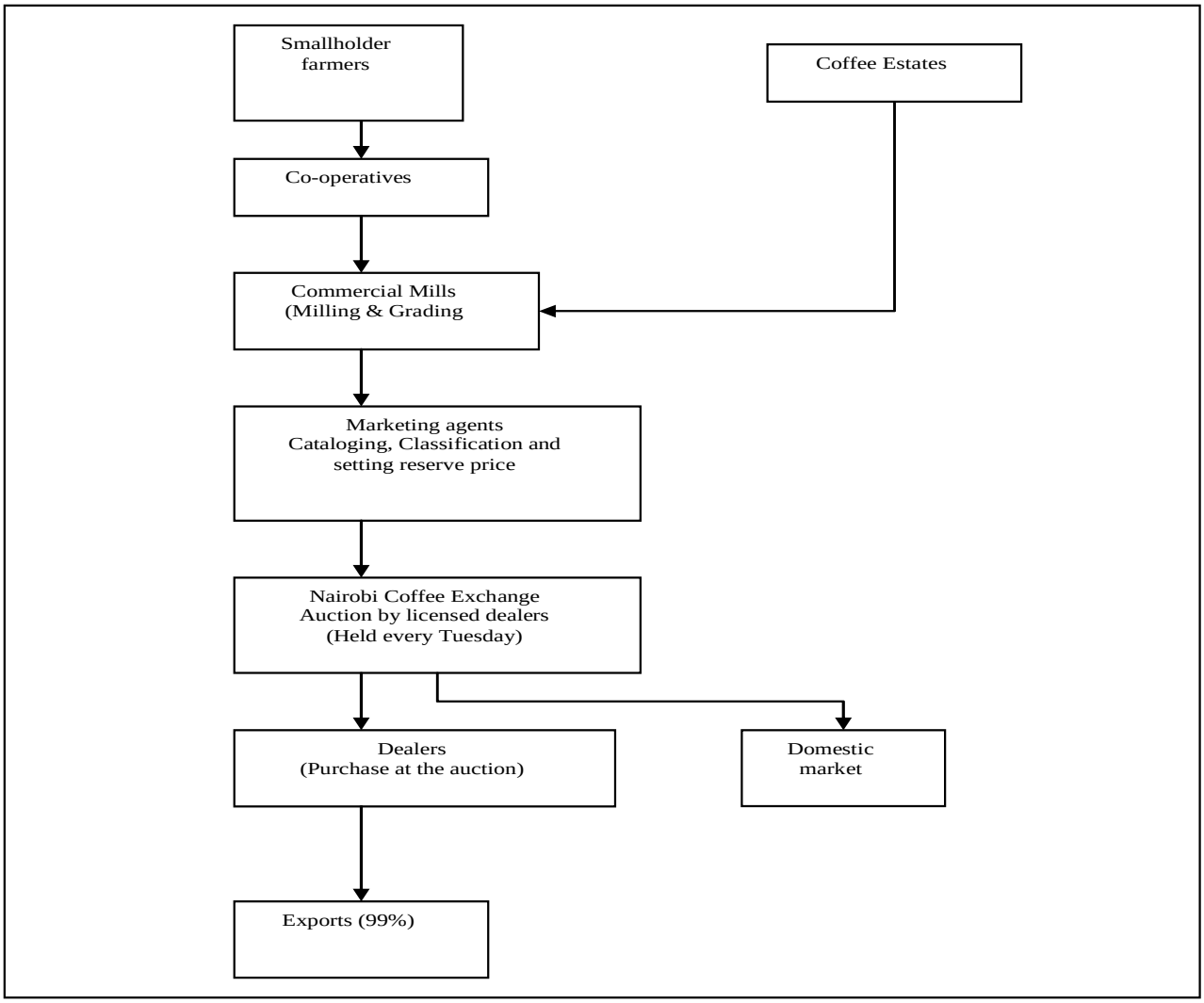
Figure 43: Dairy Marketing Value Chain – Formal and Informal – After Liberalization



2.3. Coffee Marketing and Value Chain

Coffee marketing has gradually been liberalized but the marketing chain has remained unchanged. The large-scale farmer sells parchment directly to licensed millers. Figure 44 represents the value chain in the coffee industry. The coffee value addition chain is fairly short involving production, pulping, milling and then export. The small-scale coffee farmers sell ripe cherry to a cooperative society, which pulps it to parchment and sells to KPCU or any other recently licensed millers and CMB. KPCU is owned by coffee growers and prior to liberalization, was solely responsible for milling all Kenyan coffee but currently there are five licensed coffee millers. Coffee MBoard of Kenya (CBK) holds monopoly over marketing responsibilities even after liberalization of the coffee industry. Plans for liberalization at the marketing level are underway. This will enable farmers to access the international markets directly.

Figure 44: Coffee Value Chain, Kenya



Source: Adapted from Nyangito, H.O.,2005. Tea and Coffee Industry in Kenya 2005, Export Processing Zone Authority

Overtime there has been development of middlemen between the small-scale producers and the main marketing channel and even some small-scale growers have acquired their own parchment processing facilities. Coffee is roasted, milled and then exported. Kenyan coffee is traded at the auction in the Nairobi Coffee Exchange where licensed exporters assemble and bid for various lots and exported to Europe mainly Germany, Sweden, Belgium, Finland, Netherlands, UK and Norway as well as USA. Coffee is mainly traded on the New York and London futures (terminal) markets, which exert a strong influence on world coffee prices. The major concern in coffee marketing is to establish direct coffee marketing. There are contractual agreements between small-scale coffee producers and their cooperatives; however, these agreements are watered down by delayed coffee payments. Liberalization is also undermining contractual agreements as it is freeing the market and thus encouraging farmers to sell coffee according to their choice.

There were a total of 5 auction licensees, 19 roaster licensees and 57 licensed coffee traders by 2004 (Table 21). Although coffee industry liberalization intention was to increase benefits and incentives to farmers, world prices continued to decline, resulting to decline in coffee margins.

Table 21: Number of Players per Category in the Coffee Industry, 2004

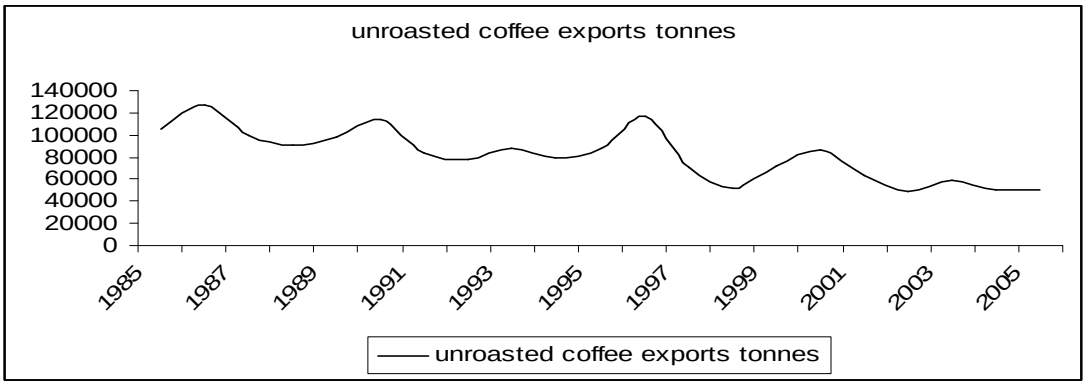
Category	Number 2004
Trade licence holders	57
Roaster license	19
Packers license holders	19
Warehouse license holders	23
Coffee auction license	5
Management license holders	6
Commercial coffee milling license	3
Private coffee milling license	2

Source: Coffee Board of Kenya

- **Coffee Exports**

In 2003, Kenya exported 58,650 tonnes of unroasted coffee valued at KShs 6,286 million. Unroasted coffee exports have been declining over the years (Figure 455). The price of Kenyan coffee has also continued to decline over the last 5 years hitting a low of KShs 117 per Kg in 2001 and Kshs 107 per kg in 2003..

Figure 45: Unroasted Coffee Exports (Tonnes)

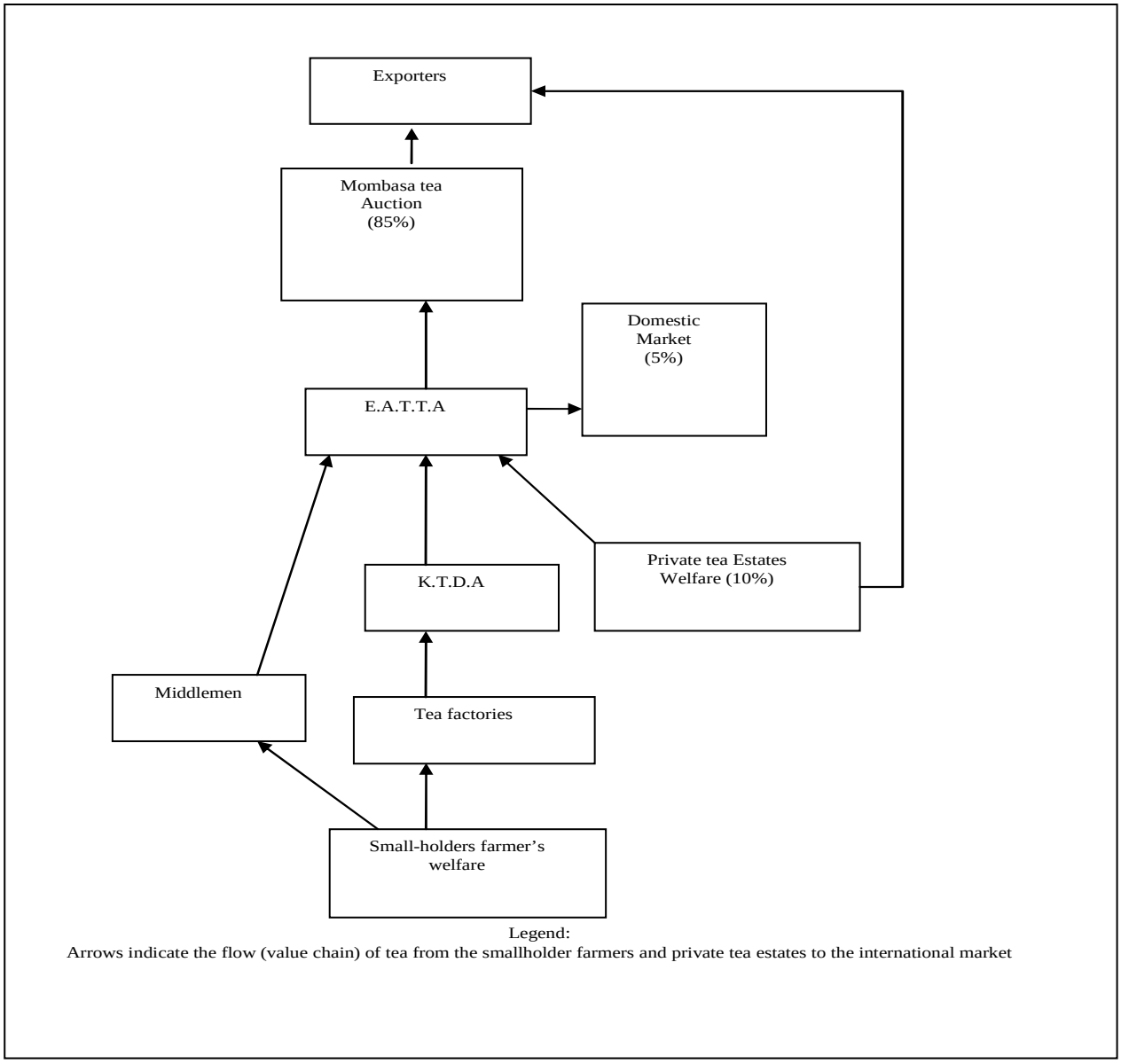


Source: Coffee Research Foundation (CRF)

2.4. Tea Marketing

The tea industry in Kenya is fully liberalized and the marketing of tea is independently carried out by tea producers and agents. Tea is either exported directly from the producers to importers in other countries or through the tea auction at Mombasa. Over 84% of Kenya tea is sold through the Mombasa tea auction, which is the second largest tea auction in the world. In 2003 over 300,000 tonnes of tea were exported earning about Kshs 41 billion in foreign exchange. The major tea export destinations were Pakistan (23.7%), Egypt (18.5%), United Kingdom (15.5%) and Afghanistan (13.7%). It is estimated that producers on through private arrangements export 10% of the total tea production with tea importers across the world (Kinyili, 2003).

Figure 46: Tea Marketing Value chain



Source: Adapted from Nyangito, H.O.,2005. Tea and Coffee Industry in Kenya 2005, Export Processing Zone Authority

Tea from the small-holders is sold as green leaf to the KTDA for processing and marketing. However there are increasing cases of green leaf tea hawking by middlemen that remains illegal. The large estates produce, process and sell tea directly to either exporters or through established commercial channels. Figure 46 shows the value chain in the tea industry. Value addition in tea is limited to processing which mainly entails fermenting and drying. There is little and insignificant packaging with packaged accounting for only 7%. Kenyan packaged tea is mainly exported to Sudan. Whereas there are increased benefits to value addition in tea, little has been done. There were 61 buyers and 62 tea packers in 2004 (Table 22). Owing to the volume of tea production in the country more players in the tea industry are required.

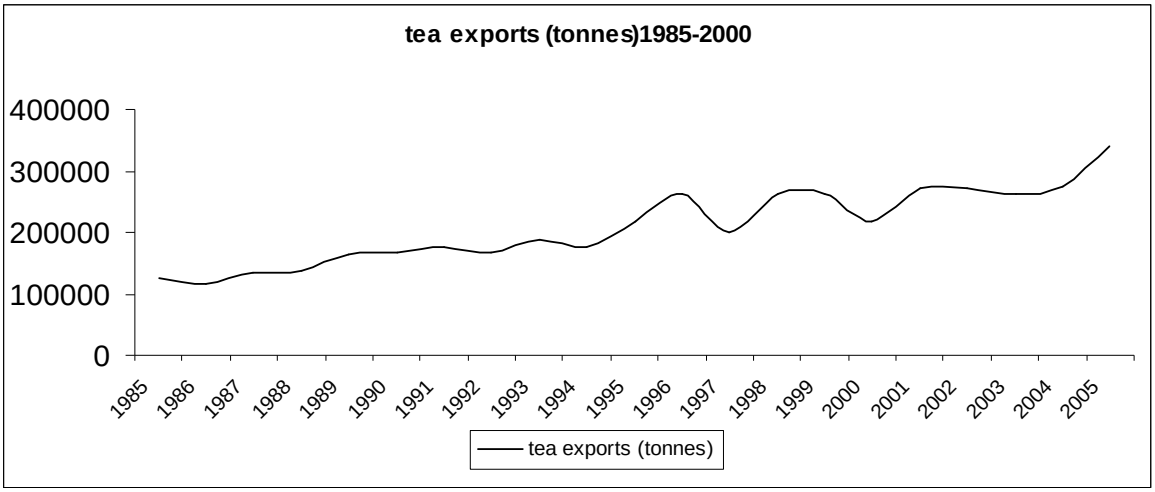
Table 22: Number of Players per Category in Tea Industry, 2004

category	Number 2004
Producers	21
Brokers	12
Buyers	61
Warehouse members	19
Associate buyers	44
packers	62

Source Tea Board of Kenya

Tea is the leading foreign exchange earner. It accounts for an estimated 20% of total export earnings (GoK, 2002-2008). The quantity of tea exports has increased over time (Figure 477) reaching 339,000 tonnes valued at 42.3 billion KSh in 2005.

Figure 47: Tea Exports



Source: CBS Economic Surveys. (Various Issues)

Reforms which took effect following commercialization of KTDA in 2000 seem to have had little influence in the tea industry with nominal prices of tea generally increasing.

2.5. Horticulture Commodity Marketing

The horticulture marketing chain is rather short with value addition mainly concentrated in packaging and labelling. The domestic market absorbs an approximated 96% of horticultural production while the rest is exported (MoA, 2004; HCDA, 2002). Large horticultural growers export their produce through exporters who own pack houses while the small-holders export through small and medium size exporters or through affiliation to large exporters as out growers. There is reported vertical integration where some exporters have moved to acquire production facilities and some large producers have also established their own export market links. Domestic markets supply comes through local traders and market vendors. Value addition is minimal in the domestic industry and a greater part of the potential lies unexploited. Focus has been on production and quality standards at the exclusion of value addition. The domestic market mainly absorb fruits and vegetables with the export markets taking more cut flowers. The domestic market is dominated by middlemen cum traders who either procure directly from the farms, at markets located in the production areas where farmers contract for green grocers, warehouses, supermarkets, institutions or directly deliver to established markets and processors (HCDA, 2002).

Horticultural marketing can be segmented into domestic and export value chains. The domestic value chain is presented in Figure 48. New entrants to the marketing chain involve the supply to supermarkets by organized farmer groups especially those dealing in indigenous vegetables. The effect of these chains has been the emphasis on quality and standards and the corresponding premium prices. However, there is still little value addition. The entry of supermarkets in the domestic horticultural industry albeit irrespective of the insignificant proportion has impacted small-holders who may not continue to operate on individual basis. While it is known that 96% of the total horticultural production enters the domestic market, the proportionate shares to the agents of the domestic market remain unknown.

The traditional marketing system, including urban wholesale markets, continues to play the dominant role in FFV (fresh fruits and vegetables) marketing in the country. The supermarket share of the FFV market in Nairobi was estimated to be below 4.4% in late 2003 with 80% being dominated by two major chains (Uchumi and Nakumatt) (Tschirley et al, 2004). Outside of Nairobi, it would certainly be lower. The supermarket's poor market share is attributed to the low number of buyers of fresh produce in these stores. Only an approximated 4.4%of the overall urban buyers buy their fresh produce from the supermarkets, with a dominant open air market accounting for 56.1% Kiosks 35.9% (Table 23).

While domestic horticultural value chain has remained entirely traditional, supermarkets have emerged but largely centered in urban areas. Their effects have largely been felt through quality and safety requirements which they impose on the fresh products and also on the development of supply-supermarket contracts. The export market is served by a few large–scale own company farms, an increasing number of contracted commercial horticultural farms, and a declining but still significant number of contracted smallholder farms (Tschirley et al, 2004).

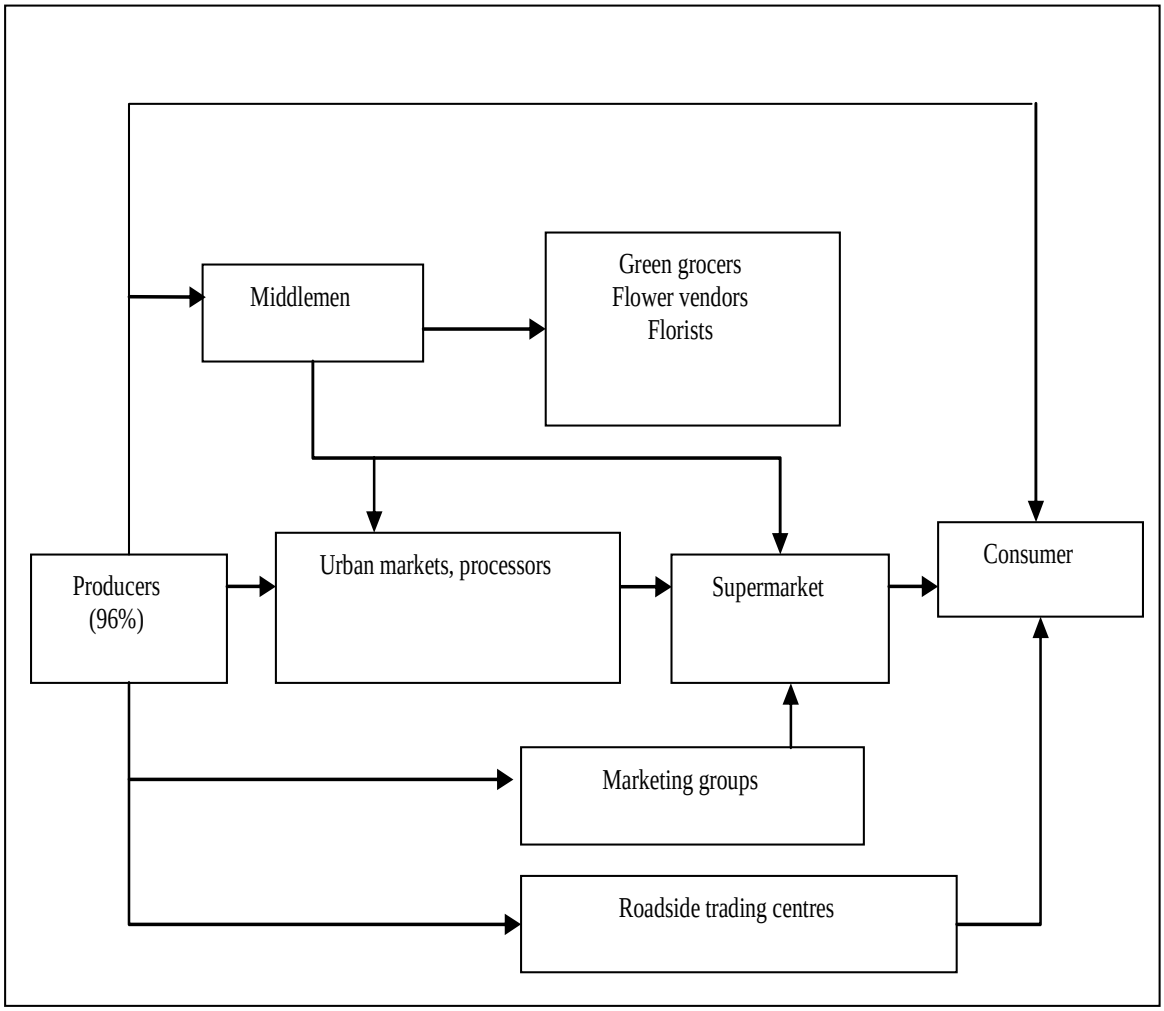
The large horticultural growers export their produce through large exporters who own pack houses while the small-holders export through small and medium size exporters or through affiliation to the large exporters as out growers.

Table 23: Domestic Fresh Produce Purchases

Per AE Income Quintile	Mean Income per AE (Ksh)	Market Outlet							
		Super-Market Chain	Small Super-Market	Duka	Open Air Market	Kiosk	Hawker	Green Grocer	Total
		---------- % of total FFV expenditure ----------							
1 (lowest)	749	0.0	0.1	0.2	54.7	41.5	3.6	0.0	100.0
2	1,890	0.0	0.0	1.3	60.6	34.4	3.7	0.0	100.0
3	3,314	0.4	0.7	0.0	60.7	36.4	1.8	0.0	100.0
4	5,599	1.7	0.1	0.3	59.5	38.0	0.4	0.0	100.0
5 (highest)	23,654	13.7	0.4	1.3	47.8	32.6	2.4	1.7	100.0
Overall	7039	4.4	0.3	0.7	56.1	35.9	2.2	0.5	100.0

Source: Ayieko et.al 2005

Figure 48: Horticultural Domestic Value Chain

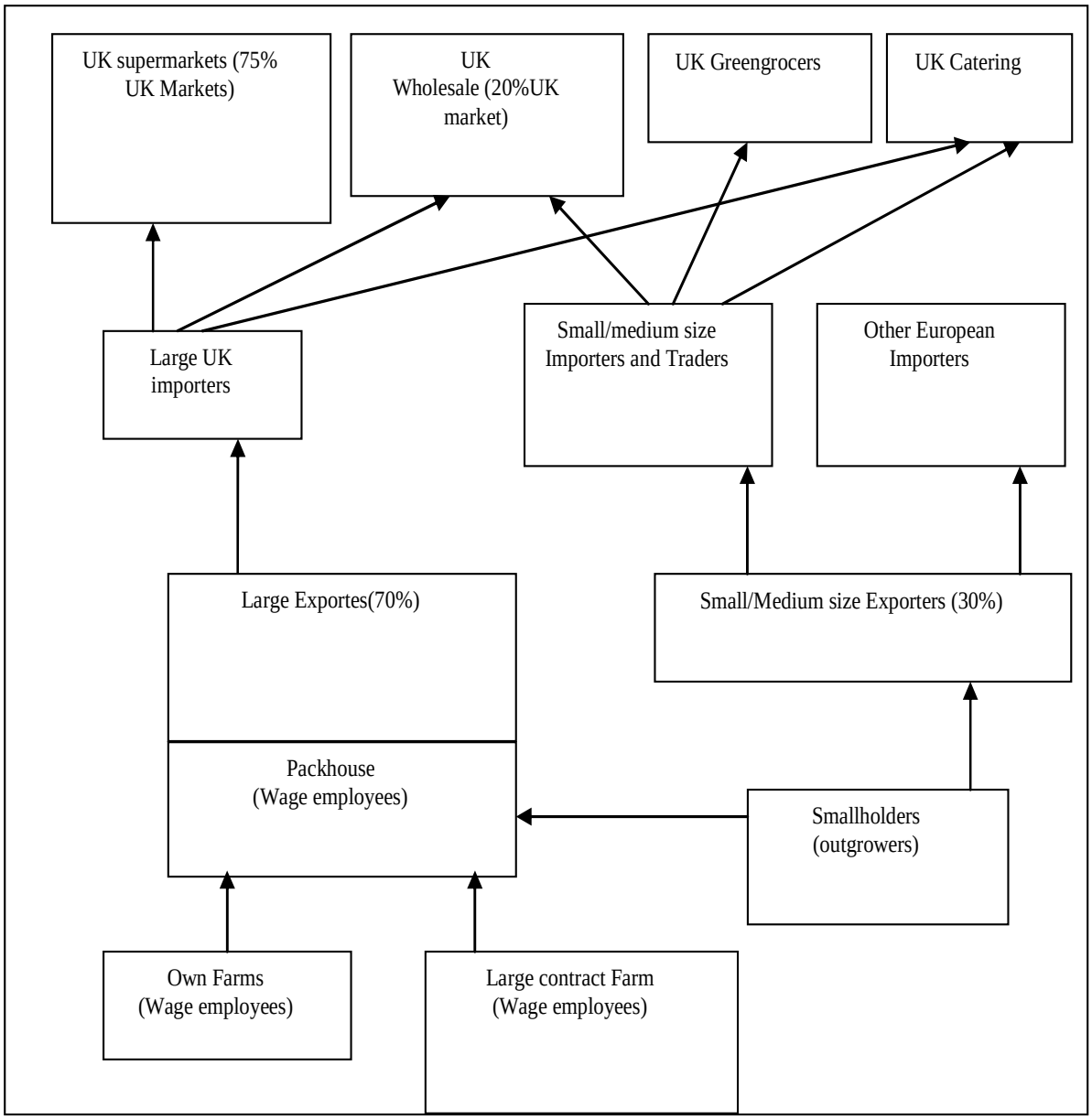


Source: HCDA

Approximately 4% of the horticultural produce is exported to the overseas markets. Value addition is also minimal in this channel with only packaging being done locally in pack houses. The horticulture export value chain is shown in Figure 49.

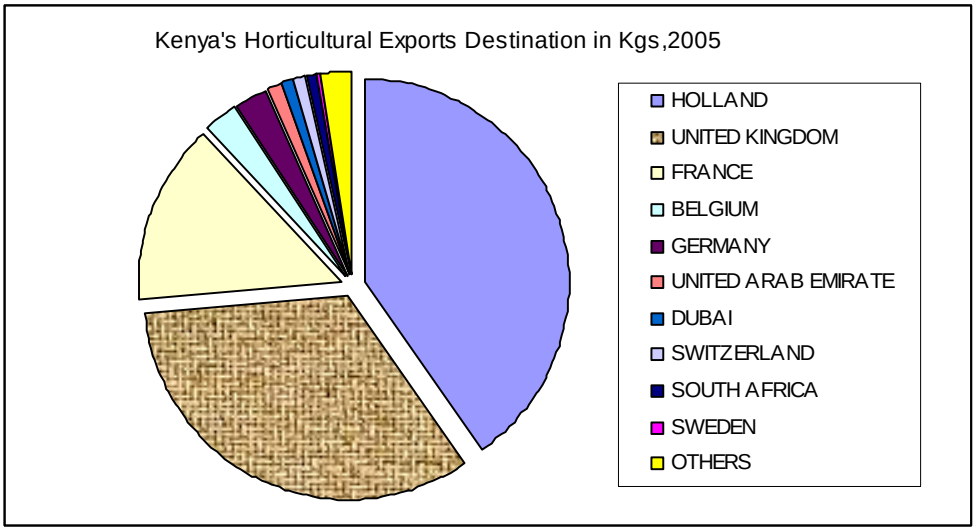
Most of the horticultural exports (89.4 percent) are destined for Europe. Figure 50 gives the destinations of Kenya's horticultural exports in 2005. In 1999 the UK's share of exports surpassed 71 percent, up from 33 percent in 1990. In 2005 UK's share was estimated at only 34% (HCDA, authors compilations). This decline was attributed to the stringent safety rules under EUREPGAP. This implies that there was diversion of horticultural produce to other market destinations with less stringent safety rules. The dependence of Kenyan horticultural exporters on the UK market has had significant consequences on the production, processing employment in the value chain.

Figure 49: Horticultural Export Value Chain



Source: Dollan, C.S and Kirsty S.(2004)

Figure 50: Horticultural Export Destinations, 2005



Source: HCDA

Total value of exported horticultural produce increased from 6.5 billion in 1995 to 38.8 billion Kshs in 2005 (Table 24). Cut flowers accounted for 59%, vegetables 35.8% and fruits 5.2% of the export value.

Table 24: Horticultural Export Values, Kenya 1994-2005

	Flowers	Fruits	vegetables	Total
Year	**Value Bn Kshs**	**Value Bn Kshs**	**Value Bn Kshs**	**Value Bn Kshs**
1994	-	-	-	-
1995	3.6	0.6	2.2	6.5
1996	4.4	0.8	2.6	7.7
1997	4.9	0.8	3	8.7
1998	5.9	3.8	3.8	13.5
1999	7.2	1.3	5.7	14.2
2000	7.3	1.1	5.5	13.9
2001	10.6	1.6	8	20.2
2002	14.8	1.5	10.5	26.7
2003	16.5	1.8	10.6	28.8
2004	18.7	1.8	12.1	32.6
2005	22.9	2	13.9	38.8

Source: HCDA

2.6. Market Integration and Concentration

As indicated by Bigsten and Durevall (2003), the extent of integration of the Kenyan economy with the world market can be illustrated by an index of price ratios (world prices to domestic producer prices) (Table 25). The period before reforms is set to unity to signify closed economy. During the transition and part of liberalization, most of the crops except coffee had higher production costs per unit than the world price (Values below 1) relative to the pre-liberalization period. As liberalization gained momentum, tea became relatively cheaper to produce owing to the increase in world tea prices.

Table 25: Price Ratios (World Price and Domestic Price)

	1966-79	1980-89	1990-95	1996-2000
Coffee	1.00	1.12	1.05	0.96
Tea	1.00	0.95	1.06	1.04
Maize	1.00	0.76	0.64	0.45
Sugar	1.00	0.96	0.79	0.62*

** Sugar price is for 1996-1999*
Source: Bigsten, A and Durevall, D 2003. Globalization and Wage Inequality in Kenya
Note: The indexes are based on the ratio between the world price and the producer price measured in domestic currency. The data are from the IFS database and various issues of Economic Survey. The world market prices are the following: the ICO index for coffee, UK prices for tea, London price for East African exports, yellow maize price in the Gulf, wheat price in the Gulf, and US sugar import price.

The process of liberalization saw coffee world prices plunge and input prices increase to make coffee relatively expensive to produce. The country has over the period remained comparatively disadvantaged in sugar and maize production. Market integration through liberalization seems to have favoured tea against coffee. While reforms reduced the relative transaction costs in tea production and marketing, the reverse happened in coffee and worsened for maize and sugar. These trends are evident in the outputs for various crop enterprises. It is evident that tea output has been increasing significantly while coffee output declined over the period of liberalisation as previously shown

Although liberalization has taken place, almost all the agricultural sub-sectors are dominated by transformed government institutions serving smallholders. This is due to these institutions already established infrastructure that enhance economies of scale relative to new market entrants. The exception is the horticultural industry that has thrived with little government involvement. The perishability of these products and the capital requirement of the business hinder the involvement of small-holders paving the way for private and large investors. Most of agricultural development efforts are centred on production with little going to processing and market promotion of the products. The SRA 2004-2014 envisaged the transformation of agriculture into a profitable activity through encouraging farmers to shift from subsistence agriculture to market oriented production by adopting modern farming practices and increase the level of integration of agriculture with other sectors of the economy.

The spatial nature of agricultural production in Kenya dictates the effects of any policy directed towards a specific sub-sector. The spatial nature of agricultural production results from climatic diversity and forms the basis for regional segmentation in supply and demand in the country. The fortunes or misfortunes of a liberalised commodity market therefore translates to regional level given the spatial nature of commodity enterprises.

3. The Factor Markets

3.1. Inputs Market

3.1.1. Fertilizer

Fertilizer importation was monopolized by the Kenya Farmers Association (KFA) from 1974 to 1984 managing 69 distribution outlets (Ariga, et. al 2006). In the 1989-1993 national development plan, the government recognized the need to make the marketing chains effective in distribution through the use of cooperatives and the private sector. During this time, donor fertilizer aid accounted for over half of the fertilizer imports. These were poorly coordinated with commercial imports leading to frequent oversupply and deficit (Kimuyu, 1994). Realizing that price controls limited fertilizer access to farmers and that they were creating a disincentive to fertilizer supply in remote areas, the government liberalized the fertilizer market decontrolled prices in 1993 and reduced donor imports to

only 5% of total consumption (Ariga, et.al., 2006). This resulted in increased private sector involvement in the fertilizer market and improved access to small scale farmers. By 1996, there were 12 major importers, 500 wholesalers and an estimated 5000 fertilizer retailers in the country. The number of retailers is estimated to have increased to 8000 by the year 2000 (IFDC, 2001).

According to Ariga et.al (2006) there are two identifiable fertilizer supply channels: (i) the commodity-based input-credit-output marketing arrangement and ii) the un-integrated system of independent importers, wholesalers and retailers operating on a cash basis. The former channel is operated by KTDA, coffee cooperatives and some sugarcane out-grower programs in Western Kenya. The model operates under exclusive rights where the firm provides fertilizer but retains the exclusive right to sell the farmers' output to recover the fertilizer cost. Liberalization has stimulated fertilizer use by small-scale farmers mainly by improving access through expansion of private retail networks hence reducing the distance to the nearest fertilizer retailer (Table 26).

Table 26: Small Farm Household's Fertilizer Usage

	Household fertilizer use behaviour 1996/97 to 2003/04				
		Not using	**Increased use**	**Decreased use**	**Using more than 50kgs/ha of maize**
Distance to the nearest tarmac road (kms)	1997	12.46	7.47	6.29	6.18
	2004	10.58	6.87	6.45	5.26
Distance to the nearest fertilizer retailer (kms)	1997	18.65	4.36	4.02	4.38
	2004	8.23	2.34	1.96	2.54

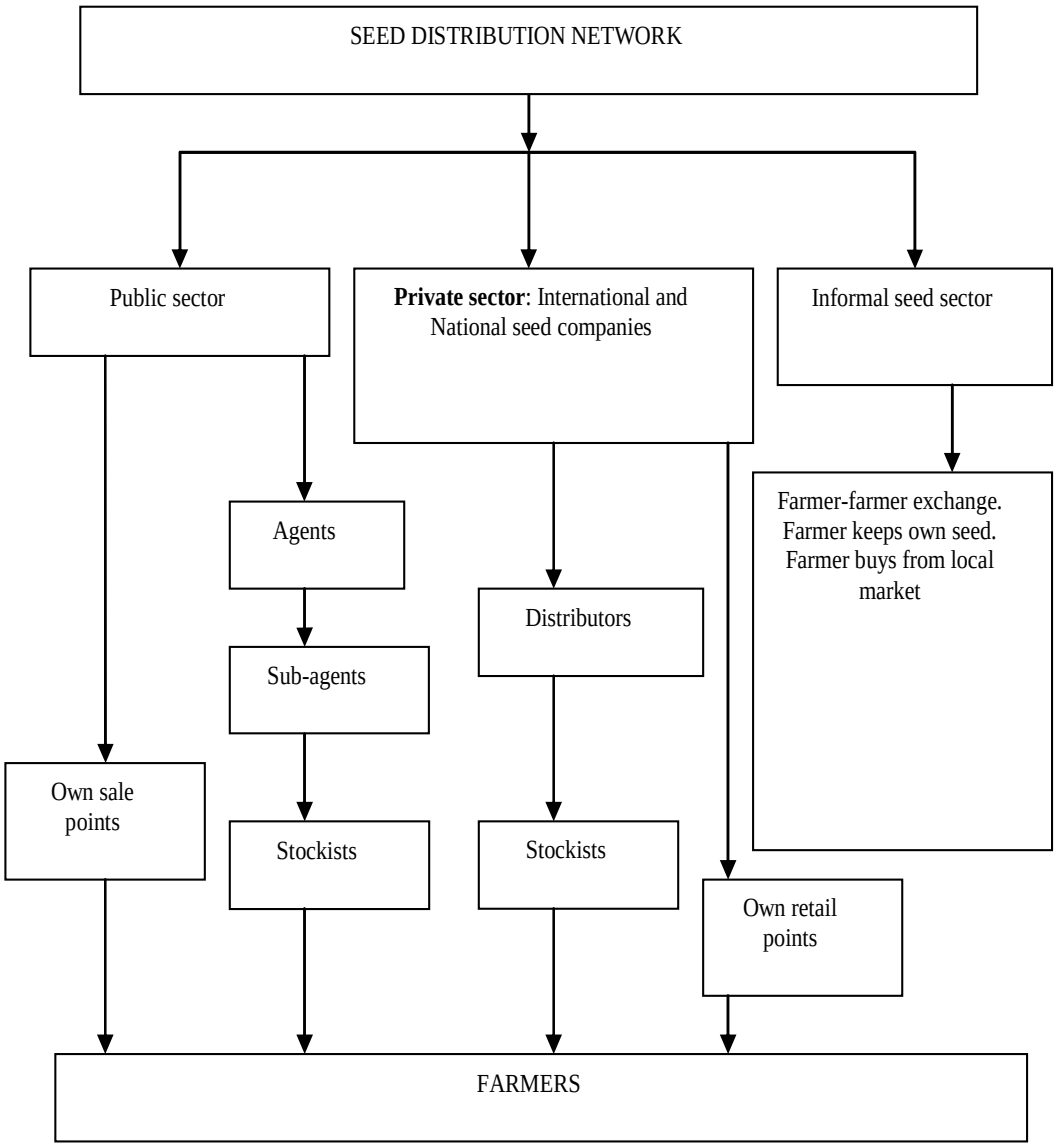
Source: Ariga et.al, (2006)

3.1.2. Seed Supply Structure

Figure 51 depicts the maize seed supply chain. Prior to liberalization, public institutes like the Kenya Agricultural Research Institute (KARI) a publicly founded organization, developed new seed varieties while seed production and multiplication was in the hands of the Kenya Seed Company (KSC). The informal distribution system involves the purchase of seeds from the local market, farmer to farmer seed exchange and farmers keeping own seed. The formal channel involved distribution agents and stockists all over the country. After liberalization, domestic and international seed firms slowly entered the seed industry. The private sector entered the market at distribution and stocking levels. However KSC still maintains the highest share in the market place. Despite the liberalization of the seed industry, use of certified high quality seed has remained low particularly among resource poor smallholder farmers mainly due to high seed prices. Seed quality is also perceived to be low by farmers.

The number of seed merchants registered per year has been fluctuating overtime while the number of active seed merchants has increased (Table 27). This is due to the stringent standards set within the industry to maintain and ensure quality of seeds.

Figure 51: Maize Seed Supply Chain



Source: Government of Kenya (Gok), 2004. The Status of Maize Seed Industry in Kenya: A Value Chain Analysis. Ministry Of Agriculture

Table 27: Merchants in the seed Industry

	Registered seed Merchants per year	No. of Deregistered seed merchants	No. of inactive merchants	No. of active Seed merchants	Total registered
1998 and below	30	None	12	18	30
1999	4	0	20	14	34
2000	10	0	21	23	44
2001	7	0	21	30	51
2002	2	21	0	32	32
2003	1	0	0	33	33
2004	13	0	1	45	46
2005	7	1	0	53	53
2006	4	0	0	57	57

Source: Government of Kenya (Gok), 2004. The Status of Maize Seed Industry in Kenya: A Value Chain Analysis. Ministry Of Agriculture

3.1.3. Seed Maize

The Kenya Seed Company (KSC) maintained exclusive rights for seed bulking from Kenya Agricultural Research institute (KARI) until the late 1990s when the industry was liberalized. This enabled entry of more players into the industry. More than ten companies were registered but the KSC maintained the largest share (86%) of the seed market and in 2003, the company accounted for about 94% of the total seed certified by Kenya Plant Health Inspectorate Service (KEPHIS) (GoK, 2004). The Kenyan maize seed system comprises participants who can be categorized according to the functions they perform along the value chain. The stages along the chain can be specified basically as research and development, seed multiplication, seed processing and storage, seed marketing and distribution. Seed marketing and distribution is undertaken by both the private and public sectors. The public seed distribution in the country is characterized by a chain of agents, sub-agents and stockists who are classified on the basis of volumes of seeds handled. Between 1964 and 2003, both local and international seed companies had released 80 maize varieties (GoK, 2004). Figure 51 shows the breakdown of Maize cultivars by public and private sectors. Between 1993 and 2003, private sector participation was high releasing 60 cultivars of maize while the public managed only 11. This followed liberalization of the maize seed industry in early 1990s hence encouraging private participation.

Table 28: Maize Cultivars by Public and Private Sectors

Period	Public sector number released	Private sector no. released	Both sectors
1964-1973	6	0	6
1974-1983	2	0	2
1984-1993	2	1	3
1994-2003	11	60	71
Type of material			
Hybrid	16	55	71
Improved OPV	5	6	11

Source: KEPHIS, 2003

3.2. The Credit Market

Provision of credit to the farmers in rural areas is widely recognized as an incentive to increase production in rural areas. During the colonial period, the shortage of working capital led to provision of credit which targeted large-scale farmers through the Land bank later the Agricultural finance Corporation (AFC) and to small farmers through the Cooperative Bank of Kenya and the Department of Lands and Settlement (Scott et.al., 1995).

In 1993, the major lenders to the agriculture sector included commercial banks (48%), AFC (20%), private non-banking institutions (18%) and combined AFC and CBK operations (14%).

While there are formal and informal models of credit supply, the government has restructured the AFC to operate efficiently. Several models were identified from Tegemeo Agricultural Monitoring and Policy Analysis (TAMPA) 2004 survey. These include: Community Owned Rural Financing Models, Private Commercial Bank Led Model, Government Led-Rural Finance Model, Donor Guarantees-Input Supply Model, Managed SACCO-Beach Banking Model and the Informal Group Based Rural Financing Model.

Despite provision of substantial credit amount, the agricultural sector did not respond positively due to massive diversion of funds to non-agricultural activities. Up to 1983, commercial banks controlled interest rates. The agricultural cooperative sector was liberalized during the late 1990s following the enactment of the Co-operative Act and session Paper No.6 of 1997 on Cooperatives in a liberalized Economy. Initially land was used as collateral to secure loans from cooperatives; however this depended on the nature of the land tenure. The consequences of eviction owing to non payment were however unbearable. This led to the shift of the collateral to the crop-secured loans which allowed money lenders more easily collectible assets. Further, the repayment of loan was complicated by erratic exchange rates and devaluation. The provision of credit services has however been increasing owing to the increasing number of savings and credit societies in the country. For instance, the savings and credit societies in the country increased from 3925 in 2001 to 4474 in 2004, with coffee, pyrethrum, dairy and sugar sub-sectors witnessing the highest number of increases (CBS, 2006). Despite the increase in number, credit access to small holders has been inadequate due to low productive capacity, mismanagement of the cooperative sector, poor performance of AFC, high interest rates, low output prices, poor management of marketing institutions, collateral requirements and high risks associated with agricultural production.(GoK, NDP.,2002-2008).

Agricultural credit has mainly been sourced from commodity based providers (Table 29). AFC has recently increased its coverage however, restructuring to cover value addition and involvement in credit wholesaling.

Table 29: Source of Agricultural Credit in Percent, 2000 and 2004

Source of credit	2000	2004
Commodity Based Credit Providers	53.5%	62.7%
SACCOs	26%	20.6%
Informal money lenders	12.1%	9.9%
Local trader/input stockists	6.8%	3.9%
AFC	0.4%	1.3%
Commercial banks	0.6%	1.0%
MFIs/ NGOs	0.6%	0.5%
	100%%	100%

Source: Tegemeo household survey, 2000 and 2004

3.3. The Agricultural Extension Market

The extension system has undergone transformation from the colonial period when the government was the sole provider. Currently, several agricultural sub-sectors and private companies carry their own extension services. For tea, extension services are carried out by KTDA, while private companies such as British American Tobacco (BAT), Kenya Breweries and East African industries carry their own extension systems for tobacco, barley and oil seed. The extension system has utilized Training and Visit (T&V) and the current Shifting Focal Area Approach (SFAA) that emphasizes Frontline Extension Workers (FEWS).

In 1993, the government reformed the agricultural extension separating funding from service and orienting agriculture extension message from the initial production husbandry practices to marketing issues. Following liberalization, extension services are moving towards multiple providers while the government retains the funding role. This will provide for increased community participation in extension resource allocation. Concerns have arisen with respect to the poor and marginalized who might suffer increased alienation due to inability to articulate their demands. Quality control aspects have also been expressed in the absence of a regulatory authority or mechanisms for self-regulation.

3.4. The Labour Market

Agriculture and related agro-industries are the largest employers in the country. The labour market in Kenya is segmented into unskilled and skilled labour markets with casual, permanent contracts or a mixture of both. The principal labour market is that of unskilled labour (Scott, et.al., 1995). The market is characterized by peaks during intensive agricultural husbandry practices such as land preparation, planting, weeding and harvesting. During the pre-independence era, a considerable gap existed between agricultural and non agricultural wages and the gap increased gradually due to market imperfections (Bigsten and Durevall, 2003). The immediate post-independence period presented an increase in earnings for skilled workers with an urban-rural gap emerging.

Prior to liberalization, the government prescribed wage guidelines under the basis of the cost of living indices, but currently various labour organizations/ trade unions review and negotiate the level of wages according to productivity and performance (Kulundu, 2002).

The most dramatic shift in pattern of labour has been the upsurge of employment in the informal sector outside agriculture (Bigsten and Durevall, 2003).The diverse regional and ecological conditions lead to different labour requirements which often influence regional wage differences. The average annual wage in the agriculture sector in 2004 amounted to 106,387 Kshs far below the services sector with Kshs 347,719. In the same period, the public and private sector wages were observed to be Kshs 178,273 and Kshs 291,460 respectively (CBS, 2006).

The price of labour has been increasing overtime. There is however variations in wages paid to labour in the different agro-regional zones as shown in Table 30 below. The difference in wages is attributed to regional labour productivity differences and market access.

The wage rates have increased in all the agro-regional zones with less variation in the hours worked.

The type of labour contract and agricultural activity also influences wage levels. It is hypothesized that casual labour contracts fetch higher wages than permanent contracts and that relatively intensive and supervised agricultural activities have higher wage rates. This may imply that households will engage in casual contracts than permanent contracts. For those who may own their few assets, casual contracts are supplements to their incomes earned from working in their own farms or businesses

Table 30: Mean Daily Wage Rate

	1997	2000		2004	
Agro-regional Zones	**Wage Rate**	**Wage Rate**	**Hours Worked**	**Wage Rate**	**Hours Worked**
Coastal Lowlands		119	7	135	7
Eastern Lowlands		79	7	100	6
Western Lowlands		56	5	61	5
Western Transitional	46	57	6	67	6
High Potential Maize Zone	63	73	6	85	6
Western Highlands		42	5	49	5
Central Highlands	66	86	7	102	7
Marginal Rain Shadow	62	92	7	110	6

Source: Tegemeo Institute/MSU Agricultural Monitoring and Policy Analysis Household Surveys, 1996/97, 1999/00 and 2003/04.

3.5. The Land Market

Land policies and legislation in Kenya gave rise to three land tenure systems: private, customary (communal) and public land tenure systems (Odhiambo and Nyangito, 2002). The full and legal ownership of land, through title deeds creates an incentive to investment, development, and necessitates efficient transfer or reallocation from one user or alternative use to another. The land market is especially thin in most areas especially those under customary tenures with very few transactions being cited (Scott, et.al., 1995). Land value varies with geographical region and potential. The degradation of land especially in marginal lands has been perpetuated by inappropriate property rights regime. This has resulted into migration from marginal areas leading to high population densities in some regions. Land tenure system creates certainty or uncertainty in the land market as it directly affects the mobility of land, its marketing and directly impacts on agricultural investment and rural development.

The general proportion of rented land to owned land indicates a declining trend. (Table 35).

Table 31: Proportion of Rented to Owned Land

Agro-regional zones	1997	2000	2004
Coastal Lowlands	0	0.03	0
Eastern Lowlands	0.02	0.01	0.01
Western Lowlands	0.14	0.09	0.06
Western Transitional	0.09	0.07	0.09
High Potential Maize Zone	0.11	0.29	0.1
Western Highlands	0.17	0.08	0.12
Central Highlands	0.16	0.03	0.05
Marginal Rain Shadow	0.15	0.16	0.05

Source: Tegemeo Institute/MSU Agricultural Monitoring and Policy Analysis household Surveys, 1996/97, 1999/00 and 2003/04.

This implies diminishing land markets. It is assumed that an increase in prices (rental values) leads to more land being offered for rent. Results however reveal the opposite, that less and less land is being rented. In all the agro-regional zones the household owned land has declined since 1997 This is attributable to the increasing population pressure leading to subdivision of land to small uneconomical plots.

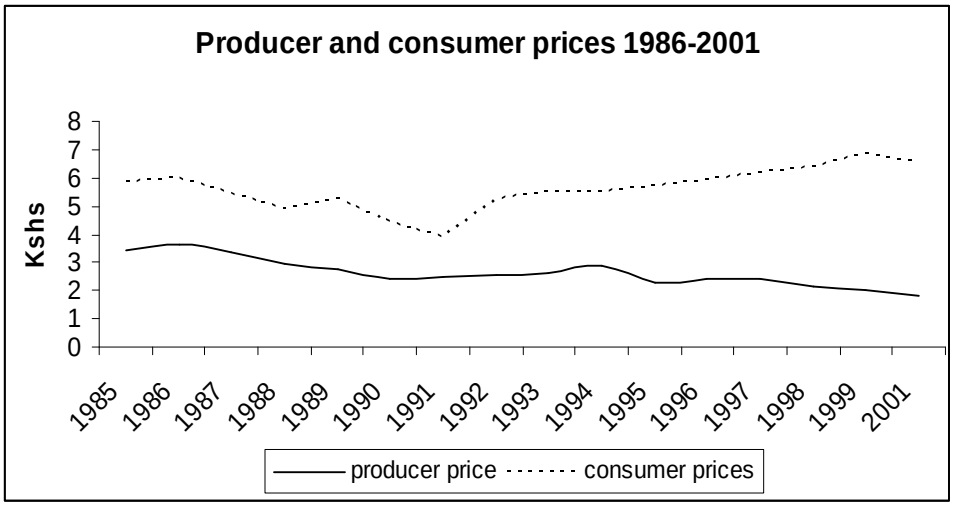
4. Evolution of Relative Prices

4.1. Main Commodity Prices

- **Dairy Prices**

While producer prices have been declining, the consumer prices for packaged milk have significantly increased from 1992 (Figure 52). Thus, the wedge between producer and consumer prices continues to widen. The price wedge per litre increased in nominal terms from Ksh 2.30 in 1985 to Ksh 35.60 in 2000. In real terms the wedge doubled from Ksh 2.44 in 1985 to Ksh 4.83 in 2000. This implies that it costs twice as much to collect, process and distribute a litre of milk as compared to its production and therefore consumers have not benefited in terms of prices from milk market reforms. The efficiency of the milk-processing channel may have deteriorated during the reform period. Further, the increasing consumer prices can explain to a large extent the declining demand for processed milk and milk products. Policies specifically targeted at enhancing efficiency in milk collection, processing and distribution are critical in improving the efficiency and competitiveness of the dairy supply chain.

Figure 52: Real Producer and Consumer Prices (1986=100)

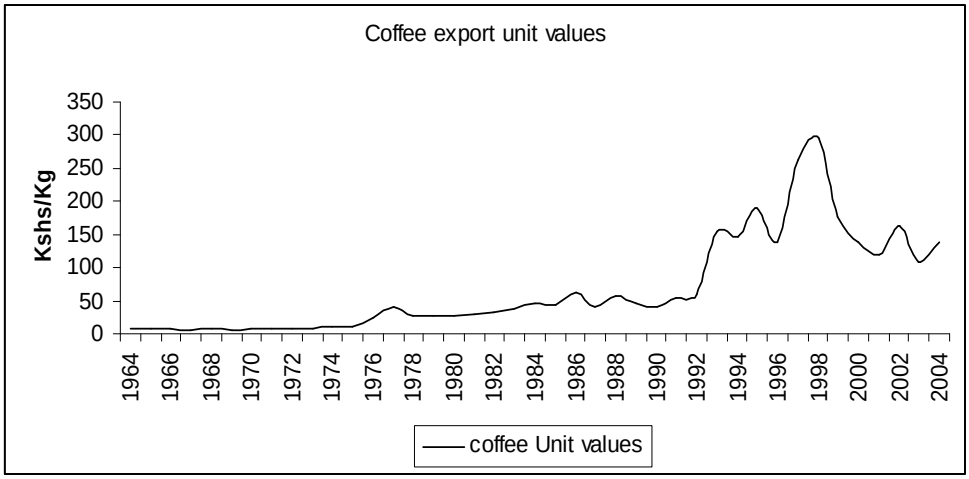


Source: KDB

- **Evolution of Coffee and Tea Prices**

Prior to the 1980s, coffee export unit values exhibited a stable trend under the import substitution policy with destabilizations only caused by booms. The prices of coffee rose during the 1976/78 boom but gradually declined afterwards only to increase marginally during the mini coffee boom of 1986. High coffee prices were witnessed in the late 1990s but these have since plunged (Figure 53). The implementation of SAPs and market reforms seems to have led to decline in coffee prices.

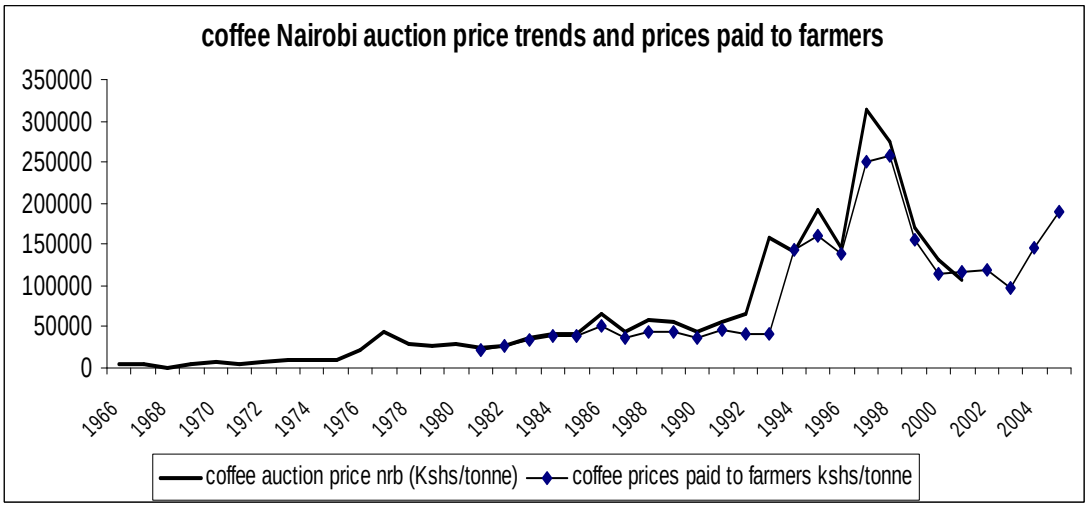
Figure 53: Coffee Export Unit Values, Kenya 1964-2004



Source: FAOSTAT

Coffee prices paid to farmers mirror the Nairobi coffee Auction prices and the export unit values with proportionate increases and declines. Gradual increase was maintained in prices until 1994/98 when drastic increases occurred. This increase was short lived and prices plummeted from 1999 (Figure 54).

Figure 54: Nairobi Coffee Auction Prices and Prices Paid to Farmers

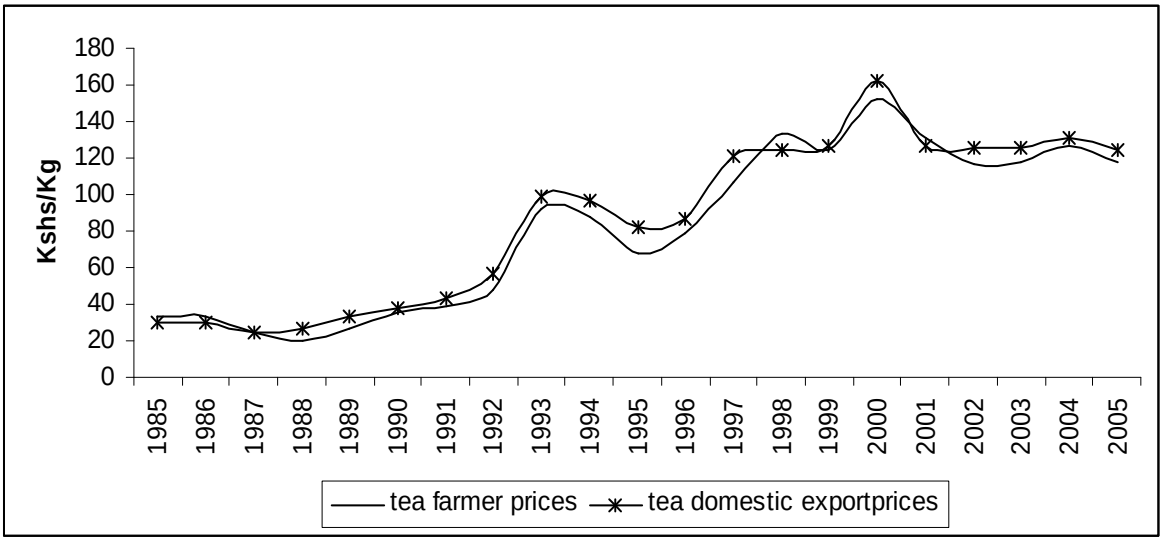


Source: CBK

This proportionate movement indicated that coffee farmers benefited from the little positive effect of liberalization. However the gains that may have been accrued during the 1990s could have been as a result of local currency depreciation against the US dollar.

Tea prices exhibited gradual increase overtime with export prices and prices received by farmers reflecting similar trends (Figure 55). The consistent trends in both prices indicate increased earnings by farmers during the liberalization period. However, in the recent past, these prices have exhibited a downward trend that is associated with oversupply in the world tea market.

Figure 55: Prices Received by Tea Farmers and Export Tea Prices

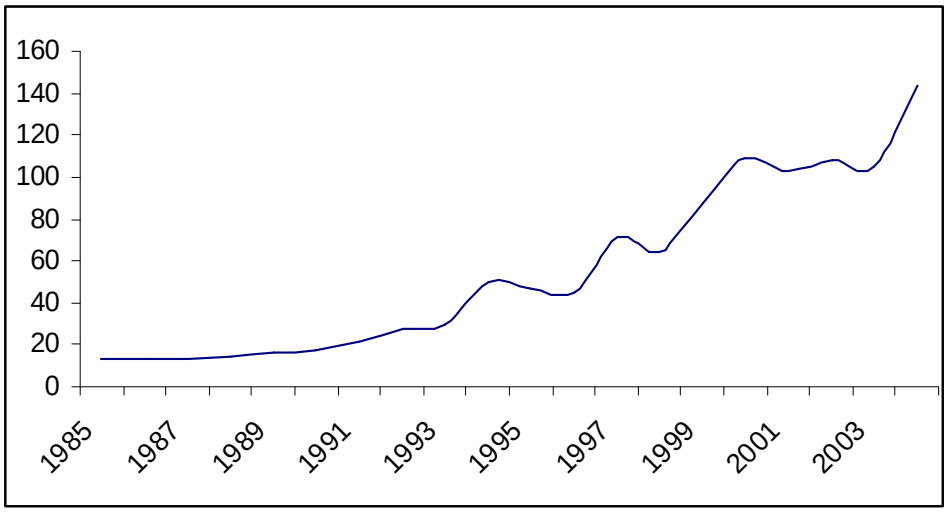


Source: CBS Economic Surveys. (Various Issues)

- **<u>Horticulture Price Trends</u>**

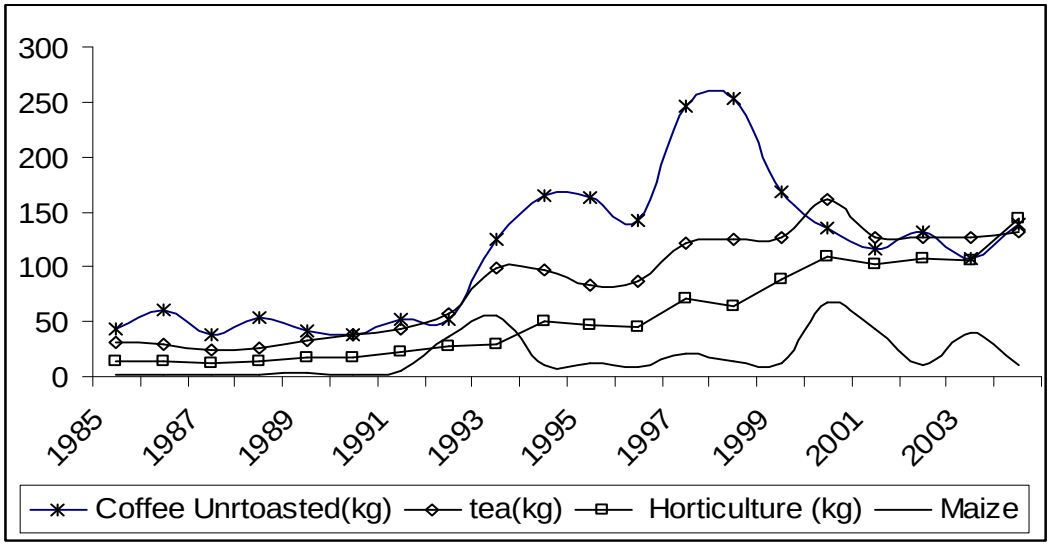
Horticultural export prices started to exhibit increasing trend in the early 1990s, and, have continued to increase (Figure 56). Liberalization of the economy appears to have triggered an increasing trend in nominal horticultural prices. The somewhat constant trend between 1999/04 coincided with imposition of quality standards through EUREPGAP. Quality standards led to decline in fruit exports quantities and decline in the export share to UK but prices seem to have been less affected. This is attributed to the fact that fruits are more demanding in quality maintenance than other horticultural products and are produced by small-scale out growers who find it difficult to comply with quality standards

Figure 56: Evolution of Export Prices of Horticulture Kenya, 1985-2005



Source: CBS Economic Surveys. (Various Issues)

Figure 57: Evolution of Export Prices of Main Productions Kenya, 1985-2005



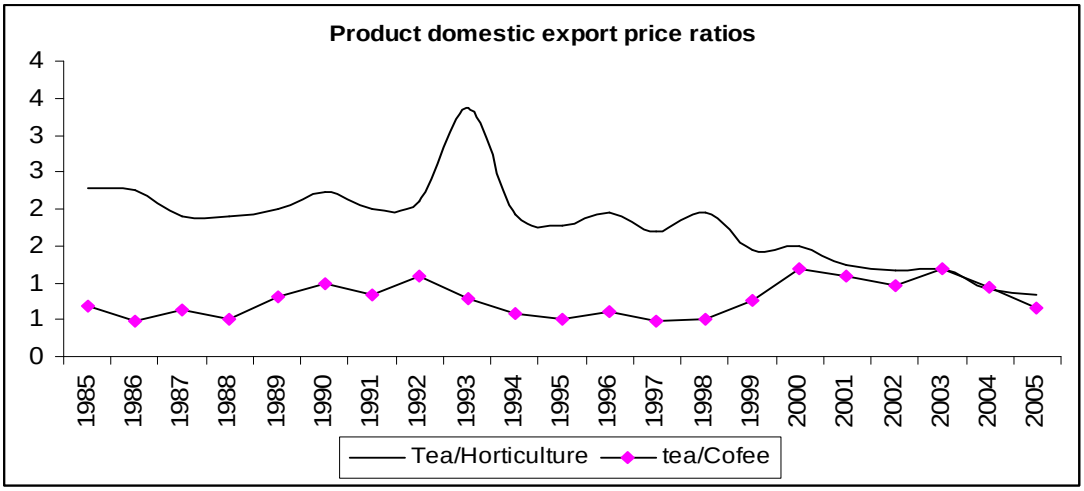
Source: CBS Economic Surveys. (Various Issues)

Main agricultural products exhibited an upward trend in prices immediately after liberalization except for maize (Figure 57). While the prices generally increased, the improved margins realized by farmers may have resulted from currency depreciation that occurred after liberalization. Decline in world prices partly explain the sudden drop in coffee prices while over supply in tea, and inception of EUREPGAP quality standards are likely causes for the tea and horticultural price trends in the post-liberalization

4.2. Major Crop Price Ratios

Tea, coffee and horticulture sub-sectors are considered in this portion due to their major contribution to the economy. The price ratio between the two most successful sub-sectors (tea and horticulture) exhibit a decline, while tea-coffee price ratio exhibit a fairly upward trend after liberalization (Figure 58).

Figure 58: Product Domestic Export Price Ratios Kenya, 1985-2005



Source: CBS. Economic Surveys. (Various Issues)

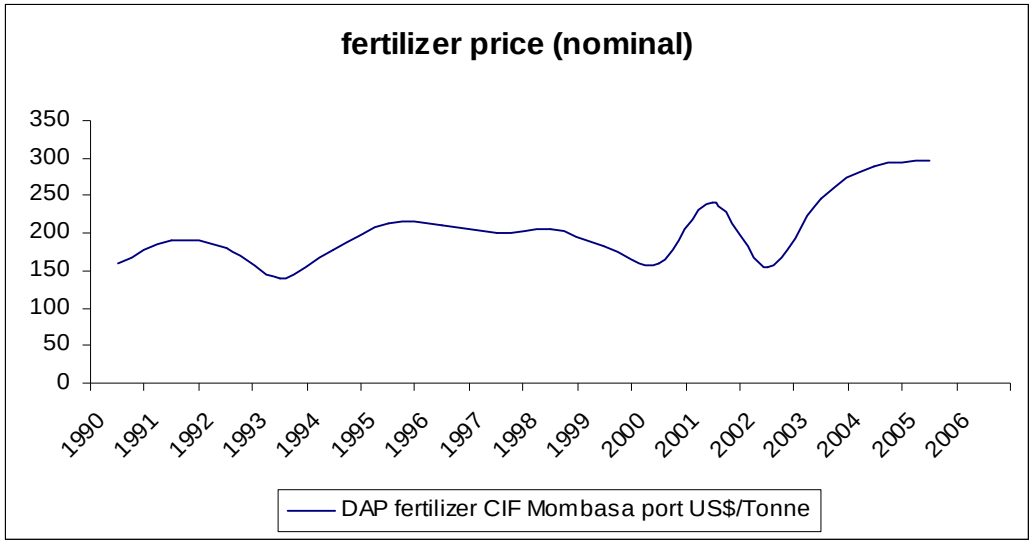
This implies that, the prices of horticulture increased faster than tea prices, while tea prices grew faster under reforms compared to coffee prices. While it is hard to ascertain the alternatives to coffee enterprise failure, the growth witnessed in horticulture, tea and the informal sector project the response of farmers toward the decline in coffee value.

- **Trends in Crop-Fertilizer Price Ratios**

Fertilizer import prices CIF Mombassa depicted a less varying trend in the early 1990s following reforms in the fertilizer market, however in the mid 1990s import prices CIF Mombassa rose sharply before declining in subsequent years. The period 2002-2005 (Figure 59) has seen a tremendous increase in prices CIF Mombassa that translated to an increase in domestic fertilizer prices (Ariga, et. al 2006). This was due to untimely government intervention in the fertilizer market through the NCPB that distorted signals to the private sector.

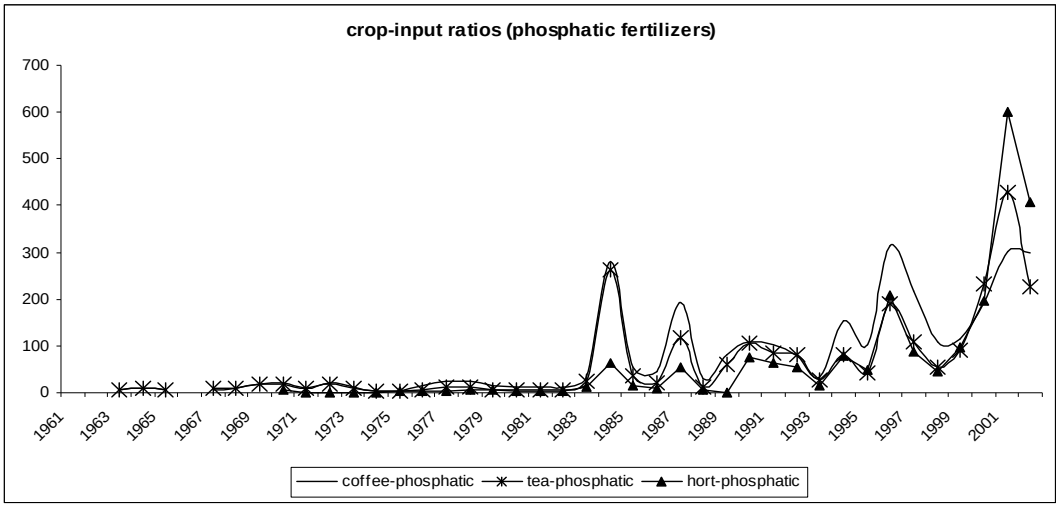
The increasing but variable pattern in crop-fertilizer prize ratios through the reforms and post reforms (Figure 60)is an indication that commodity prices have continued to fall behind fertilizer prices. This implies that fertilizer prices are rising at a faster rate compared to commodity prices and squeezing producer margins due to higher production costs.

Figure 59: Fertilizer Price Trends



Source: Ariga et.al, (2006)

Figure 60: Crop-input Ratios (Crop-Phosphatic fertilizers) Kenya



Source: FAOSTAT

Conclusion

Prior to liberalization, agricultural markets were characterized by strong government controls through various parastatals inherited from the colonial period. Prices and movement controls were imposed for grain foods while export crops production and marketing were tightly controlled by boards such as the KTDA, CBK, and KDB for tea, coffee and dairy respectively. Markets were monopolized and entry was barred through regulation, licensing and even economies of scale that constrained entry resulting into major market distortions. The import substitution policy which envisaged and perpetuated these monopolistic tendencies however did not spur much economic benefits and reforms were therefore initiated to streamline commodity markets. The gradual implementation of the reforms package led to government withdrawal from markets setting competitive forces into play. This has has stimulated the intereand encouraged the entry of multiple market players enhancing efficiency in production and correcting market distortions that existed.

Increase in the number of players and agents along the chain puts enforcement of food safety regulations and maintenance of quality standards in a precarious situation such as in dairy industry where informal chain has dominated. Due to emergency of multiple players in the marketing chains, enforcement and maintenance of contracts is now a major challenge especially with regard to observance of business ethics. The ensuing competition has forced the exit of some players resulting in industry concentration. Those who remain, grapple with readjustment and the new entrants have to contend with massive initial capital investment. Within the input sector, concerns are emerging particularly in line with diminishing land markets, poor quality of inputs and regulatory inefficiency. For instance, the quality of seeds, extension information, fertilizers and the continued alienation of the marginalised in provision of these services are raising concerns. Integration and differentiation along the chains has also taken place. While it is expected that integration minimises transaction costs and creates synchronization in demand and supply, some sectors such as dairy have witnessed widening gap between producer and consumer prices and a corresponding decline in welfare.

PART III -

SEGMENTATION PROCESS OF PRODUCTION STRUCTURES: TRENDS AND OBSERVED CONSEQUENCES

CHAPTER 6 - SEGMENTATION PROCESS OF PRODUCTION STRUCTURES

The nature and evolution of agriculture and rural development in Kenya has perpetuated segmentation. This geographical and climatic dimension of segmentation was aggravated by a similar trend in public infrastructure development that is manifested in constrained regional development and agricultural enterprise.

Economic reforms initiated in the early 1980 were intended to open up the economy. The exposure of the economy to external market forces and new policies implemented in the absence of the necessary infrastructure, institutional and legal framework led to varied results. Some sectors of the economy such as manufacturing were favoured by the reforms while others, notably agriculture were not. During the colonial period production was biased to the large-scale sector which was managed by white settlers. By early independence (1960s), the land policy advocated for land subdivision and reallocation. With increasing population pressure and demand for agricultural products during this time, the small-holder sectors emerged as the major contributor to the economy.

In the course of liberalisation, previously performing sub-sectors such as coffee, sugar and cotton gradually declined in growth and economic contribution. However, some such as tea and horticulture thrived during and after the reforms. This regional imbalance in the effects of the reforms resulted to the supply segmentation in these major crops. There were increases in tea and horticultural production while coffee and sugar declined with cotton becoming unsustainable. Alternative livelihood strategies in the losing regions had to be adopted. This included pursuance of alternative income sources within the rural setting. New crops, technologies, cropping systems, alternative off-farm income sources and even migration within and outside are some of the strategies adopted. The flexibility of the rural households to adopt these strategies is pegged upon their natural and physical resource endowment and income levels. While data to elucidate segmentation succinctly may be limiting, hypothesis and inferences can be made from the available national data on major sectoral and sub-sectoral productions.

1. Indicators of Change

1.1. Fertilizer Use

Fertilizer use especially nitrogenous and phosphatic fertilizers have increased over the years (Table 32). The trend has however been erratic indicating the unstable nature of the availability and access exhibited by the fertilizer markets. Total fertilizer use has risen from 11100 tonnes in 1961 to 142758 tonnes in 2002 (FAOSTAT). The total fertilizer consumption during the pre-liberalization era was 38802.4 tonnes increased to 96851.7 tonnes and eventually 131705.3 tonnes during the transition and liberalization eras. The increased use of fertilizers indicates the extent of soil deterioration that has taken place. However, it may also indicate the increasing demand for intensive production technologies or their adoption and extent of use as opposed to extensive production.

Table 32 Average Annual Fertilizer Use (MT)

Fertilizer category	Pre-liberalization 1961/80	Transition 1981/92	Liberalization 1993/02
Nitrogenous fertilizers	16465.4	44008.2	55398.6
Phosphatic fertilizers	18765.55	44901.8	66628.5
Potasic fertilizers	3571.45	7941.7	9678.2
Total fertilizer use	38802.4	96851.7	131705.3

Source: FAOSTAT

- **Large- and Small-scale Fertilizer Use**

Using the Tegemeo panel survey data, Ariga, et.al., (2006) explored the sources of increased fertilizer use in 24 districts in Kenya. The panel covered 1,364 small scale farmers. The findings indicate that small-scale farmers' use of fertilizer in 2003/04 was almost double the amount used in 1996/97 (Table 33).

Table 33: Small-scale Fertilizer Use in Kenya

Year	Tonnes
1995/96	225
1996/97	237
1999/00	447
2003/04	424

Source: Tegemeo Institute/MSU Agricultural Monitoring and Policy Analysis household Surveys, 1996/97, 1999/00 and 2003/04.

The analysis further revealed the fertilizer use over the diverse ecological conditions of the country as follows (Table 34).

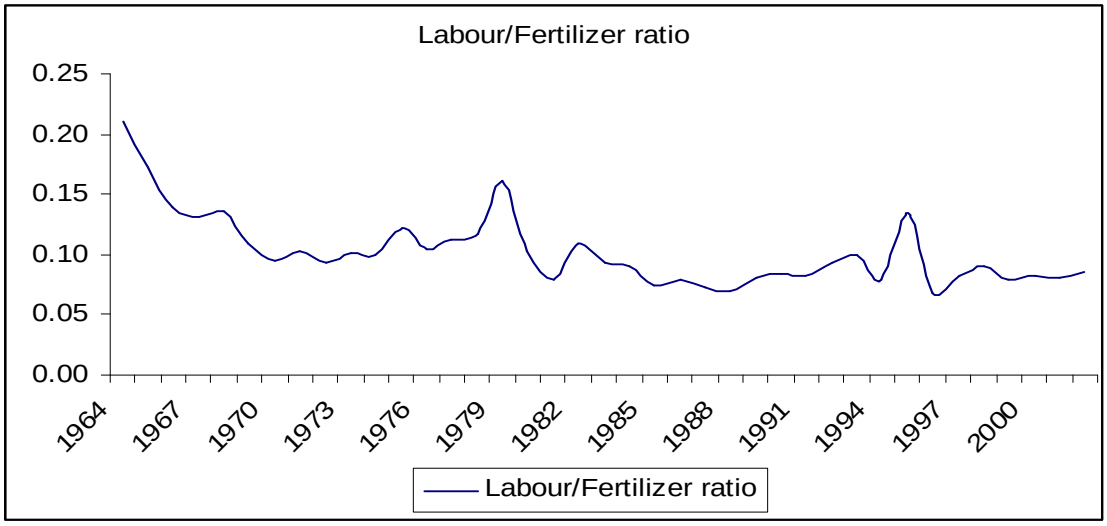
Table 34: Small-scale Households Using Fertilizers by Ecological Region

Region of Kenya	1995/96	1996/97	1999/00	2003/04
Coastal Lowlands	2%	3%	5%	6%
Eastern Lowlands	19%	30%	37%	46%
Western Lowlands	2%	3%	4%	8%
Western Transitional	29%	32%	59%	61%
High potential Maize Zone	67%	69%	86%	90%
Western Highlands	52%	57%	73%	74%
Central Highlands	63%	78%	90%	93%
Marginal Rain Shadow	12%	20%	22%	27%
Nationwide Sample	43%	51%	64%	69%

Source: Tegemeo Institute/MSU Agricultural Monitoring and Policy Analysis household Surveys, 1996/97, 1999/00 and 2003/04.

Although the number of small scale farmers using fertilizers has increased, it has varied widely depending on the agro-ecological zone. The high potential areas have exhibited increasing number of small-holders using fertilizers while the arid and semi arid areas have been constant. Despite increased number of small-scale farmers using fertilizers, the total quantities have declined to 424 tonnes in 2003/04 from 447 tonnes in 1999/00. The reduced use of fertilizer among sections of small holders provides the basis for the governments' unwillingness to fully free the market thereby contributing to uncertainty in the fertilizer market. This is in addition to the high concentration and cartel behaviour experienced by the fertilizer market. For instance, in 2003, 75% of the fertilizer imports were acquired through only top three firms. The top importer controlled a significant 40%. Liberalisation in this case appears to have enhanced rather than reduced concentration (Ariga et al, 2006) a clear legal framework and organized stakeholder institution is also yet to be realized. Labour-fertilizer use intensity ratio trend has been on a decline (Figure 61) showing increased use of fertilizers compared to labour.

Figure 61: Labour Fertilizer Use Ratio

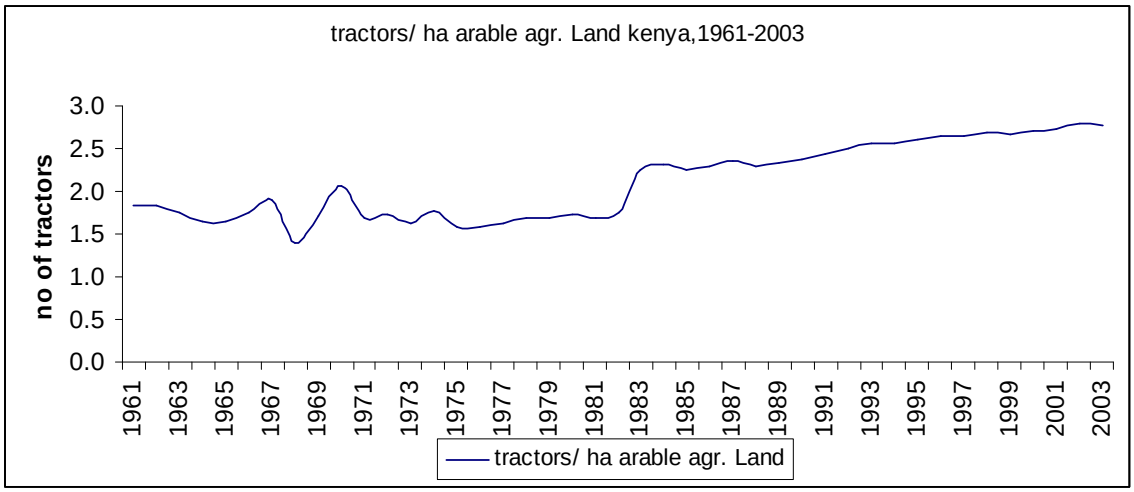


Source: FAOSTAT

1.2. Mechanization

The use of tractor power in farm operations has increased as indicated by the number of tractors in use per unit area. Although there was less fluctuation in tractor utilization in the pre-liberalization era, a drastic increase in tractor use occurred in early 1980s and later stabilized (Figure 62). Whereas tractor use has been on an upward trend, it is notable that the liberalization era has not registered significant increase.

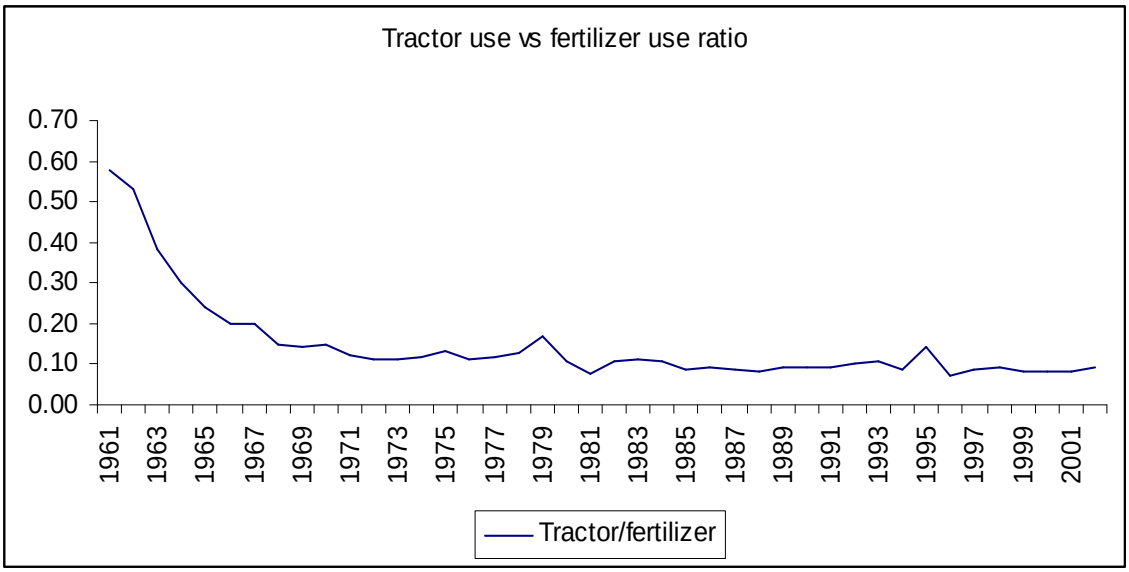
Figure 62: Tractor Use Intensity



Source FAOSTAT

Tractor-fertilizer use ratio indicates declining tractor use against the fertilizers (Figure 63). This may indicate the increasing substitutability of fertilizer for tractors.

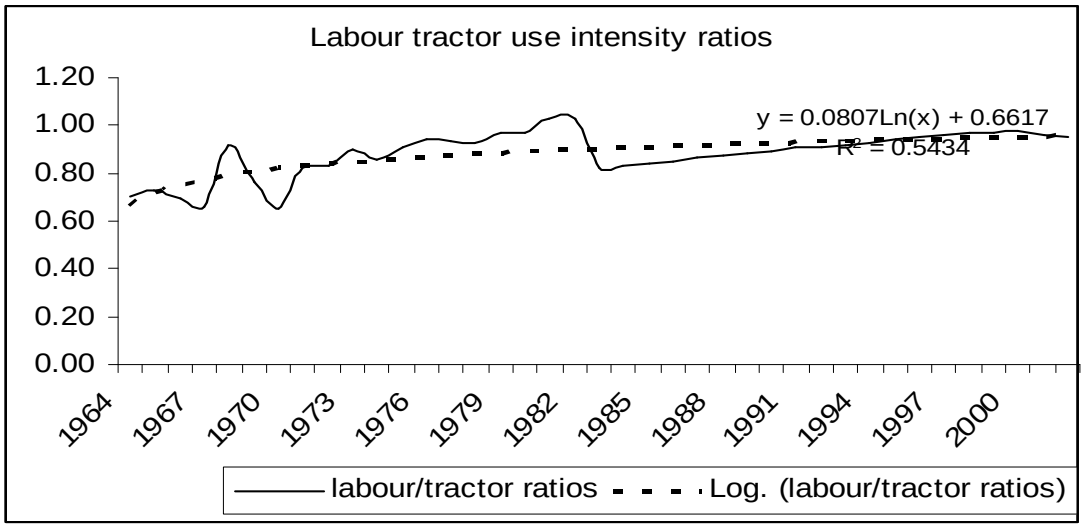
Figure 63: Tractor-Fertilizer Use Intensity Ratio



Source FAOSTAT

The ratio of labour to tractor use indicates that agricultural evolution has adopted more manual labour than mechanized labour (Figure 64). The reduced use of mechanized labour can be attributed to the increased cost of agricultural machines as compared to manual labour that is supported by increasing population.

Figure 64: Labour-Tractor Use Intensity Ratio



Source FAOSTAT

1.3. Evolution of Farm Sizes

Land subdivision was adopted after independence to ensure equitable distribution and access to productive land. Rapid population increase continued to exert pressure on land prompting continued subdivision and less economies of scale. According to Pearson teal, 1995, immediately after independence families resettled, were allocated between 33 and 23 acres of high density plots depending on the agricultural production capacity of the household. To date small-holder land holding per capita is estimated to be 0.4 ha (Gamba and Mghenyi, 2004). The decline in land holding per capita links favourably with the decline in the share of income from agricultural related enterprises (Table 10).

Analysis from Tegemeo panel data indicates a declining trend of land holding sizes across all agro-regional zones (Table 35).

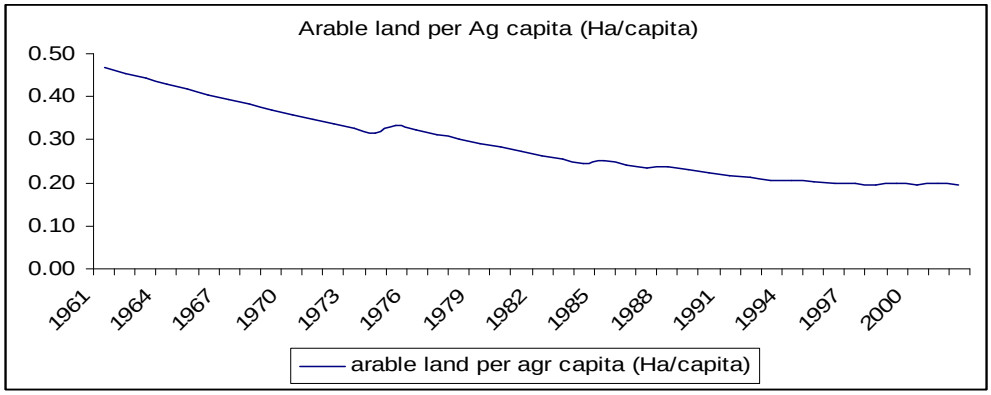
Table 35: Mean Land Owned and Rented in Acres

	Mean land owned and rented in acres					
	1997		**2000**		**2004**	
Agro-regional zones	**Owned**	**Rented**	**Owned**	**Rented**	**Owned**	**Rented**
Coastal Lowlands	5.3	1.5	4.1	1.2	4.0	1.0
Eastern Lowlands	6.5	1.4	5.1	1.0	4.4	1.9
Western Lowlands	3.8	1.4	2.8	1.3	2.8	2.0
Western Transitional	5.8	2.8	4.7	2.0	4.0	2.0
High Potential Maize Zone	11.0	5.7	6.4	6.6	4.6	2.5
Western Highlands	2.1	1.0	2.3	0.6	1.8	1.0
Central Highlands	2.9	0.8	2.6	0.6	2.2	2.1
Marginal Rain Shadow	6.1	1.6	1.6	1.4	1.8	1.3

Source: Tegemeo Institute/MSU Agricultural Monitoring and Policy Analysis household Surveys, 1996/97, 1999/00 and 2003/04.

Agricultural land per capita has also declined overtime (Figure 65). In 1956 the agriculture land available was estimated to be 2.81 hectares per capita and has since decreased to 0.47 hectares in 2003 while arable land estimated at 0.36 hectares per capita in 1965 decreased to 0.20 hectares in 2003 (FAOSTAT). This is an estimated decline of 68% and 60% for agricultural and arable lands availability respectively.

Figure 65: Arable Land per Capita



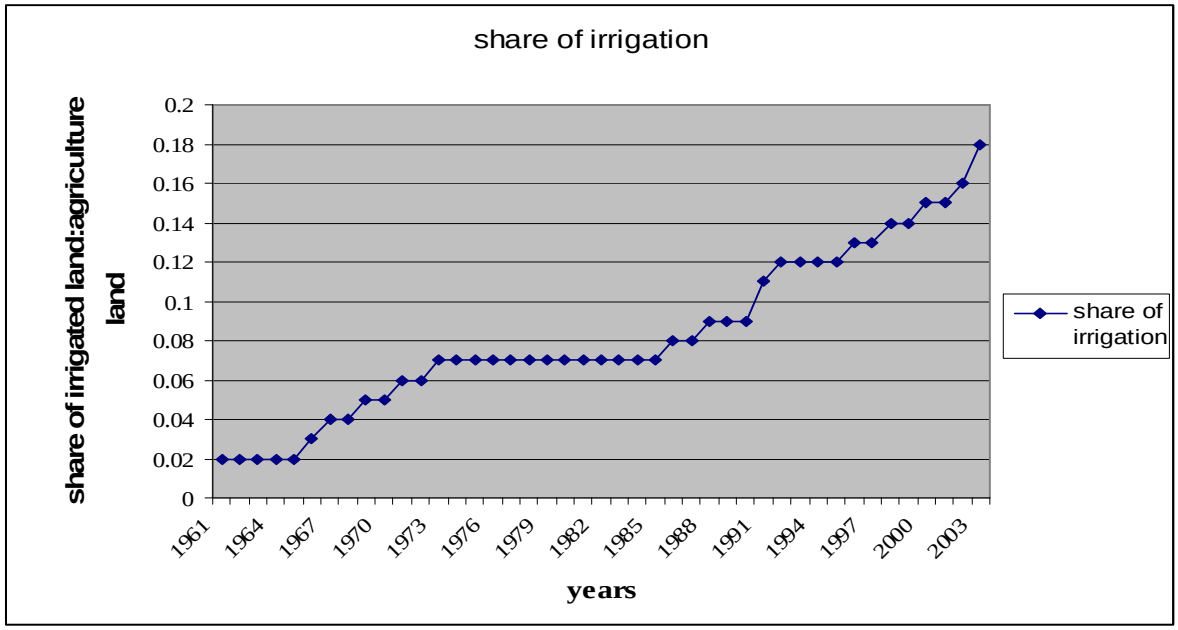
Source: FAOSTAT

Land markets are disappearing as shown by declining land being offered for rent. While this scenario is dictated by the perceived returns from production at a given season, rental values can be increased if land productivity is increased. Adoption of improved production technologies may prompt release of land for renting. However this is yet to be realised given the low proportions of rented land.

1.4. Share of Irrigation

Kenya has scarcely embarked on irrigation as a means of filling the gap between food demand and production. Currently only 1.8 % of agricultural land is under irrigation (FAOSTAT). However irrigation seems to have increased from late 1980s. While prior to 1989 the government irrigation policy favoured large-scale irrigation schemes, policy shifts occurred as from late 1990s towards supporting small-holder irrigation. This policy shift was driven by the urge to generate employment, increase incomes and assure local food security (GoK, NDP, 1989-1993). This was partly the reason for the growth in area under irrigation from 1985 (Figure 66).

Figure 66: Share of Irrigation



Source FAOSTAT

2. The Processes of Change and their Explanatory Factors

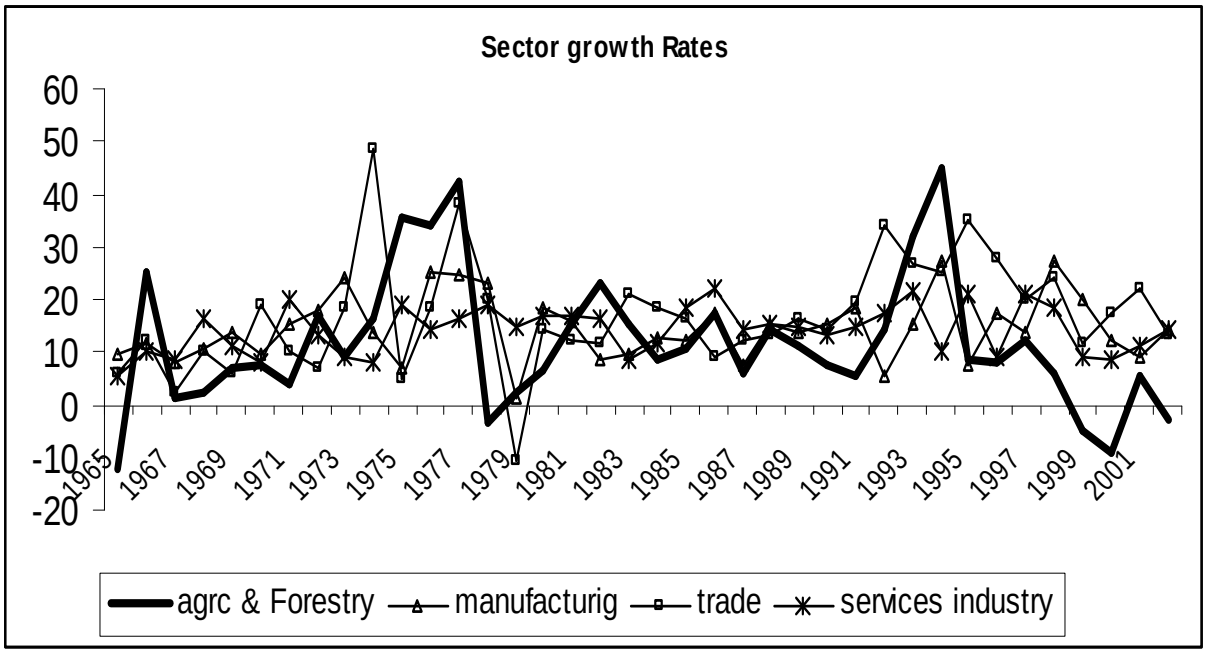
2.1. Evolution of the Rural Area

The rural areas have been grappling with underdevelopment, increasing inequity and resource depletion that have led to increasing poverty levels. The rural population was estimated to have reached 23.3 million in 1999 according to the 1999 population census (CBS, 2002).With declining arable land, increasing population and dwindling agricultural income rural poverty levels were estimated to have risen to 59.6% compared to the urban area 51.48% in 2000 (Gitu, 2004). Rural households derive their livelihoods from agriculture and related activities. The post-colonial government perpetuated the segmentation by developing the potential areas under the assumption that

people will migrate from low to the high potential areas. While this segmented the rural development the population increased rapidly leading to decline and subdivision of farm land to uneconomical units. Agricultural reforms that took place as from 1980 under the initiation of the SAPs partly favoured some economic sectors. These reforms further created economic segmentation besides the regional segmentation. The rural areas have been characterized by varying phenomenon- pestilence, floods and dry spells. The 1984 drought, the El Nino 1998/99 and La Nina (post El Nino) are the most significant natural calamities to have occurred. These have adversely affected the agricultural sector leading to declining yields. Other periods have been characterized by average rainfall with generally poor distribution. Environmental degradation is also high as indicated by declining per capita arable and agricultural land. Observed shifts in income sources, migration within the country and increasing unemployment are outcomes of this segmentation. Agriculture presents a major opportunity to counter these problems and improve the welfare of rural households.

2.2. Change in the Economic and Institutional Environment

Figure 67: Sector Growth Rates, 1987-2001



Source: CBS Economic Surveys 1990-2002

Economic and institutional reforms entailed in the liberalization package have led to redefinition and restructuring of the roles and responsibilities of various economic agents. As economic development continued economic dominance changed also. The once dominant agricultural sector continued to decline in contribution to the country's GDP relative to other sectors. Although, agriculture is still dominant in economic contribution (Figure 1: Nominal Sector contribution, 2005), its growth rate has been surpassed by manufacturing, trade and services sectors (Figure 67) the trend evident from mid 1990s, indicate the shift from an agricultural oriented economy towards a manufacturing sector-led economy

To conform to the requirements of liberalization and for efficient implementation of the reforms, various institutions and parastatals underwent restructuring, merging, privatization and commercialization. KTDA was renamed, commercialized and its roles and organizational structure redefined to enhance its competitiveness. The rural areas have witnessed formation of more groups especially marketing associations to exploit economies of scale and exert market influence. This has

been evident in the horticulture and dairy sub-sectors where more groups have been formed. Coffee cooperatives increased from 129 to 583 and dairy from 139 to 248 between 1985 and 2005 (CBS, Issues 1989-2006). The players in dairy (Table 20), coffee (Table 21) and tea (Table 22) sub-sectors increased. Liberalization of the dairy industry has led to the emergence of a strong informal sector but raised concerns on food safety, health, quality and standards. This requires strengthened and effective legal structure and regulatory agencies for enforcement. The new entrants must also be versed with food safety standards. While marketing networks have been improved through reforms, the resulting competition between the informal and formal sector has however resulted in higher consumer prices and declining producer prices. This widening gap signifies the increasing erosion of consumer and producer surpluses in the market. The economies of scale that would have accrued due to deregulation of prices have been absorbed by raising milk processing costs due to inefficient processing transport costs. The dairy, tea and horticultural are gradually integrating and rapidly concentrating. This is happening in the wake of weak and ineffective business ethics regulatory environment. This is manifested in the lack of or low utilization of contractual arrangements in the major commodity sub-sectors. Attempts have been made in the case of sugar and dairy through enacting legislation but these are hardly implemented exposing the concerned parties to major losses in cases of default. Sub-sector arbitration mechanisms are largely moribund. The monopolies and prices commission despite restructuring is still incapable of tracking and reigning in errant anti-competitive economic behaviour.

Rural financial institutions have also evolved. Formal and informal organizations have come into place to avail working capital to the rural poor. However, financial services and products available to the rural households are still poor, inappropriate and inaccessible. Input supply structures that were initially government dominated have embraced reforms allowing more suppliers and stockists of inputs. This has enhanced distribution and access therefore enhancing productivity. However rapid increases in input prices hinders utilization and limits increased agricultural production.

2.3. Technical and Economic Change

There has been increased adoption of intensive production technologies such as increased use of fertilizers in all agro-regions and high yielding varieties owing to increasing land pressure and declining arable land. These changes have taken place in the milieu of liberalization process. Agricultural productivity, in this section defined as output per unit area has been influenced by these changes which also result into reallocation of productive resources. The effects of these changes upon various crops enterprises are discussed below.

2.3.1. Tea

The national tea land productivity has been on increase since independence. It increased from an annual average of 572.38 Kgs/Ha in 1974-80, 1193.61 Kgs/ha in 1981-90 and 2058.77Kgs/Ha and 2290.28 Kgs/Ha in 1991-2000 and 2001-05 respectively. The small-holder tea grew from 1664.41Kgs/Ha to 2103.6 Kgs/Ha in 1991-2000 and 2001-05 respectively while the estate tea dropped from 3017.85 Kgs/Ha to 2663.19 Kgs/Ha in the same period (Table 36). This improved land productivity was due to the growth rate in output that surpassed area growth rate (Figure 36). The increase in output may be attributed to the adoption of high yielding varieties coupled with good agricultural practices and the services rendered by KTDA to small-holder farmers. Liberalization seems to have had little impact on national tea productivity, but seems to have had negative influence on the estates productivity. Favourable international prices under liberalization have been the impetus for improved tea productivity. However, the growth in productivity has been more favourable to the small-holder tea sector than the estate tea. Recently estate tea productivity has had a declining trend. The productivity gap between estates and smallholders has continued to narrow down in the post-liberalization period.

Table 36: Tea Productivity in Kenya

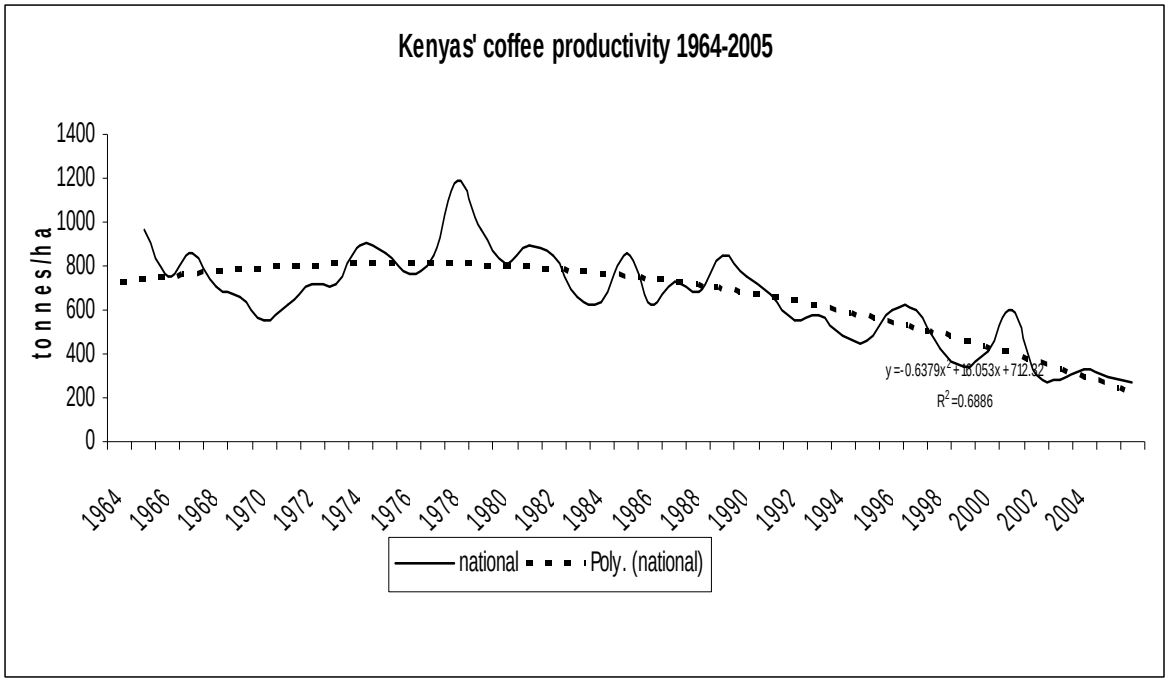
	Pre-liberalization 1970/80	Transition 1981/92	Liberalization 1993/02	Post-liberalization 20031/05
National productivity (Kgs/Ha)	572.38	1193.61	2058.77	2290.28
Small-holder productivity (Kgs/Ha)	-	-	1664.41	2103.6
Estate productivity (Kgs/Ha)	-	-	3017.85	2663.19

Source: CBS Economic Surveys. (Various Issues)

2.3.2. Coffee

Whereas the area under coffee bushes and output may have increased over some periods, coffee yields/productivity (tones/Ha) has been on decline since the reform era started in early 1980s As indicated in Figure 34, coffee production hitherto rising started to decline in late 1990s to date. Productivity had already started declining long after the coffee boom 1976/78 (Figure 68).

Figure 68: Kenya's Coffee Productivity (tonnes per ha)



Source: Coffee Research Foundation (CRF)

Despite the large area and output by co-operatives, smallholder productivity remains far below that of estates. High estate productivity can be attributed to intensive use of high yielding coffee varieties, fertilizer, timely harvesting, sufficient labour, optimum husbandry practices and access to international markets.

Coffee productivity by cooperatives has continued to decline while estates productivity increased from independence but has been declining since reforms in early 1990s (Table 37). Smallholder productivity decline is attributed to declining world coffee prices, high input costs, erratic weather, diseases and pests upsurge, mismanagement of the coffee co-operative societies, poor marketing and disharmony between the current policy and the legal framework (GoK, 2002-2008).

This decline in coffee land productivity translates into reduced returns and has resulted in further differentiation of coffee producers. Varying human capital and resource endowment has differentiated coffee producers into those able to transition from coffee growing to new enterprises such as horticulture, floriculture and medicinal plants against those unable to transition. In this case there are challenges emerging in terms of intervention measures necessitated by the increasing marginalisation of those producers unable to transition from enterprises

Table 37: Productivity of Coffee by Producer

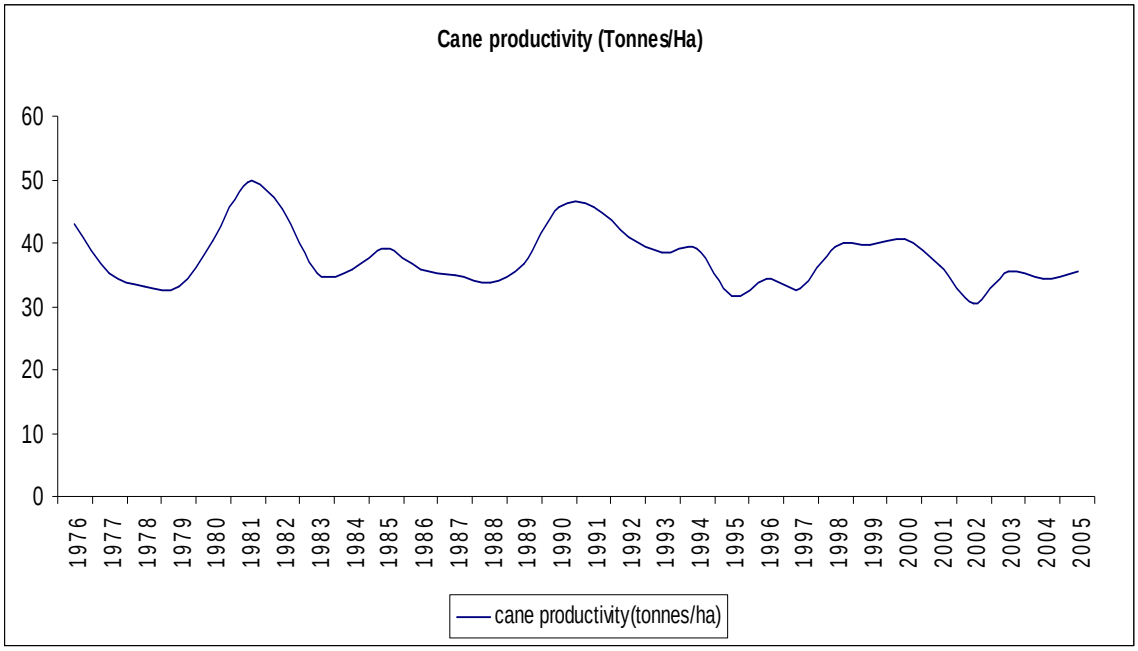
	Pre-liberalization 1970/80	**Transition 1981/92**	**Liberalization 1993/02**	**Post-liberalization 20031/05**
Coffee productivity growth rates				
Co-operatives growth rates	-1.08	-2.79	0.32	-3.2
Estates growth rates	6.79	2.18	-4.35	-0.18

Source: Coffee Research Foundation (CRF)

2.3.3. Sugar

Despite the growth in area under cane and cane output (Table 16), cane productivity (per unit area) has been fluctuating and tending to decline (Figure 69). Fluctuations in area of cane planted and the area of cane harvested (Figure 39) contribute to this slight decine in productivity. This can be attributed to inefficiencies in cane harvesting, poor cane varieties, lack of inputs, mismanagement of public run sugar factories and changing weather regimes.

Figure 69: Cane Productivity Trend



Source: CBS. Economic Surveys. (Various Issues)

Sugarcane farmers are undergoing differentiation in process in view of competitive forces emanating from COMESA regional trading block's common external tariff. A moratorium won by Kenya comes to an end in 2008. Further, cane producers are undergoing differentiation through varying production costs that designate competitive versus uncompetitive farmers. Cane producers with uncompetitive production costs may be forced to exit the industry and engage in alternative enterprises that will

require development of human capital. The extent of this adjustment can only be ascertained in a subsequent sub-sector study.

2.3.4. Horticulture

Horticulture has experienced a phenomenal growth in area and production. Favoured by increasing international prices, the industry has had a record performance in exports. The drastic decline in exports in early 1990s (Figure 37) was attributed to the decline in canned beans export (Ayieko and Gamba, 2005). While the industry is composed of both large and small scale farmers, it is dominated by large-scale producers who have the capacity to comply with the stringent standards and quality requirements. The investment burden imposed by international requirements as described in EUREPGAP favour large-scale over small-scale farmers. The cost of compliance is prohibitive to individual small-scale producers. It is hypothesized that these restrictions have led to the exit of small scale farmers. It is anticipated that further restrictions in the markets will lead to exit of more small-scale producers, reduced earnings and decline in employment in the absence of a concerted effort and mechanism to accommodate and enhance their capacity to meet the requirements. This requires institution of both technical and financial services programme to affected smallholders.

2.3.5. Milk

The quantity of milk processed has increased rapidly owing to reforms in the dairy sector. In 2005 an estimated 10% was processed with the remaining percentage marketed through the informal sector (Table 38). The informal sector continues to dominate the marketing chain in dairy industry. The challenges posed by the segmentation of the dairy industry into informal and formal channels continue to pre-occupy the policy, legal and regulatory frameworks of the dairy industry. The proportion of unprocessed milk in the market is such that its marginalisation ids not tenable and requires efforts to incorporate it into the forma channel. Standards and quality issues isolate and marginalise the most vulnerable portion of milk producers.

Table 38: Processed and Unprocessed Milk

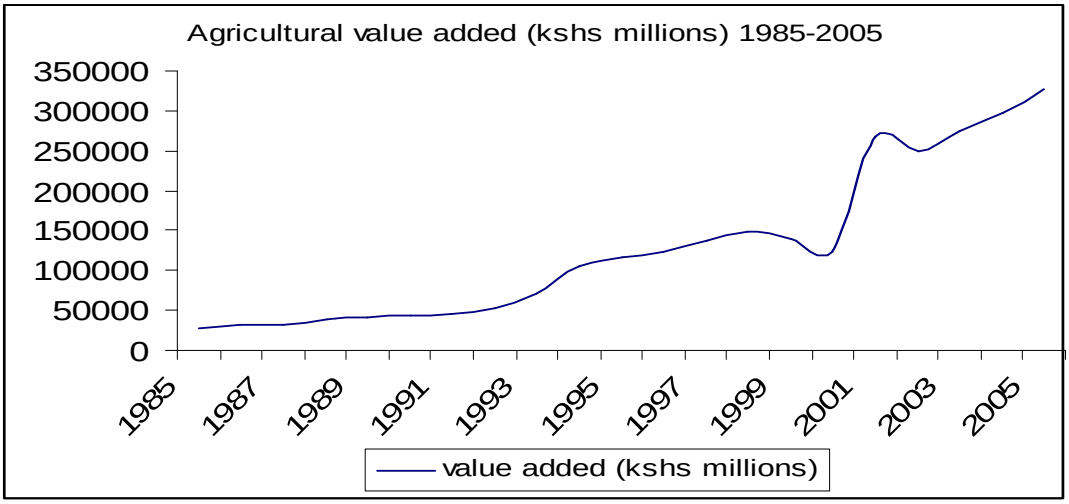
Year	Milk production (Mn litres)	Milk processed (Mn litres)	% of processed
2001	3,051	152	5
2002	3,129	144	5
2003	3,207	197	6
2004	3,323	274	8
2005	3,455	340	10
Total	16,165	1,107	7

Source KDB

2.3.6. Agricultural Value Addition

Agricultural value addition has increased over the period (Figure 70). This is an indicator of growing processing and manufacturing infrastructure in the agro-industrial sector in the country. While the domestic market favours value addition, the international markets have retarded this development especially through tariff escalation phenomenon that imposes corresponding increases in tariffs associated with increasing levels of value addition. This process further differentiates the producers in accordance with the capacity to engage in value addition services

Figure 70: Agricultural Value Added

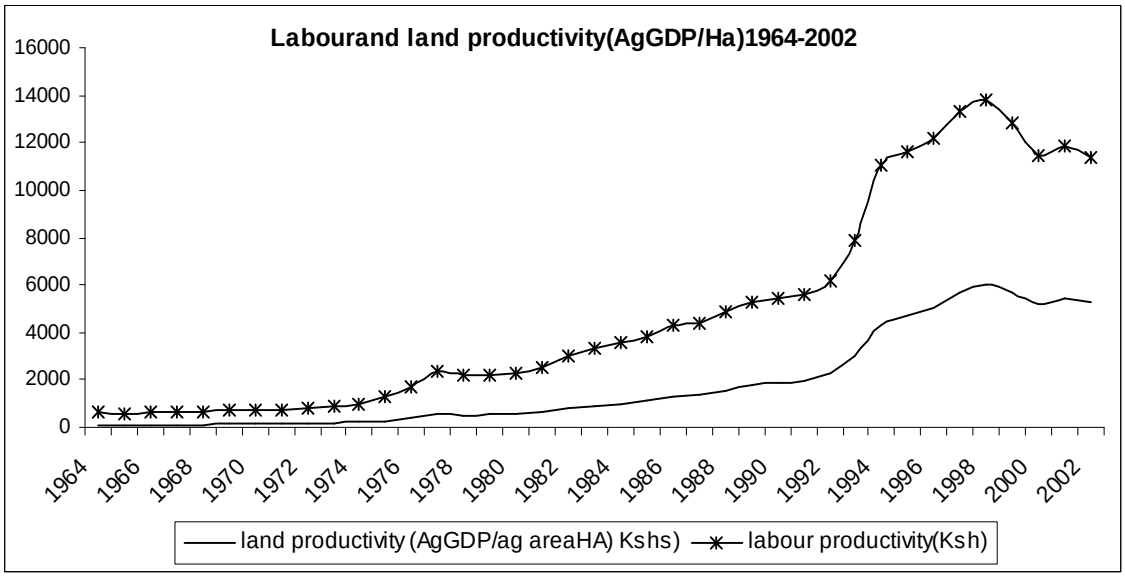


Source :CBS Economic Surveys(Various Issues)

2.3.7. Land and Labour Productivity

Land and labour productivity has increased over the period since 1964 reaching a maximum in the late 1990s (Figure 71). Land productivity increase is owed to the increased adoption of intensive production technologies spurring the output in agriculture while the labour productivity trend is attributable to growth in agricultural GDP, exchange rates and the decline in agricultural labour force. It is hypothesised that agricultural sector will only be sustainable if land and labour productivity maintains an upward trend in the presence of competitive and marketing costs.

Figure 71: Labour and Land Productivity (AgGDP/ Ag Arable Land (Ha))



Source: CBS Economic surveys (various Issues) and FAOSTAT

3. Structural Consequences

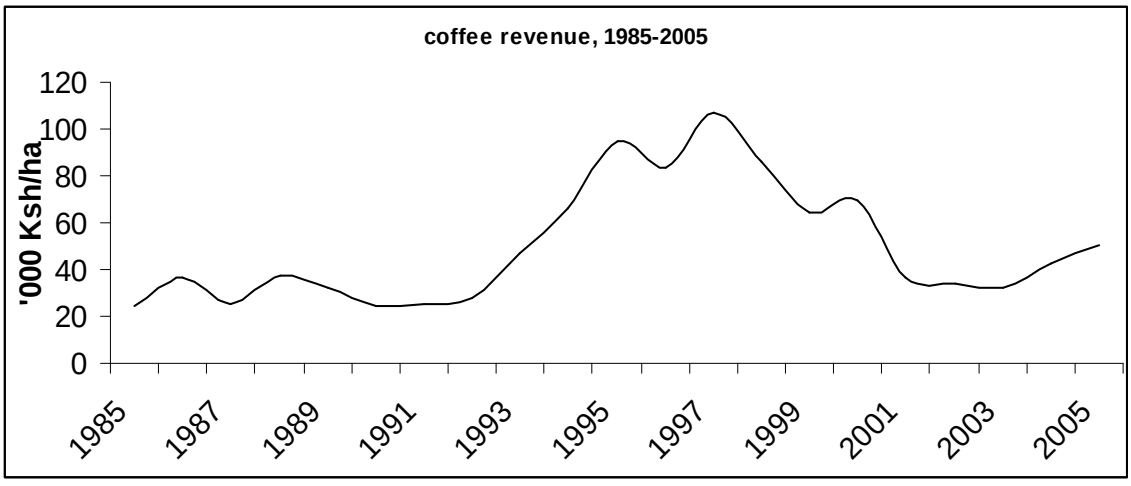
3.1. Economic Performance of Farming Activities

The agricultural sector and rural economies have certainly changed over the last decade. However, it is futile to attempt to ascribe these changes entirely to liberalization. Empirical analysis is critical to establish causal relationships. A brief review of performance of specific commodity enterprises can nonetheless provide an indication of the effects transmitted through liberalization.

3.1.1. Coffee

Producer prices were used estimate coffee revenue per hectare. The increase in the coffee revenue as from the early 1990s is attributed to increases in producer prices and the depreciation of Kenya shilling against the US$ (Figure 72). The world coffee price decline was not immediately passed on to the producer as the exchange rates absorbed the shock. Farmers have responded to poor incentives by either uprooting or abandoning their coffee and using the land as pasture or for planting beans, vegetables, napier grass in addition farmers have turned their efforts towards off-farm opportunities and other non-farm related income generating activities. The extent of these changes needs further scrutiny particularly with respect to contribution to poverty. Also a shift in labour force to alternatives including migration to urban areas has occurred and warrants an empirical assessment.

Figure 72: Coffee Revenue

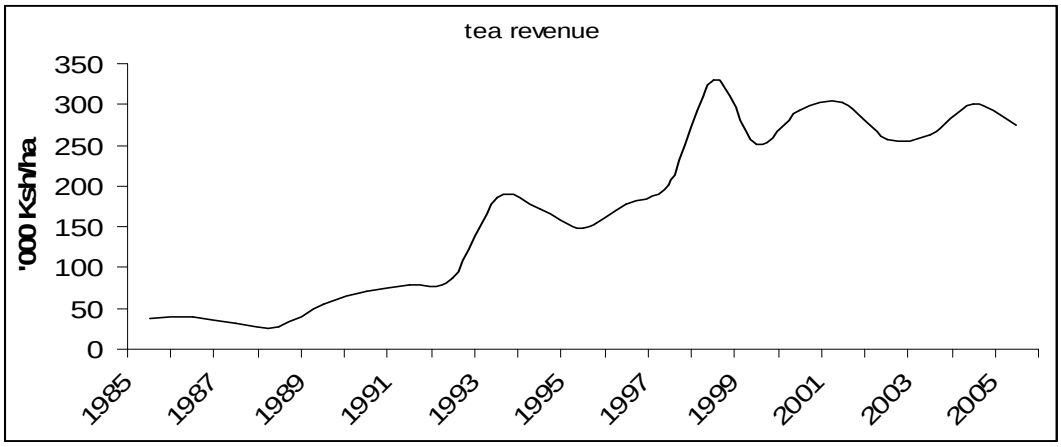


Source: CRF

3.1.2. Tea

Tea revenue has been on the increase in the recent past reaching 301,251 Ksh per hectare per year in 2003 from 10,825 Kshs/ha/year in 1981 (Figure 73). The revenue reached a peak of 329720ksh/ha/yr in late 1999 coinciding with the sectors liberalization. However, the revenue trend declined and has been fluctuating slightly below the peak. The entry of illegal green tea hawkers collaborating with some tea estates has led to decline in the prices paid to tea farmers and poses a threat to quality.

Figure 73: Tea Revenue

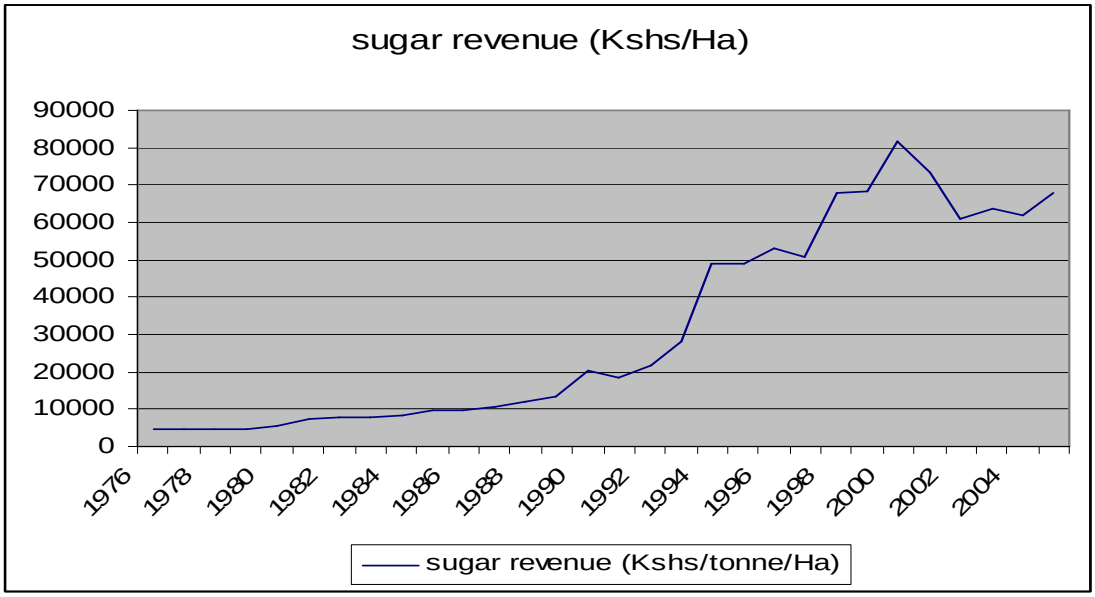


Source: CBS Economic Surveys. (Various Issues)

3.1.3. Sugar

Despite declining sugarcane productivity, farmers' revenue from sale of cane has been increasing (Figure 74). This is attributed to the increasing producer price which seems to have counteracted the effects of the decline in productivity.

Figure 74: Sugar Revenue



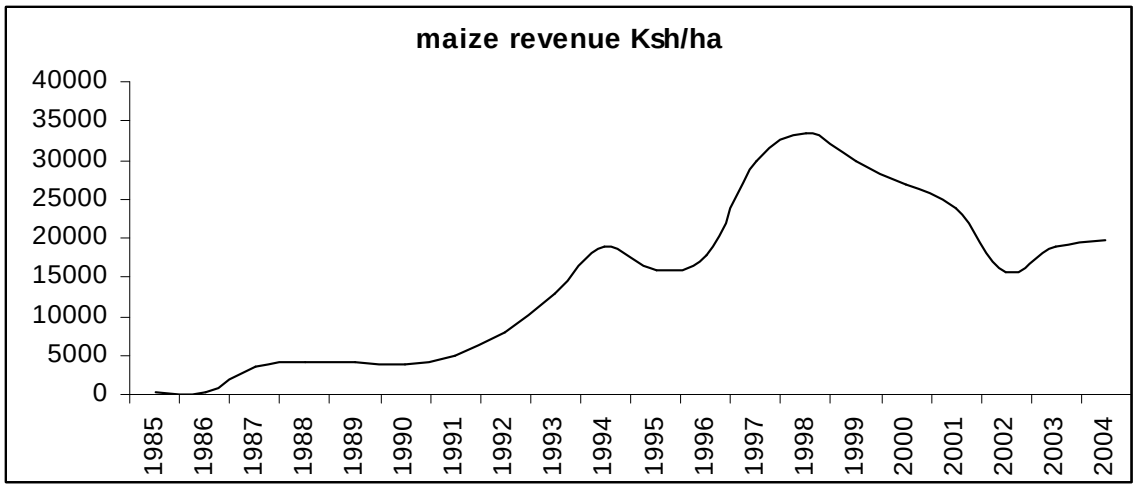
Source: CBS Economic Surveys. (Various Issues)

The decline in sugar revenue in 2001-2003 coincided with the opening up of the COMESA regional cooperation which may have led to cheaper sugar imports. Given the opening of the domestic markets to imports from COMESA and the favoured treatments, the high producer prices are unsustainable. The country has been offered a moratorium up to 2008. The impending lapse of the moratorium implies the sugar industry has to reform to remain competitive. There is an increasing likelihood that inefficient cane producers and millers will exit the sugar industry. The precise consequence of the lapse of the moratorium on the rural livelihoods is an issue that needs an empirical assessment.

3.1.4. Maize

Maize revenue increased since liberalization of the industry in early 1990s but has been declining since late 1990s (Figure 75). This is perhaps due to dry spells that have reduced productivity.

Figure 75: Maize Revenue



Source: CBS Economic Surveys. (Various Issues) and FAOSTAT

The decline in returns resulting from these major crops may result to further decline in production. Productive resources such as labour may shift from these enterprises to alternative high returning enterprises. Decline in maize revenue may result to reduced production and consequently increased food insecurity.

3.2. Economic Performance of Agricultural Households

With the evolution of reforms, rural households have over the period shifted from agricultural activities towards non-farm income sources such as businesses and salaried labour. While income from crops declined by 40% and livestock by 57%, income from businesses grew by 20% with salaries and remittances growing almost twofold4 (Table 39). Table 10 indicated the declining trend of agricultural performance against the off-farm income. The results which are regional can be compared with the decline in returns accruing from major regional crop productions. For instance in the high potential maize zone which is mainly the country's maize production area, overall crop contribution to total household income declined from 52% in 1997 to 33% in 2004 while off-farm activity contribution increased from 18% to 45% in 1997 to 2004. In the central where coffee dominated the crop contribution declined from 47% in 1997 to 35% in 2004. Appendix II further presents the distribution of net value of income sources by region. Remittances have increased in high maize potential zones, central highlands and the western highland regions between 2000 and 2004 as shown in Appendix II. Whilst these are mainly crop dominated areas, the increase in remittances signifies dependence from outside the regions owing to decline in crop returns. Despite increase income from off-farm activities, the ensuing loss from crops and livestock enterprises could not be offset. Therefore, this resulted to household economic performance decline by 6% between 1997 and 2004.

Table 39 Rural Household Gross Income in Ksh. by Source, 2000 and 2004

Source of Income	1997	2000	2004	% change
Crop	90958	62,828	54,383	-0.40
Livestock	65140	24,002	28,028	-0.57
Business	51386	24,991	61,724	0.20
Salary and domestic Remittance	24899	27,774	74,292	1.98
Total Income	232383	139595	218427	-0.06

Source: Tegemeo panel data 1997, 2000 and 2004

Conclusion

Rural segmentation started during the settler period. This was perpetuated by post-independence governments that emphasised on the development of high potential regions at the expense of the low potential areas. Economic reforms further segmented even the high potential areas triggering within region segmentation and producer differentiation. Today, segmentation can be viewed in terms of agricultural production that is dictated by climatic conditions and public infrastructure investment. Economic reforms had varied effects in different regions depending on the type of agricultural production of the area. The effects flowed from the macro to micro level and to the rural households influencing livelihoods, and either forcing a change in livelihood strategy or a diversification of income source. Changes in strategies are indicated by increased use of fertilizers, increased irrigation area, decline in mean land owning sizes and changes in the economic performance of the households.

While livelihood strategies have changed, household farm size and arable land per capita has declined as the gap between the high and low potential regions fertilizer use widened. At the same time non-agricultural enterprises growth has surpassed agricultural growth fostering a shift from agriculture to other non-farm sources of income. Land and labour productivity are currently on a declining trend prompting decline in agricultural revenue. Although, there has been a shift from agricultural income sources to non-agricultural income sources, the scenario has culminated into decline in overall income levels of the households. Concerns are emerging on continuation of this trend especially increased declining land sizes prompting increased fertilizer use, which may result to environmental degradation. This coupled with decline in income will put the households in a poverty-environment spiral complex. The phenomenon needs to be immediately addressed through review of the existing structures and elements that may be perpetuating the trend since it threatens to bring more anguish and poverty to the affected households in the near future.

PART IV -

IDENTIFIABLE RISKS AND IMPASSES, POSSIBILITIES FOR AFDAPTATION AND RESTRUCTURING PROCESSES

CHAPTER 7 - IDENTIFIED POSSIBLE RISKS AND IMPASSES AND EXISTING EXIT OPTIONS

1. Vulnerable Social Groups and Places

The liberalization process was intended to restructure the economy and improve social welfare. However, liberalization has manifested itself in varied forms in the different sub-sectors and regions. As observed some sub-sectors and regions have thrived while others have continued to experience increased marginalization. This varied impact of liberalization makes it critical to identify vulnerable groups and regions to further develop social safety nets and complementary policies. Vulnerable groups are here by defined as those losing their economic performance over time especially due to decline in agricultural performance. It is estimated that 80% of Kenya's population live in rural areas and derive their livelihood largely from agriculture through crop and livestock production, forestry and exploitation of other natural resources (Strategic Plan, 2006-2010). While previously the vulnerable in Kenya have been defined as those in marginal and non-productive lands, overtime even those in the high potential regions have become vulnerable especially due to the market driven economy. According to Tegemeo data different agro-regional zones experienced varying economic losses in the period 1997 to 2004. Table 40 indicates how various agro-ecological zones have lost in income gain between 1997 and 2004.

Table 40: Regional Vulnerability as Indicated by Change in Overall Net Income

	Crops	**Livestock**	**Off-farm**	**Net total**
Agro-regional Zones	**1997-04**	**1997-04**	**1997-04**	**1997-04**
Coastal Lowlands	1.13	-0.47	0.16	0.16
Eastern Lowlands	-0.26	-0.64	1.44	0.26
Western Lowlands	-0.09	-0.68	0.50	-0.10
Western Transitional	-0.34	-0.57	0.99	-0.14
High Potential maize zone	-0.53	-0.47	0.86	-0.26
Western Highlands	-0.03	-0.60	0.91	-0.10
Central Highlands	-0.47	-0.69	0.81	-0.29
Marginal Rain Shadow	0.42	-0.61	1.78	0.31
Total	**-0.42**	**-0.58**	**0.84**	**-0.17**

Source: Tegemeo panel data 1997, 2000 and 2004

In this case negative changes in overall net income were identified in western lowlands, western transitional, high potential maize zone, western highlands and central highlands while positive income gains were witnessed in coastal lowlands, eastern lowlands and the marginal rain shadow. However, the overall total income net income declined for the whole country. The vulnerable are located in these regions experiencing negative gains in their incomes since this translates to decline in welfare of the people.

2. Evolution of Rural Poverty

The rural population has recorded increasing poverty levels. The Welfare Monitoring Surveys (WMS) established to monitor the incidence of poverty under SAPs were conducted in 1992, 1994 and 1997 using the National Sample Survey and Evaluation Programme (NASSEP) III Frame. During these years, rural poverty increased from 46.33%, 46.75% and 52.93% while urban poverty also rose from 29.29%, 28.95% and 49.20% respectively (GoK, 2000a). The extent of poverty incidence is dependent on specific factors of the survey and administration, geographical coverage and survey

instrument among others (Gamba and Mghenyi, 2004). While there is no conclusive indicator of poverty incidence, a standard and widely used indicator is based on the purchasing power parity, exchange rates and country poverty lines. These form the basis of the 1$/day poverty line currently under use. However, this indicator is vulnerable to exchange rates since consumption is expected to respond to the fluctuations in inflation due to currency depreciation (Gamba and Mghenyi, 2004).

The poor in rural areas are characterized by low education levels, low asset ownership, reduced fertilizer use, reduced high yielding variety seeds usage, poor access to agricultural credit and low crop land productivity. The rural poor have limited access to land which also constrain productive capacity. Rural poverty is aggravated by the land tenure system that does not provide an incentive to investment. Further property rights play an important role in making investment and land markets more efficient. while property rights through land title deeds provides legal access to land, it also guarantees security to investment and claim and acts as a form of collateral to access credit facilities which are deemed important for rural development.

Variations in poverty levels as dictated by income are related to regional ecological differences that determine comparative agricultural productivity. According to the findings by Gamba and Mghenyi, 2004, geographical categorical variables increased in their explanatory power as administrative units got smaller and smaller to the village level. Table 41 presents the distribution of poverty categories within agro-ecological zones.

Table 41: Distribution of Poverty Categories within Agro-ecological Zones

Zone	Non-poor	Transitory poor	Chronic poor
Coastal lowlands	15.2	50.6	34.2
Eastern lowlands	36.5	35.3	28.2
Western lowlands	9.1	22.3	68.6
Western lowlands	22.4	41.8	35.8
High potential maize zone	39.2	24.4	36.4
Western highlands	10.8	30.9	58.3
Central highlands	64.3	23.6	12.0
Marginal rain shadow	22.9	43.8	33.3
Country total	33.1	30.0	36.9

Source: Gamba and Mghenyi, 2004

3. Characteristics of Vulnerable Farms

Defining poverty under the standard 1$ dollar per day/per capita expenditure allows vulnerability to poverty to be defined as the capability/capacity of the households to deal with poverty related risks. Vulnerability increases as economic opportunities and access to basic social services dwindle. The most vulnerable are the chronic poor and transitory poor rural households. Rural households live in environments characterized by physical, economic and institutional features which together determine risk level. They are also confronted with natural calamities such as drought and floods, macro-economic, external and internal shocks, as well as theft and insecurity (Christiaensen, 2001). Thus the poor and vulnerable are characterized by low education levels, low incomes, poor access to social amenities and infrastructure, poor access to factors of production, limited access to credit facilities and poor climatic conditions. Vulnerability varies from region to region as defined by the agro-ecological zones of the country. The level of vulnerability in the ASAL regions is higher than in the non-ASAL lands.

Land tenure in pastoral areas is mainly communal and inhibits the process of development due to insecurity. The tenure also encourages land degradation and thus reduces its productivity thereby limiting income growth.

4. Existing exit options: A Reality of Limited Options

Marginalized communities are faced with limited exit options when their source of livelihood, agriculture becomes uncertain. Currently the informal sector in the country is providing the exit option for agricultural communities in some regions. This sector has rapidly grown since early 1990s. Table 39 present the trends of employment in informal sector. Informal wholesale and retail trade, informal manufacturing and communication and social services have dominated this sector

Table 42 Informal sector number of people engaged by activity '000

Year	Manufac-turing	Construct-ion	W / sale R / trade	Trans-port / commun	Communi-cation, social & personal serv.	Urban	Rural	Total
1992	342.7	17.9	777	19	80.7	817.1	420.3	1237.3
1993	418.3	20.6	909.9	23.6	94.1	960.8	505.7	1466.5
1994	492	26	1126	28.9	118.9	1165	627	1791.8
1995	616.9	31.6	1405.5	35.1	151.6	1470	770.2	2240.7
1996	710.9	36	1675.5	41.5	179.9	1692	951.8	2643.8
1997	803.1	40.7	1893	46.8	203.3	1939.7	1047.2	2986.9
1998	897.6	51.2	2121.1	52.8	230.8	2173.7	1179.8	3353.5
1999	992.1	58.9	2366.5	58.8	262.5	2397.7	1341.1	3738.8
2000	1075.1	62.3	2652.4	66.4	294.7	2673.2	1477.7	4150.9
2001	1039.7	140.9	2717.1	136.9	422.3	1596.3	3072.4	4456.9
2002	1121	150.2	2986.4	150.8	467.9	1730.4	3377.9	4876.3
2003	1199.3	158.9	3252.6	165	514.5	1889.1	3657.3	5290.3
2004	1281	168.8	3529.1	181.2	560.5	2030.9	3961.9	5720.6
2005	1386.1	183.9	3760.9	191.3	593.8	2171.4	4235.8	6116.0

Source: CBS Economic Surveys (various Issues)

The growth of these enterprises within the informal sector can be attributed to liberalization. Liberalization of the telecommunications sector has brought about rapid changes in access to communication prompting timely access to information especially market information that enhance bargaining power. Money transfer services in the rural areas have also enhanced access to working capital and coincided with the emergence of non-agricultural opportunities provided by liberalization. For instance, financial service associations have increased from 34 in 1999 to 70 in2005. There are also more than 330 money transfer outlets distributed all over the country (Kibaara, 2006). The marginalized have also migrated within the country either to the urban or other better endowed areas in search of employment or have migrated to in search of economic opportunities.

The rural informal sector has since its inception in 1992 grown to accommodate a significant labour force (4.2 million) while the urban employed 2.1 million people. The total population employed nationally under the informal sector is estimated as above 6 million. Whole sale and retail trade accounted for 61.7% while manufacturing accounted for 22.4% in 2004 (CBS, 2006). Migration of labour from the rural to urban areas can be detected through the increasing urban populations. However, job seekers per se do not compose the migrating population but constitute the major proportion. According to Dolan and Sutherland 2004 migration of labour force within the country is evident with insecurities in asset holding, coupled with limited farming and income opportunities in original home areas providing the impetus for migration.

4.1. Developing Rural Non-farm Activities

Christiaensen, 2001, recognized the importance of non-farm activities as promising sources of incomes and as risk-mitigation strategies for the rural dwellers in ASALs of Kenya. Currently non-farm activities are gaining prominence in income contribution to the rural households. Table 40 indicates that the overall change in non-farm income was positive in all the agro-ecological regions. The government and private sector have been on the forefront of developing alternative sources of income to agriculture. This has been envisaged in the several police and strategy documents by the government.

The DFRD focused on developing the rural areas by decentralizing development services and job creation through rural industrialization. This was however not implemented owing to lack of adequate administrative and legal institutional framework, political support and capacity building for stakeholders participation in the development process (GoK, National Development Plan, 2002-2008). The Economic Recovery Strategy for Wealth and Employment Creation (ERS-WEC, 2003), a post liberalization government strategy to combat economic decline and unemployment, emphasizes the development of both agriculture and non-agricultural activities in rural areas. The Constituency Development Fund (CDF) initiative undertaken by government provides for the devolution of development finance to constituency level. This permits the financing of community development options.

4.2. Rural Depopulation

Rural population growth in Kenya has taken place amidst demographic transition, family planning programs, the devastating HIV/AIDS and other diseases such as malaria and tuberculosis. However, it is some specific age categories that have now become the majority in the rural areas. These are mainly the old and the very young. The working age population has migrated to more productive areas, urban areas and other countries or decimated by HIV/AIDS. Centralization of development programs in urban areas in Kenya especially industrial development has formed a pull factor for rural populations. Rural depopulation in this case is seen in terms of the migration of the agricultural labour force to other regions within or outside the country. The push factor in rural areas is the decline in agricultural job creation in the rural areas while the pull factor is the availability of job opportunities in other regions especially in urban areas. However, even though the job absorption in urban areas surpasses that of rural areas, it cannot accommodate the entire rural population outflow. An estimated one million people are employed in various economic sectors in urban areas with 22.9% in manufacturing, 36.4% in the communication and services and 17% in whole sale and retail trade sector (CBS, 2005). There has been rapid growth in these sectors in urban areas unlike in rural areas where only the informal sector has thrived. A mitigation factor has been the rapid increase in urban centres that has relieved the rural urban migration directed at the existing major urban centres

4.3. Migrations

4.3.1. Inter-regional/ Rural-urban Migrations

The movement from the rural areas by specific age categories to other rural areas, urban areas or other countries has been increasing. The push factors of this movement emanate from economic, political, environmental and institutional factors that affect their livelihoods. These pull factors are mainly regional resource imbalances and politically based development of social services. Despite the limited data on either rural-urban or international migration in Kenya, the causes of observed patterns of movement and population concentrations can be discerned.

The characteristics of those who migrate, the families and communities may be distinct from those who do not migrate. The destination of choice is pegged upon the returns to labour and the cost of

migration and settling. It is often assumed that educated people have a higher probability of migration than the less educated.

Human capital theory predicts that educated human capital migration is conditional on school outcome, information about foreign labour markets and migration costs that makes educated workers relatively more productive when outside their homes.

In Kenya the initial increasing growth trend in 1960s and early 1970s in rural-urban migration could be attributed to the existence of job opportunities and better social amenities in the urban areas. It is also plausible to assume that it is the elite who have migrated to the urban areas in search of better employment. This poses a labour force depolarization to the rural based agricultural sector and results into lower productivity consequently food and nutritional insecurity and plays a major role in perpetuating the vicious cycle of poverty.

4.3.2. International Migrations

The number of people migrating to other countries indicates increasing perception that foreign countries provide economic opportunities, best social amenities and infrastructure. International migration declined mainly due to economic growth and political stability until late 1980s. The rise in political instability in the early 1990s triggered migrations only to reverse in the mid 1990s after the political calm was restored. Those who migrate internationally are mainly the educated, the well endowed in assets and those with relations abroad.

While migration may take place internationally or regionally, the season and duration of migration may vary with the objectives of the individuals. However, it is observed that people who migrate do not sever their relationships with their rural relatives. This maintenance of relations has triggered the phenomenal growth in remittances from international sources.

CHAPTER 8 - RESHAPING RURAL ECONOMIES

1. Reconstructing a Rapidly Changing Agricultural and Rural Economy

The reconstruction of rapidly changing agricultural and rural economies as a result of liberalization and the diminishing rural urban population boundaries due to globalization demands a more integrated approach to development initiatives and poverty reduction objectives. The decline in agricultural inputs productivity puts the rural livelihoods in a precarious situation since it implies declining marginal returns. The result of this changing scenario has been reallocation of available land, labour and capital to other economic alternatives which seem to be more certain and sustainable within the changing economic situation. While this is geared towards increased incomes to the rural populations, results show that the overall net income has declined between 1997 and 2004, migration from rural areas has increased and that more non-farm income sources have been adopted by the rural population (Table 39).

Over the elapsed period of liberalization, the Kenyan government has recognized the need to reshape the rural economies which are critical sources of agro-industrial raw materials. Several strategies targeting alleviation of rural poverty and improvement of both agricultural and non-agricultural income sources have been formulated. The Poverty Reduction Strategy Paper 2001 (PRSP), the Economic Recovery Strategy for Wealth and Employment Creation, 2003 spells strategies for rural economy reconstruction focusing on sectors that have an impact on rural economy through industrialization, employment creation and decentralization of the social infrastructure and amenities. Other policy strategies include The Ministry of Livestock and Fisheries Development Strategic Plan 2003 –2007 Strategy for Revitalizing Agriculture, 2004 – 2014, The National Development Plan (NDP), 2002-2008, The Kenya Rural Development Strategy (KRDS), 2002-2017 and the Ministry of Agriculture Strategic Plan (2006-2010) which all seek to address the ailing agricultural sector.

While these are seemingly well planed strategies, their implementation demands strong political will and financial support. Reconstruction programs should not only focus on those who are disadvantaged but also maintaining and enhancing the trend of the positive effects of reforms. For instance, the downstream vertical integration by some horticultural export companies and dairy farms links rural farmers directly to local and international markets.

2. Manifestations of a New Configuration of Rural Economies

While it is important to note that different farming regions have experienced different growth rates, the policy changes adopted since early 1980s have had significant effects on rural economies. Aggregate analysis show that agricultural sector is becoming more flexible in some parts of the country as indicated by the changing land use patterns, intensity of land use in high potential areas, over cropping in marginal areas, decline in land holding sizes, shifts in cropping patterns and the relative decline or increase in productivity has elicited new challenges. While the government and private bodies are waking up to reverse the trend of rural economies, their programs have not taken effect to the household level. Currently observable rural manifestations are those triggered by economic reforms. Table 10 shows an off farm orientation as the significantly growing income source within the rural setting. Information provided in Table 40 supports this orientation by assessing the rates of change for various income sources. Regions which have had positive crop incomes are coastal lowlands and the marginal rain shadow while the other regions have experienced negative crop income changes. Livestock enterprise has in all the regions have been dwindling while off-farm registered positive growths in income in all. The increase in informal labour employment further as shown in Figure 17 and Table 42 further emphasizes on the ongoing re-orientation of the rural

economies currently. The central region is dominated by coffee production, while coffee has declined in income generation, enterprise mix has been witnessed and sometimes even uprooting of the coffee bushes to create space for other crops. The growth of dairy sector in the central in the recent past may be attributed to the reallocation of resources from coffee this dairy enterprise. This has taken place due to lack of farm based enterprise alternatives and strategies to absorb the exiting production resources such as labour force. Rural economies are basically agricultural since few non agricultural options exist. While it is important to address agriculture related issues to safeguard rural livelihoods, strategies formulated from such should not be devoid of linkages to non-agricultural issues.

Both non-agricultural and agricultural sources of income have been recognized as important in sustaining rural livelihoods however, their importance has been eroded over time by the uncertainties emerging from changing international scenes. In some cases the effects of liberalization have been manifested in income losses (declining revenue per hectare) prompting shifts in economic activities and reallocation of productive resources. For, instance, labour, capital and land under agricultural enterprises deemed uncertain under reforms have been abandoned in favour of more economically viable ones

- **Informal Activities**

The growth of Informal activities involving manufacturing, trade and services in rural areas has offset losses in agriculture. The net effect has been increased migration into high potential regions and over exploitation of marginal areas. This movement has denied marginal areas the required labour force and resources rendering the areas more prone to poverty through reduced productivity.

- **Land Use Patterns**

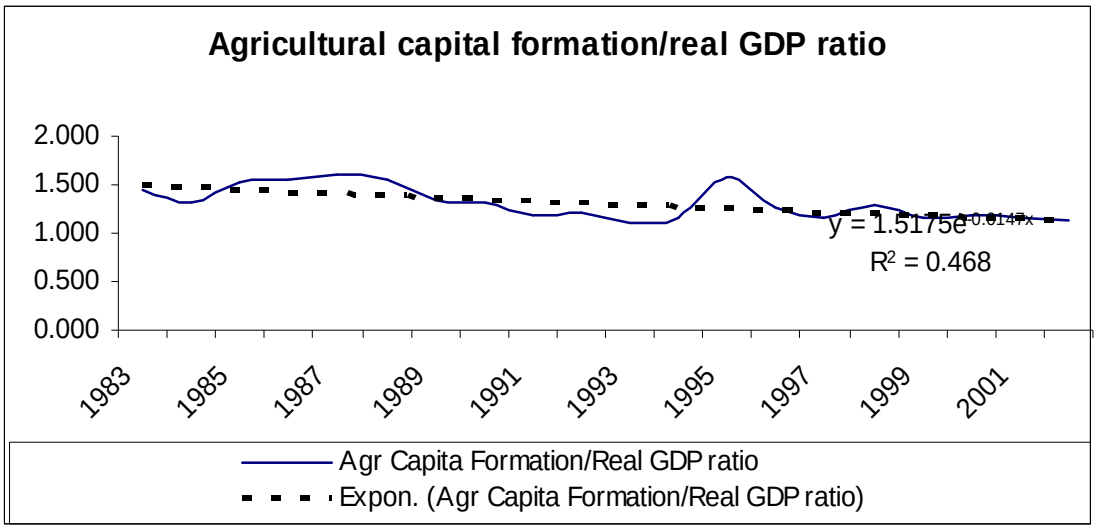
Land use pattern and systems have changed in response to changes in climatic conditions, real prices of agricultural commodities and certainty in production. for instance, it is evident that some pastoral communities in the ASALs are transforming changing from livestock to crops production. Rural households try to reallocate their productive resources to the most productive enterprises. Intensity of production has been noticed through increased yields in tea but declining trends in productivity in sectors such as coffee and sugar but increasing in tea. The response follows changes in input prices relative to the output prices of the agricultural commodity prices. The gross revenue also realized has been responding to the changing input and output real prices. Enterprise mix in coffee growing zones is evident as farmers seek to cater for the losses from coffee.

- **Governments' Agricultural Expenditure**

The trend of agricultural capital formation has been on decline since early 1980s (Figure 76).

Agricultural capital formation declined from 1.4% of the real GDP in 1983 to 1.14% in 2002. The decline was high during the liberalization period until in mid 1990s when a slight change occurred.

Figure 76: Government Agricultural Expenditure Trends



Source: CBS Economic Surveys. (Various Issues)

- **Decline Prices Paid to Farmers and Domestic Export Prices**

Producer prices for some agricultural prices such as coffee and sugarcane have decreased constraining the cost price squeeze as input prices increase

- **Productivity and Income**

Labour and land productivity shows slight increase; however specific sub sectors such as coffee and sugarcane have had declining trends in their land productivities. Tea has increased in land productivity. Decline in productivity may be the cause of shifts from crops observed between 2000 and 2004. Between 2000 and 2004 contribution by crops enterprise to total household income declined from 45% to 25%, livestock, 17% to 13% while salary and remittances increased from 20% to 34% and business 18% to 28% (Table 39) . Non-farm income sources such as salaries, businesses, and remittances have become more significant income sources while agricultural income from livestock and crops has dwindled. Revenue from crops and livestock has reduced significantly with sometimes farmers experiencing missing markets. The result of this scenario is abandoning of agricultural activities for the best alternative. For instance, decline in coffee prices has resulted decline in profit margins causing the farmers to neglect coffee bushes and in some areas they have been uprooted.

- **Input Use Intensity**

There is increased use of inputs in agricultural production. Higher levels of fertilizers are being used compared to labour and tractors as shown by their respective ratios in use intensities. Agriculture has been labour intensive as indicated by the labour-tractor use intensity ratio. This indicates that although agricultural employment has declined over the period, the sector still remains labour intensive which is economically viable for the huge labour available in the country.

- **Institutional and Organisational Structures**

Economic reforms have brought changes in institutional setting, organizational structures roles and responsibilities. The reorganization of institutions was to align them to the requirements of reforms make them more efficient in service delivery, allow for increased private sector participation and

competition. Such institutions which were restructured include KTDA, KPCU, CBK, Telecommunications industries, NCPB amongst others.

- **Migration**

Although in the recent past rural urban migration seems to have slowed down, migration from disadvantaged regions is still hypothesized to be taking place owing to loss of agricultural livelihood especially in the high potential areas. This is supported by the increase in remittances in these areas relative to the low potential areas.

- Destination and Impact of Funds Transferred by Migrants

Tracing the allocation of money transferred by migrants to specific economic activity is constrained by data. However, it is evident that migrants transfer funds to their rural relatives for consumption or investment. The proportion of remittances to the total income of households in various agro-regional zones is shown in appendix II. It is evident that remittances contribution to the total income increased in the high maize potential zone, central highlands and the western highlands between 2000 and 2004. It is apparent that with increasing efficiency in communication and money transfer networks, more funds and timely transfers have been made. This could be one of the factors leading to the recent population growth trends witnessed in the rural areas in which alternatives to agricultural activities such as informal trade and manufacturing are being observed. Communication is also transforming the rural areas by informing the populations on national developments in terms of economics and politics.

Conclusion

Vulnerable groups are distributed all over the country. However, liberalisation has had a major impact on agricultural producers and rural households. The most visible category is that group that has lost livelihoods due to economic changes and inability to adjust to a liberalised market environment. When agricultural productivity declines and returns per unit output reduce due to increased input prices, rural households are forced to turn to new agricultural enterprises or non-agricultural ventures. The unavailability of non-farm income sources, inflexibility of skills, cost and time of attaining new skills pose an impasse in the rural economy and impede transition from traditional enterprises.

Vulnerability increases with regional disparities in resource endowment, agricultural potential and public investment as illustrated by the correspondence of poverty levels to agro-ecological zones. Rural economies are reshaping in view of exit options provided in the form of internal and external migration, domestic and international remittances and the development of non-farm activities including increasing value adding processes. The new configuration of rural economies will be dictated by the growth of informal activities, land use pattern, government expenditure on agriculture and investment choices for domestic and international remittances

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ACRONYMS

ACP-EU	African Caribbean Pacific-European Union
AFC	Agricultural Finance Corporation
ATC	Agreement on Textiles and Clothing
ATI	African Trade Insurance Agency
BAT	British American Tobacco
BFI	Non-Bank Financial Institutions
CBK	Coffee Board of Kenya
CBS	Central Bureau of Statistics
CDF	Constituency Development Fund
CIF	Cost Insurance and Freight
CMA	Capital Markets Authority
COMESA	Common Market for Eastern and Southern Africa
EAP	Economically Active Population
EPA	Economic Partnership Agreements
ERS-WEC	Economic Recovery Strategy for Wealth and Employment Creation
EU	European Union
FDI	Foreign Direct Investment
FOREX-Cs	Foreign Exchange Bearer Certificates
GATS	General Agreement on Trade and Services
GATT	General Agreement on Trade and Tariffs

GDP	Gross Domestic Product
GoK	Government of Kenya
IGAD	Intergovernmental Authority for Development
KARI	Kenya Agricultural Research Institute
KDB	Kenya Dairy Board
KEPHIS	Kenya Plant Health Inspectorate Service
KFA	Kenya Farmers Association
KTDA	Kenya Tea Development Agency
MFIs	Micro Finance Institutions
MOCO	Mumias Outgrowers Company
MSE	Micro Small Enterprises
MUB	Manufacture Under Bond
NCPB	National Cereals and Produce Board
PTA	Preferential Trade Area
SAL	Structural Adjustment Loan
SAP	Structural Adjustment Program
SPS	Sanitary and Phytosanitary
SRA	Strategy for Revitalizing Agriculture
TRIPS	Trade- Related Intellectual Property Rights
WB	World Bank
WDI	World Development Indicator
WMS	Monitoring Surveys

WTO	World Trade Organization

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ANNEXES

ANNEX I

Table 43: Distribution of households' members by sex and poverty (%)

Region	poor			Non-Poor		
	No.	Male %	Female %	No.	Male %	Female %
Central rural	1122485	48.4	51.6	2464684	48.4	51.6
Kiambu	280632	49.7	50.3	840901	48.4	51.6
Kirinyaga	156168	50.7	49.3	288218	48.9	51.1
Murang'a	381363	47.1	52.9	607052	46.5	53.5
Nyandarua	100867	46.0	54.0	273334	51.4	48.6
Nyeri	203455	48.4	51.6	455179	49.0	51.0
Coast rural	860701	46.9	53.1	535947	53.2	46.8
Kilifi	422397	45.9	54.1	220742	58.8	41.2
Kwale	264477	48.1	51.9	173229	50.8	49.2
Lamu	25725	55.0	45.0	39723	50.3	49.7
Taita Taveta	132526	45.9	54.1	70215	45.5	54.5
Tana River	15576	50.7	49.3	32038	48.9	51.1
Eastern Rural	2260472	48.9	51.5	1631738	48.8	51.2
Mbeere	76058	51.6	48.4	80659	51.6	48.4
Embu	105419	50.0	50.0	82420	52.9	47.1
Kitui	462117	49.0	51.0	254281	48.1	51.9
Machakos	523045	45.1	54.9	301121	48.7	51.3
Meru	134150	49.9	50.1	203847	49.7	50.3
Makueni	450261	51.8	48.2	161595	50.9	49.1
Tharaka Nithi	168566	45.8	54.2	139091	50.7	49.3
Nyambene	340856	51.0	49.0	408723	46.2	53.8
Rift Valley Rural	2656993	49.0	51.0	2673712	50.9	49.1
Kajiado	77396	48.3	51.7	203183	53.7	46.3
Kericho	312299	49.4	50.6	287701	49.0	51.0
Laikipia	90523	46.3	53.7	180748	52.4	47.6
Nakuru	388617	48.6	51.4	478946	51.0	49.0
Nandi	340556	51.9	48.1	185561	52.7	47.3
Narok	138052	44.8	55.2	125999	54.6	45.4
Bomet	342807	51.6	48.4	205436	51.7	48.3
Transmara	135831	46.2	53.8	101974	38.0	62.0
Baringo	133866	49.8	50.2	234861	50.6	49.4
Elgeyo Marakwet	124742	49.0	51.0	143876	50.6	49.4
Trans Nzoia	233219	46.0	54.0	193161	49.0	51.0
Uasin Gishu	181674	48.2	51.8	254633	52.6	47.4
West Pokot	157412	50.4	49.6	77631	50.4	49.6
Western rural	1690036	49.2	50.8	1192420	47.0	53.0
Bungoma	447484	49.7	50.3	374909	49.7	50.3
Busia	280364	46.9	53.1	142365	45.3	54.7
Kakamega	584760	51.8	48.2	446301	47.8	52.2
vihiga	377428	46.5	53.5	228845	42.2	57.8
Nyanza Rural	2630855	47.4	52.6	1580697	45.6	54.4
Kisii	403082	49.1	50.9	309261	50.3	49.7
Kisumu	340659	46.8	53.2	186963	48.5	51.5
Siaya	391839	46.1	53.9	289401	41.4	58.6
Homa Bay	379742	47.8	52.2	114747	42.5	57.5
Migori	498054	45.9	54.1	372541	43.9	56.1
Nyamira	617479	48.4	51.6	307785	46.0	54.0
Total Rural	11221542	48.4	51.6	10079199	48.8	51.2

Source: Second Report on Poverty in Kenya

ANNEX II

Table 44: Panel Mean Rural household income in KES. by Source, 1997, 2000 and 2004

	Net value crop production			Net value livestock income			Net value of off-farm income			Total net hh income		
Agro-regional Zones	**1997**	**2000**	**2004**	**1997**	**2000**	**2004**	**1997**	**2000**	**2004**	**1997**	**2000**	**2004**
Coastal Lowlands	15,556	35,130	33,083	24,532	4,119	12,920	105,547	68,613	122,823	145,635	107,862	168,826
Eastern Lowlands	50,424	51,703	37,316	42,595	17,843	15,368	54,229	68,535	132,184	147,248	138,081	184,868
Western Lowlands	25,906	15,735	23,476	36,092	8,283	11,645	35,234	24,187	52,818	97,232	48,205	87,939
Western Transitional	81,915	100,055	54,100	44,647	13,365	19,411	30,722	46,375	61,219	157,284	159,795	134,730
High Potential Maize Zone	156,997	63,950	73,936	92,084	44,462	48,643	54,728	63,639	101,944	303,809	172,051	224,523
Western Highlands	42,243	63,661	40,820	55,489	17,514	22,060	25,225	32,802	48,214	122,957	113,977	111,094
Central Highlands	127,240	94,008	67,083	91,820	25,524	28,850	53,893	54,188	97,729	272,953	173,720	193,662
Marginal Rain Shadow	26,167	16,136	37,220	68,545	24,372	26,767	40,438	65,100	112,610	135,150	105,608	176,597
Total	91,340	63,474	53,318	66,513	24,279	27,905	48,811	52,853	89,855	206,664	140,606	171,078

Table 45: Percentage changes in the composition of rural household incomes by sources, 1997/2000 and 2000/2004

	Crop		Livestock		Off-farm		Total	
Agro-regional Zones	**'97/00**	**'00/04**	**'97/00**	**'00/04**	**'97/00**	**'00/04**	**'97/00**	**'00/04**
Coastal Lowlands	56	-6	-496	68	-54	44	-24	35
Eastern Lowlands	2	-39	-139	-16	21	48	0	22
Western Lowlands	-65	33	-336	29	-46	54	-68	43
Western Transitional	18	-85	-234	31	34	24	9	-31
High Potential Maize Zone	-145	14	-107	9	14	38	-60	15
Western Highlands	34	-56	-217	21	23	32	11	-6
Central Highlands	-35	-40	-260	12	1	45	-40	1
Marginal Rain Shadow	-62	57	-181	9	38	42	-6	37
Total	-44	-19	-174	13	8	41	-31	11

Table 46: Proportion of gross income by various sources and regions

	Crop income			livestock			informal income			salary			remittances	
	1997	2000	2004	1997	2000	2004	1997	2000	2004	1997	2000	2004	2000	2004
Coastal Lowlands	16	35	20	8	4	4	38	38	45	38	17	27	6	4
Eastern Lowlands	40	42	30	22	12	12	12	20	25	26	17	27	10	6
Western Lowlands	43	39	36	19	20	14	18	19	22	21	12	18	9	9
Western Transitional	57	64	43	25	8	26	9	16	15	9	10	12	3	3
High Potential Maize Zone	59	33	48	23	32	20	8	19	15	10	15	14	2	4
Western Highlands	50	53	44	26	20	18	8	8	11	16	16	19	4	7
Central Highlands	54	55	49	25	15	18	7	13	13	14	13	15	4	5
Marginal Rain Shadow	2	18	34	34	23	22	41	28	13	27	26	27	5	4

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